

# aTOMos v3.6.1.11

## Exact operator seeds and literal state evolution

One-bit words

Exact expressions before numerical output

Madgwick IMU / MARG · orbital dynamics

ASA/NA and JK · UGTS · receiver equations

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Preserved parent 3.6.1.10 ORBIT-SEED-R1

Method Logical derivation and critical review

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The authoritative state consists of exact operator expressions and their input lineage. A finite-precision output does not replace that state. This subversion specifies the mathematics and records strategy improvements; it does not claim a newly implemented engine or new experimental accuracy.

Editable formalization with the complete inherited equation corpus. No new algorithm tests, simulations or benchmarks. Prepared with AI assistance.

## Release method and status

The user requests full formalization and improvement of the strategy through logical reasoning, without testing the proposed system. Accordingly, new claims in this volume are definitions and conditional mathematical deductions. Document rendering and source inspection establish readable delivery, not empirical algorithm performance. No new numerical experiment is reported.

The scope includes the exact operator representation; original Madgwick attitude recurrences; orbital forces, frames and time evolution; station and body geometry; literal ASA/NA and synchronous JK; coupled mechanics; both original UGTS layouts; full calendar/winding; and the retained satellite-navigation receiver equations. The exact arithmetic definition is relative to its supplied inputs and constants. It cannot restore information absent from those inputs.

The complete 3.6.1.10 equation corpus follows the new formalization as dated reference material. Its existing measured results are inherited. They do not test or validate EXACT-OPERATOR-SEED-R1. The existing CPU/CUDA runtime remains available in its preserved parent release; no R11 executable is claimed here.

**Evidence vocabulary.** A definition fixes the profile. A proposition gives a conclusion under stated hypotheses. A logical review is a readable argument, not a proof-assistant certificate. An implementation obligation remains a future engineering requirement. A physical assumption does not become measured truth because the arithmetic is exact.

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## 1 Authoritative operators and the revised strategy

The requirement is to preserve the mathematical construction before repeatedly projecting it to finite-precision numerical states. The representation contains the operators, exact input literals, binding of their arguments and roots naming the current state. The bits in a packed word are literal Boolean storage lanes. Word width limits a storage unit, not the mathematical precision of a value represented by several units or by an expression.

Write a seed as

$$\mathcal{S} = (\mathcal{V}, \mathcal{T}, \mathcal{C}, \mathcal{G}_0, \mathcal{R}_0, \mathcal{I}, \mathcal{P}),$$

where  $\mathcal{V}$  fixes the version and operator meanings,  $\mathcal{T}$  fixes types, units and frames,  $\mathcal{C}$  contains exact constants,  $\mathcal{G}_0$  is an operator graph,  $\mathcal{R}_0$  names its state roots,  $\mathcal{I}$  fixes the input schema, and  $\mathcal{P}$  records assumptions and source provenance. For the measured input event  $u_n$  and exact elapsed time  $h_n$ , the new root is

$$\mathcal{R}_{n+1} = \text{Build}(\mathcal{V}, \mathcal{G}_n, \mathcal{R}_n, u_n, h_n).$$

Its denotation is the selected recurrence  $F$ :  $\llbracket \mathcal{R}_{n+1} \rrbracket = F(\llbracket \mathcal{R}_n \rrbracket, u_n, h_n)$  on the declared domain. Build creates references and applies justified exact transformations; it does not define the next state as the rounded display of the previous state.

The word state  $q_{JK} \in \{0, 1\}^{32}$ , attitude quaternion  $Q_{\text{att}}$ , orbital state  $(r, v)$  and absolute time have distinct types. Equal spelling of a letter in earlier sources does not merge their meanings. A state feedback edge crosses an explicit time delay; an instantaneous circular definition is a separate fixed-point problem and requires its own existence/selection rule.

### 1.1 Exact relative to which input?

A recorded integer sensor count is exact as a count. A stated decimal calibration ratio  $a/10^k$  has that rational denotation. An imported finite IEEE value has its exact dyadic denotation and retains a raw-bit identity if requested. These are different imports. Converting a previously rounded input to an arbitrary-precision number does not recover the original physical quantity. A symbol such as  $\pi$  has a declared mathematical meaning, while a fitted physical coefficient retains its measurement/model provenance.

Units and frames participate in typing. An angle supplied in degrees requires the exact factor  $\pi/180$  when mapped to radians; calling a finite binary approximation to that factor exact would change the source contract. Noise, calibration uncertainty and missing inputs are physical information, not lost bits that can be restored by serialization.

### 1.2 The output projection never becomes an implicit checkpoint

For a defined real-valued expression  $E$ , an output request can seek a rational enclosure  $[a, b]$  with  $\llbracket E \rrbracket \in [a, b]$  and  $b - a \leq \epsilon$ . A finite decimal or IEEE display can be derived from it with a stated rounding rule. Exact midpoint decisions may need additional reasoning; general transcendental comparisons are not promised to terminate. The retained root remains  $E$ .

**Proposition 1.1** (No recurrent arithmetic roundoff in the denotational state). *Suppose every imported literal has its declared exact denotation, every demanded operator is defined, and every rewrite preserves denotation and definedness. Then after any finite number  $n$  of input events, the retained state root denotes exactly the  $n$ -fold selected recurrence on those inputs, independent of the precision of any separately requested displays.*

*Proof.* The initial root has the specified denotation. If the claim holds for event  $n$ , the construction rule applies  $F$  to that denotation and the exact next input. Denotation-preserving rewrites leave

this result unchanged. Display projections are not arguments of the recurrence. Induction proves the finite-prefix claim. The proposition is about the specification; implementing its premises is a separate obligation.  $\square$

A controller may deliberately consume a rounded actuator command or a measured response. Such a value then enters through a declared input port. It cannot be silently treated as an exact internal state checkpoint. Checkpoints must preserve the reachable expression graph, equivalent exact sufficient state, or enough original inputs and operator definitions to reconstruct it.

### 1.3 Useful simplifications justified before execution

For  $s = \sum_i v_i^2 > 0$  and the positive square root, let  $q_i = v_i/\sqrt{s}$ . Then

$$\sum_i q_i^2 = \frac{\sum_i v_i^2}{(\sqrt{s})^2} = \frac{s}{s} = 1.$$

This identity permits an exact unit-norm certificate without numerically summing rounded quaternion components. It does not permit normalization of the zero vector. Likewise  $x/x = 1$  requires  $x \neq 0$ , and  $\sqrt{x^2} = |x|$ ; replacing the last expression by  $x$  requires an additional nonnegative-sign premise.

For a nonzero quaternion  $p = (w, u)$ , define  $s = w^2 + u \cdot u$ . Its rotation of a vector  $v$  can be written without first computing a normalization square root:

$$\mathcal{R}_p(v) = v + \frac{2}{s} [w(u \times v) + u \times (u \times v)].$$

**Proposition 1.2** (Homogeneous rotation readout). *For  $p \neq 0$ , this expression equals the vector part of  $p(0, v)p^*/s$ , and  $\mathcal{R}_{\lambda p}(v) = \mathcal{R}_p(v)$  for every real  $\lambda \neq 0$ .*

*Proof.* Expanding the Hamilton product gives  $((w^2 - u \cdot u)v + 2u(u \cdot v) + 2w(u \times v))/s$ . The vector triple-product identity converts this to the displayed form. Multiplying  $p$  by  $\lambda$  multiplies both the numerator correction and  $s$  by  $\lambda^2$ , which cancels because  $\lambda \neq 0$ .  $\square$

The adopted improvement removes a redundant root from a rotation *readout*. It does not delete normalization from the Madgwick recurrence: its source objective and Jacobian assume the declared unit-quaternion state, and its gain sets a correction magnitude. A projective recurrence would need a separate derivation proving the same dynamics.

Shared operators retain their types, scopes, assumptions and registry version. Two nodes with identical complete records can share one stored record. A digest is an index for finding a candidate record; equality of the records, rather than an assumption that hashes never collide, establishes exact identity.

### 1.4 Definedness and exact branch decisions

Algebraic expressions admit exact sign/equality procedures in principle, although their cost can grow sharply. General expressions involving transcendental functions, floor boundaries and implicit differential-equation solutions do not inherit that decision guarantee. A comparison can remain a symbolic predicate with both alternatives retained, or produce a declared unresolved result. It must not silently select a branch from an insufficient finite approximation.

Rewriting must also preserve which expressions are demanded. In  $\text{if}(P, 1/x, 0)$ , the inverse need only be defined when  $P$  is true. An eager transformation that evaluates  $1/x$  before deciding  $P$  changes the domain at  $x = 0$  when  $P$  is false. Sharing a lazy node is valid; forcing its evaluation outside its original demand condition is not justified by algebraic textual similarity alone.

## 1.5 Exact recurrence and exact physical flow are separate specifications

The current orbit implementation defines finite RK4 updates. Its exact lift retains the same stage formula with exact operators. The continuous model instead defines an initial-value problem  $\dot{x} = f(t, x)$  and, under appropriate local existence and uniqueness assumptions, a flow  $\Phi(t, t_0, x_0)$ . A seed can name that exact problem compactly. Naming  $\Phi$  does not supply a closed-form solver or a constant-cost evaluation algorithm.

For an exact model trajectory  $x_M$ , exact discrete trajectory  $x_D$ , displayed trajectory  $\tilde{x}_D$  and physical state  $x_*$ , the identity

$$x_* - \tilde{x}_D = (x_* - x_M) + (x_M - x_D) + (x_D - \tilde{x}_D)$$

separates physical/model discrepancy, discretization discrepancy and output evaluation discrepancy. Exact internal arithmetic addresses the intermediate roundoff term that an inexact computation of  $x_D$  would additionally introduce. No independence or statistical variance assumption is used in this identity. None of the other terms becomes zero by changing the packing representation.

For sufficiently smooth dynamics, fixed-step forward Euler and RK4 have their usual first- and fourth-order global truncation behavior under the regularity and stability hypotheses of the corresponding convergence theorems. Discontinuous shadow switches, impulses and unmodelled input changes require an explicit piecewise/event treatment. Exact arithmetic cannot supply absent event data.

## 1.6 Finite representation, growing work and causal information

**Proposition 1.3** (A finite-prefix graph bound). *If each input event adds at most  $K$  fixed-template nodes and existing nodes are shared by reference, a prefix of  $n$  events has at most  $N_0 + nK$  nodes before deleting unreachable nodes. This does not bound the total number of literal bits or the cost of evaluating the denoted expression.*

*Proof.* Each event increases the count by at most  $K$ ; induction gives the bound. Arbitrary-size numerators, denominators, references, retained input records and algebraic degrees are not bounded by the count of nodes. Their sizes and exact evaluation cost require separate accounting.  $\square$

Algebraic extensions formed by  $r$  successive square-root adjunctions have degree at most  $2^r$  over the initial rational field, with reductions possible when roots are dependent. This explains why a small expression graph does not imply a small expanded algebraic representation. Keep shared operator structure and delay expansion until a demanded conclusion needs it.

There are  $2^n$  possible unrestricted  $n$ -bit input histories and only  $2^B$  fixed  $B$ -bit containers. For  $n > B$ , no injective encoding can retain all such histories in those containers. A model may instead retain an exact sufficient state or an explicitly supplied deterministic input generator. Neither permits reconstruction of an arbitrary unknown future measurement stream from a fixed initial seed.

## 1.7 A strategy that improves through deductions

The strategy revision record contains the proposed change, its hypotheses, a proof or counterexample argument, and the resulting decision. Accepted changes must preserve denotation, definedness, units, frames, input chronology and source variant. A profile that intentionally changes  $F$  receives a new semantic version and is described as a changed model rather than an equivalent optimization.

The reasoned improvements adopted here are:

1. Keep operator expressions authoritative; do not emulate IEEE rounding and relabel that arithmetic exact.
2. Extend rational arithmetic with named roots and transcendental terms where the source equations require them; do not replace roots by rational literals.
3. Share typed subexpressions and prove local identities before requesting numerical projections. Preserve lazy branch demand.
4. Use homogeneous rotation readout to remove a provably redundant root, while retaining the source attitude recurrence's normalization semantics.
5. Pin the original Madgwick equation variant and magnetic-reference update timing; appendix code order is a distinct source choice.
6. Treat a fitted nine-entry frame polynomial as a matrix approximation unless orthogonality is independently a mathematical consequence of its definition.
7. Keep discrete recurrence, continuous flow, measured inputs and output projections as explicit, different objects.

These are changes to the formal strategy justified by deductions in this volume. They are not measured speedups, generated empirical results or claims that a particular implementation already satisfies every premise.

Exact arithmetic and lazy expression representation are established mathematical approaches; see [CGAL's exact computation paradigm](#) and its [exact expression number types](#). R11 specifies the required denotations and proof obligations for this kernel. It does not infer a universal physical accuracy bound or a complete transcendental decision procedure from those sources.

## 2 Exact operator seeds: a mathematical contract

This chapter defines the proposed XOP-R1 operator-seed profile for release 3.6.1.11. Definitions specify a representation and its meaning; propositions establish conditional mathematical consequences. An implementation must still discharge the stated parsing, arithmetic and proof-checking obligations. No new runtime, machine-checked proof, test, simulation or measured improvement is asserted. The implemented R10 orbital seed and its binary64 arithmetic keep their existing format and interpretation.

### 2.1 Bits, values and exact expressions are distinct types

Write  $\text{Bits}(w) = \{0, 1\}^w$  for a finite raw bit string. Its unsigned interpretation is an explicitly requested map

$$\text{uint}(b) = \sum_{i=0}^{w-1} b_i 2^i.$$

A binary64 payload belongs to  $\text{Bits}(64)$  until an explicit conversion is applied. It does not become an arbitrary-precision real by being packed in one-bit planes. A finite binary64 value can be converted to the exact dyadic rational denoted by that value; this preserves its rounded input value, not an unknown physical quantity. Signed zeros are distinct raw payloads but the same rational zero. Infinity and NaN have no value in the exact real scalar types.

The scalar hierarchy is

$$\mathbb{Z} \hookrightarrow \mathbb{Q} \hookrightarrow \mathbb{A} \hookrightarrow \text{RealTerm}, \quad \mathbb{A} = \overline{\mathbb{Q}} \cap \mathbb{R}.$$

Here  $\text{RealTerm}$  is finite syntax for a possibly partial real expression, not a claim that every real number has a finite description. Coercions preserve denotation and are explicit in the typed graph.

Booleans, finite words, tuples, vectors, matrices, functions and streams have separate type constructors. Dimensioned scalars also carry a seven-component rational SI exponent vector; addition requires equal dimensions, multiplication adds dimensions, and transcendental functions require dimensionless arguments. Angle conventions and coordinate frames are declared metadata, not inferred from an untyped number.

An integer is  $(s, a_0, \dots, a_{k-1})$  with  $s \in \{0, 1\}$  and

$$z = (-1)^s \sum_{i=0}^{k-1} a_i \beta^i, \quad \beta = 2^{64}, \quad 0 \leq a_i < \beta.$$

Zero has  $s = 0, k = 0$ ; otherwise  $a_{k-1} \neq 0$ . There is no format-level bound on  $k$ . A resource-limited implementation must return an explicit resource result, never silently discard high limbs. A rational is  $(p, q)$  with  $q > 0$ ,  $\gcd(|p|, q) = 1$ ; its zero is  $(0, 1)$ .

An algebraic real has a primitive irreducible polynomial  $P \in \mathbb{Z}[X]$  with positive leading coefficient and an index selecting one of its distinct real roots in increasing order. This fixes its value and canonical mathematical identity. A rational isolating interval is a separate witness, not part of that identity. Checking irreducibility, isolating roots and obtaining a canonical minimal polynomial are explicit exact-algebra obligations. A general operator term has canonical *syntax*; different syntax may denote the same real.

## 2.2 Exact arithmetic on finite machine words

The literal full adder takes three bits  $a, b, c \in \{0, 1\}$  and returns

$$s = a \oplus b \oplus c, \quad c' = (a \wedge b) \vee (c \wedge (a \oplus b)), \quad a + b + c = s + 2c'.$$

For subtraction with incoming borrow  $d$ , its bit and outgoing borrow are

$$r = a \oplus b \oplus d, \quad d' = (\neg a \wedge b) \vee (d \wedge \neg(a \oplus b)), \quad a - b - d = r - 2d'.$$

Feed the outgoing carry or borrow of position  $i$  to position  $i + 1$ , starting with zero. Signed values are handled by separate sign and magnitude, not by discarding an infinite sign extension. A product begins with partial bits  $p_{i,j} = a_i \wedge b_j$  in position  $i + j$ ; finitely many full-adder chains sum these shifted rows, retaining the final carry. Exact magnitude comparison scans from the most significant padded bit, with equality flag  $e = 1$  and greater flag  $g = 0$ , updating simultaneously

$$g' = g \vee (e \wedge a \wedge \neg b), \quad e' = e \wedge \neg(a \oplus b).$$

This comparison supplies the branch in division; no approximate sign oracle is needed for integers. The bit equations also establish the limb implementation:  $a + b = (a \oplus b) + 2(a \wedge b)$  and the two carry terms are disjoint, giving the full-adder identity by substitution. Summing the bit identities with powers of two telescopes all internal carries and borrows.

Limb storage does not force modular arithmetic. For nonnegative addition, pad the shorter operand with zero limbs, set  $c_0 = 0$ , and define

$$u_i = a_i + b_i + c_i, \quad r_i = u_i - \beta \lfloor u_i / \beta \rfloor, \quad c_{i+1} = \lfloor u_i / \beta \rfloor.$$

Append the final nonzero carry. A native implementation may realize  $u_i$  with two-word operations or bitwise carry logic; it cannot assume that  $u_i$  fits one limb. Subtraction with  $A \geq B$  uses  $c_0 = 0$  and

$$u_i = a_i - b_i - c_i, \quad c_{i+1} = \lfloor u_i < 0 \rfloor, \quad r_i = u_i + \beta c_{i+1}.$$

Signed addition selects addition or magnitude subtraction according to signs and exact magnitude order, then normalizes zero.

Multiplication is finite convolution followed by carrying, starting with  $c_0 = 0$ :

$$v_j = c_j + \sum_{i+\ell=j} a_i b_\ell, \quad r_j = v_j \bmod \beta, \quad c_{j+1} = \lfloor v_j / \beta \rfloor.$$

Continue carrying until it vanishes. The accumulator  $v_j$  is itself an exact multiword quantity; using one overflowing hardware word would violate the rule. For division by  $D > 0$ , process dividend bits from most to least significant:

$$Q_0 = R_0 = 0, \quad u_i = 2R_i + b_i, \quad e_i = \lfloor u_i / D \rfloor, \quad R_{i+1} = u_i - e_i D, \quad Q_{i+1} = 2Q_i + e_i.$$

The dividend prefix processed after step  $i$  is  $DQ_i + R_i$ , with  $0 \leq R_i < D$ . Signed Euclidean division is the unique  $z = dq + r$  with  $d \neq 0$  and  $0 \leq r < |d|$ ; sign conversion must preserve this convention, including negative dividends. Integer powers with nonnegative exponent use finite repeated multiplication; negative powers produce a rational only when the base is nonzero.

**Proposition 2.1** (No mathematical fixed-width overflow). *For finite integer operands, the preceding finite procedures return their exact sum, difference, product, or Euclidean quotient and remainder, provided every intermediate carry is retained and sufficient storage is available.*

*Proof.* For addition and subtraction, multiply each defining equation by  $\beta^i$  and sum: the carry terms telescope, leaving the represented operand relation. Convolution represents the product by distributivity, and carrying preserves each coefficient's value. For division, the prefix invariant follows by substitution, while  $0 \leq R_i < D$  implies  $0 \leq 2R_i + b_i < 2D$  and hence preserves the remainder bound. These arguments are conditional on retaining every term; they establish no bound on time or allocated memory.  $\square$

### 2.3 Positive roots and elementary functions

$\text{SQRT}(x)$  denotes the unique  $y \geq 0$  satisfying  $y^2 = x$ , with domain  $x \geq 0$ .  $\text{ROOT}(n, x)$  requires integer  $n \geq 1$ : for even  $n$ , it is the nonnegative root and requires  $x \geq 0$ ; for odd  $n$ , it is the unique real root. An arbitrary polynomial root uses the algebraic root index above, or an explicit existence-and-uniqueness predicate for a real-term root. No unnamed branch is selected. Thus  $\sqrt{x^2} = |x|$  for real  $x$ , and cancellation to  $x$  requires  $x \geq 0$ . Complex branches are outside this real profile.

The real elementary declarations have ordinary mathematical denotations:

$$\begin{aligned} \exp x &= \sum_{n=0}^{\infty} \frac{x^n}{n!}, & \sin x &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \\ \cos x &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}, & \log x &= \int_1^x \frac{du}{u} \quad (x > 0), \\ \text{atan } x &= \int_0^x \frac{du}{1+u^2}, & \pi &= 4 \text{atan}(1). \end{aligned}$$

These definitions permit finite operator terms for infinite-process values. They are not a request to store an infinite series or an assertion that a finite limb evaluator can return all digits. A certified finite approximation needs separate truncation and rounding bounds. A raw approximate evaluation cannot serve as an exact equality proof.

ATAN2( $y, x$ ) is undefined at  $(0, 0)$  and uses range  $(-\pi, \pi]$ :

$$\text{atan2}(y, x) = \begin{cases} \text{atan}(y/x), & x > 0, \\ \text{atan}(y/x) + \pi, & x < 0, y \geq 0, \\ \text{atan}(y/x) - \pi, & x < 0, y < 0, \\ \pi/2, & x = 0, y > 0, \\ -\pi/2, & x = 0, y < 0. \end{cases}$$

The literal source OTAN2 remains a distinct named template with its own equation and domain; it is never an alias silently replaced by this operator.  $\tan x$  requires  $\cos x \neq 0$ ; inverse sine and cosine use ranges  $[-\pi/2, \pi/2]$  and  $[0, \pi]$  and domain  $[-1, 1]$ . Floor is the unique integer  $n$  with  $n \leq x < n + 1$ ; wrapping is  $x - P \lfloor x/P \rfloor$  for  $P > 0$ , with the integer winding retained separately.

## 2.4 Decisions, partiality and honest result states

Exact rational order is decided by integer cross multiplication with positive denominators. Algebraic order and equality can be decided by exact polynomial arithmetic, common-root analysis and real-root isolation. For unequal algebraic roots, sufficiently refined disjoint rational isolating intervals determine the order; common roots are handled algebraically instead of waiting forever for interval separation. Supplying those algorithms is an implementation obligation.

There is no general equality or sign oracle for unrestricted transcendental terms. COMPARE returns one of LT, EQ, GT, or UNKNOWN, each decided result carrying a valid justification. FLOOR, branch selection and nonzero checks may likewise remain unresolved. Outcomes distinguish a value or a retained term, an unmet domain obligation, an unknown decision, and exhausted resources. These are not numeric zero. IF is lazy: only the selected branch must be defined once its predicate is proved; an undecided predicate preserves the conditional term.

An exact finite decimal input denotes a rational with its declared decimal scale. A measurement with error is a value plus an explicit uncertainty model; giving its central value an exact rational spelling removes no measurement error. This distinction applies equally to station coordinates, timing and force coefficients in an orbital seed.

## 2.5 Scoped operator graphs and named dependencies

An operator seed contains finite typed records. A node is

(opcode, result type, ordered operand references, opcode payload).

The fixed XOP-R1 catalog in `formal/OPERATOR_CATALOG.md` supplies each opcode's arity, type relation, denotation, domain and payload fields. An unknown opcode is rejected; a user-defined formula is a named template built from catalog operators, with an explicit interface and body. There is no implicit "evaluate arbitrary text" opcode or undeclared exact solver.

Each scope has an ordered node table and ordered exported references. Local node operands refer only to earlier nodes. Lambda, finite sum/product, integral, derivative and solution declarations carry nested scopes. Bound variables use de Bruijn indices: zero denotes the nearest binder, and larger indices count outward through lexical binders. The type environment fixes every index. Substitution shifts indices when passing a binder and is capture-avoiding. Nested scopes cannot reference an outer local node by an accidental numeric ID; captured values enter through explicit binders. Names are display metadata; renaming them does not change variable binding.

References have distinct tags for a local node, bound variable, old state slot, current sample slot, or dependency export. A template application additionally identifies its template and supplies all parameters in declared order. Dependency records carry the complete canonical seed/template

bytes in the self-contained form. A digest is a lookup aid; acceptance requires the expected complete bytes and interface. A dependency-only manifest is explicitly an open package until all required bytes are supplied. Dependency and template-call graphs are finite and acyclic in this profile. Finite summation and delayed state do not imply unrestricted recursive template calls.

Sample declarations identify their type, absolute time convention, provenance and binding mode. An inline sample is an indexed literal accompanied by its absolute timestamp; an external sample is a declared input requirement. `SAMPLE` selects the value bound to the current logical step, with its timestamp retained. Missing data is unresolved input, not a guessed next sample. A modular phase, spatial key, hash or template identifier does not supply the state, timestamp or sample bytes it refers to.

## 2.6 Canonical variable-length wire contract

The canonical semantic stream begins with the eight ASCII octets `XOPSEED1`, followed by profile numbers (1, 0) and the ordered tables specified in the catalog. Every natural number uses minimal unsigned base-128 encoding: digit payloads occupy seven low bits, the high bit marks continuation, the least significant digit comes first, and the final high bit is zero. Zero is one zero octet; a multi-octet number cannot have a zero final payload. Thus counts and references are not limited to a fixed machine word.

A byte string has a natural-number length followed by exactly that many octets; a sequence has an element count followed by exactly that many schema-defined elements. An integer has a sign octet, a limb count and that many little-endian 64-bit limbs. Rational and algebraic literals use the canonical value forms already defined. Strings used as identifiers are restricted ASCII byte strings; free-form provenance is opaque bytes and does not participate in name lookup. Types, references and opcode payloads are encoded by their declared tagged schemas. Nonminimal counts, redundant leading limbs, trailing bytes, nonzero padding, invalid references and payload fields outside the selected schema have no canonical decoding.

Canonical here means one encoding for the same ordered, scoped syntactic record. It does not mean that all mathematically equal expressions, all sharing patterns, or all valid topological node orders receive one universal byte string. Structural sharing is permitted only for identical typed syntax under the same bindings and immutable dependency identities. The strategy history records any rewrite that changes the syntax while preserving its meaning.

**Proposition 2.2** (Unique literal and record decoding). *Given the catalog, a finite well-formed canonical stream has one decoded record and its canonical re-encoding returns the original stream.*

*Proof.* The continuation terminator uniquely delimits each minimal natural-number encoding. Lengths and fixed-width limbs uniquely delimit strings and literals. The record schema fixes field order and the selected tag fixes each variant. Induction over finite sequences and nested scopes therefore determines every field and its boundary. Rejecting alternative zero forms, redundant lengths and trailing bytes makes re-encoding identical. This is a structural proof of the specified grammar, not evidence about a parser implementation.  $\square$

To pack literal one-bit planes, let  $C$  be the canonical octet stream, of length  $L$ . Pad its last block with zero octets to a multiple of 512 octets. Interpret each block as 64 little-endian words  $w_0, \dots, w_{63}$  and set

$$a_{r,c} = ((w_r \gg c) \& 1), \quad p_c = \sum_{r=0}^{63} a_{r,c} 2^r.$$

The envelope is eight ASCII octets `XOPPLAN1`, then the minimal encoding of  $L$ , then the plane words in increasing block and plane order. Its size fixes the required block count; after inverse

transpose, padding must be zero and the first  $L$  octets must decode as XOPSEED1. Empty input has no seed grammar, although the underlying byte permutation can act on an empty byte string.

**Proposition 2.3** (Bit-plane involution and information accounting). *The padded block transform  $T$  satisfies  $T(T(W)) = W$ . Including  $L$  makes packing injective on canonical seeds. The transform changes organization, not information content or the number of bits required by its uncompressed padded blocks.*

*Proof.* The output bit at  $(c, r)$  is the input bit at  $(r, c)$ . Transposing again restores every coordinate. The explicit length removes padding ambiguity. Consequently distinct canonical inputs cannot share an envelope. A bijective permutation cannot itself identify multiple arbitrary inputs with a smaller code space.  $\square$

This envelope is newly named. It neither changes nor retroactively reinterprets R10's uint64/binary64 codec. A conversion from R10 must declare whether it preserves raw bits, converts finite values to dyadic rationals, or replaces an approximation by a separately justified symbolic model.

## 2.7 Causal recurrence and the literal two-mask kernel

The state table declares each slot's type, initial expression and next-state expression. Reading slot  $j$  at logical step  $t$  is precisely

$$s_j(0) = \text{init}_j, \quad s_j(t+1) = E_j(s(t), d(t)); \quad s_j = \text{DELAY}(\text{init}_j, E_j).$$

All right-hand sides read the same old state and the same input snapshot. The new tuple commits simultaneously after all required domains and decisions are resolved. A self-loop in an instantaneous operator DAG is inadmissible: every recurrence cycle must cross a declared DELAY. An algebraic fixed-point equation is a separate solution declaration needing existence, uniqueness and branch obligations; it is not an execution order.

For  $w$ -bit masks, define the retained whole-word operation

$$\begin{aligned} a &= x \wedge A \wedge V, & n &= a \wedge N, & h &= \text{popcount}(n \wedge B), \\ \text{ASA}(x; V, A, N, B) &= \begin{cases} 0, & h > 0, \\ n, & h = 0. \end{cases} \end{aligned}$$

The two ordered sets and literal J–K update are

$$\begin{aligned} x_t &= E_X(q_t, d_t), & y_t^A &= \text{ASA}(x_t; V_A, A_A, N_A, B_A), \\ y_t &= \text{ASA}(y_t^A; V_B, A_B, N_B, B_B), & K_t &= E_K(q_t, y_t, d_t), \\ J_t &= E_J(q_t, y_t, d_t), & & \\ q_{t+1} &= [(J_t \wedge \neg_w q_t) \vee (\neg_w K_t \wedge q_t)] \wedge V_q. \end{aligned}$$

Here  $\neg_w q = (2^w - 1) \oplus q$  and every expression has  $\text{Bits}(w)$  type. Both intermediate  $(a, n, h, y)$  tuples are observables. Absorption zeroes the whole output of its stage; it neither clears selected bits individually nor silently substitutes a hold transition. The mathematical width  $w$  is explicit; the retained executable orbital kernel still uses its specified 32-bit word profile.

**Proposition 2.4** (Causal uniqueness and conditional finite periodicity). *For fixed initial states and an input stream, a typed finite instantaneous DAG whose required operations are defined determines at most one next state at each step. If all operations are total on the reachable states, it determines a unique state stream. A fixed autonomous transition on  $m$  active Boolean state bits is eventually periodic, with preperiod plus period at most  $2^m$ .*

*Proof.* Topological induction determines every instantaneous node from old-state and input values; simultaneous commitment then determines the next tuple. Induction on steps gives the stream. For the finite autonomous case, among  $2^m + 1$  states two coincide; determinism makes their subsequent tails coincide. Changing the input, strategy, timestamp, unbounded integer state or physical continuous state does not meet that finite-autonomous hypothesis.  $\square$

## 2.8 Vectors, calculus and orbital expressions

Vector and matrix operators are typed compositions of scalar arithmetic:  $u \cdot v = \sum_i u_i v_i$ ,  $\|u\| = \sqrt{u \cdot u}$ , and  $(AB)_{ij} = \sum_k A_{ik} B_{kj}$ . Cross product is restricted to three components. A normalized vector requires nonzero norm; a matrix inverse requires a square matrix and nonzero determinant. Polynomial, Chebyshev, Legendre, finite Fourier and rational expressions expand to declared finite templates. The catalog includes finite sum/product, differentiation, gradient and Jacobian binding, definite integration and an initial-value solution declaration.

$\text{DIFF}(f, x)$  denotes a derivative only where the stated limit exists.  $\text{INTEGRAL}(f, a, b)$  denotes an oriented Riemann integral when the finite interval integrand is integrable; an improper integral has a separate limit declaration.  $\text{ODE}(F, t_0, s_0, t, I)$  denotes the value at  $t \in I$  of a unique differentiable solution on the declared interval. A local existence theorem or local regularity condition alone is not a certificate of a global interval. If existence, uniqueness or continuation is unresolved, the node remains a declaration with obligations, rather than an evaluated exact orbit.

A finite RK4 template can be represented exactly as expressions, for example

$$\begin{aligned} k_1 &= F(t, s), & k_2 &= F(t + h/2, s + hk_1/2), \\ k_3 &= F(t + h/2, s + hk_2/2), & k_4 &= F(t + h, s + hk_3), \\ \text{RK4}_h(t, s) &= s + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4). \end{aligned}$$

Exact evaluation of this finite expression does not make it equal to the continuous ODE solution. Discretization, uncertain initial conditions, unmodelled forces and discontinuous external events remain separate mathematical or physical questions. All force and station expressions must retain their units, frames, absolute time and explicit singular domains. A finite symbolic term is not an error bound on its physical denotation.

## 2.9 Proof rules and strategy self-improvement

Let  $\Gamma$  contain typed assumptions and proven domain predicates. A permitted rewrite carries the judgement

$$\Gamma \vdash e \simeq e' : \tau, \quad \text{Dom}_\Gamma(e) = \text{Dom}_\Gamma(e'), \quad \llbracket e \rrbracket = \llbracket e' \rrbracket \text{ on that domain.}$$

Domain equality is essential: simplifying  $x/x$  to 1 without  $x \neq 0$  changes the defined inputs. The proof vocabulary contains assumption, reflexivity, symmetry, transitivity, typed congruence, capture-avoiding substitution, definition expansion and explicit ordered-field/domain lemmas. Branch rules require their predicates; differentiation and integration rewrites additionally require their analytic hypotheses. A rule name or a source digest alone is not a proof of its side conditions.

Structural common-subexpression sharing, exact constant folding and eliminating a pure unused node preserve the denotation of the demanded exports when the typing, scope and domain obligations are satisfied. Reassociation is justified in exact integer/rational arithmetic but is not silently applied to R10 binary64 execution. Eliminating an observed ASA trace, changing a sample read, or changing query cadence is not observationally neutral merely because a final word matches.

For stateful replacement use a relation  $\mathcal{R}$  between old and new states. Its proof must relate the initial states and, for every admitted input, preserve definedness, all declared observations and the relation after one synchronous step. Induction then gives equal observable streams. Absolute timestamps and selected sample identities belong to the observations whenever a query schedule is exported; a change in those requires a separately named strategy contract.

A strategy version is an immutable tuple

$$S_v = (\text{rule catalog, priority policy, cost model, proof system ID, resource policy}).$$

It may propose a new expression graph or a new strategy  $S_{v+1}$ . Acceptance requires a finite proof object under the already accepted proof system, a preserved input/output contract, and the stated termination/resource argument. The proposal cannot justify itself by replacing its own checker or dropping an unmet premise. A proof-system change creates a new explicit trust boundary. Strategies and proposal histories may be unbounded, so finite Boolean periodicity does not apply to the combined evolving system.

**Proposition 2.5** (Conditional preservation under strategy replacement). *Suppose the accepted proof rules are sound for the declared denotation, each accepted replacement carries a valid domain-preserving proof or stream simulation relation, and proof checking admits only valid derivations. Then any finite chain of accepted replacements preserves its declared observations.*

*Proof.* A sound proof preserves one replacement's domain and observations. Compose these equalities, or the stepwise stream relations, along the finite chain. Induction gives preservation after every accepted replacement. This establishes a conditional theorem; it supplies neither a proof-checker implementation nor an unconditional claim that proposed derivations are valid.  $\square$

Logical strategy improvement can now be stated precisely. Sharing an identical pure subgraph reduces duplicated graph storage; memoizing it avoids a repeated evaluation only when the complete binding, state step and dependency identity are the same. A strictly decreasing natural-valued rewrite measure proves termination of that rewrite phase; a finite proposal budget also terminates a search without proving it finds a better expression. A predicted reduction in operator count is a claim under the declared cost model, not a measured speedup or a guarantee of lower big-integer time, memory use, physical error or global optimality. These are the available reasoning-based improvements in this subversion; execution and machine proof remain future obligations.

### 3 Original Madgwick recurrence as an exact operator graph

**Status.** This chapter defines a mathematical profile and gives conditional proofs. It does not supply a new runtime, machine-checked proof, simulation, benchmark or physical accuracy result. Its purpose is to retain the chosen operators and their dependencies before a rounded numerical state is materialized. The verified R10 runtime remains a separate preserved implementation.

#### 3.1 Source identity and discrete profile

The source is Sebastian O. H. Madgwick's 2010 report, *An efficient orientation filter for inertial and inertial/magnetic sensor arrays*. We select its simplified recurrence, equations (25)–(34), (42)–(51), under the name MDG2010-EQ-SIMPLIFIED-R1. Normalization, sample ordering and otherwise undefined branches are specified below. The finite-mixture family from equations (35)–(38) is retained as a distinct alternative, not an identity used to rewrite the simplified recurrence.

The current [x-io Fusion library](#) describes a revised thesis algorithm. It is not silently substituted for the selected 2010 profile. Here “fusion” in the finite-mixture equations names the source’s combination of estimates, not that later software package.

Equation (45) forms the magnetic reference from the previous attitude and the current normalized magnetic sample. The report’s appendix C code instead stores a magnetic reference after its new attitude update for use at the next sample. It also prepares `twom_x/y/z` before normalizing the magnetic input and later uses those aliases for the flux calculation. These orders have distinct literal meanings. This chapter pins the equation profile; it asserts neither bit identity with that C code nor equivalence to its unnormalized aliases.

### 3.2 Typed state, units and frame convention

Let  $B$  be the calibrated right-handed sensor/body frame and  $E$  the right-handed filter reference frame. The positive  $E$ - $z$  axis is the declared accelerometer reference direction after sensor-sign calibration; its horizontal  $x$  axis is the chosen magnetic or heading reference. The alignment of  $E$  with station ENU is an explicit transform, not an assumed identity.

The scalar-first Hamilton quaternion

$$q_{\text{att}} = q = (w, x, y, z) = (w, v), \quad q^T q = 1$$

maps body vectors to  $E$  vectors. The separate ASA/JK word is denoted  $q_{\text{word}} \in \{0, \dots, 2^{32} - 1\}$ . The quaternion and Boolean word have different types and no shared state fields. For quaternions  $a = (a_0, \mathbf{a})$  and  $b = (b_0, \mathbf{b})$ , define

$$a \otimes b = (a_0 b_0 - \mathbf{a} \cdot \mathbf{b}, a_0 \mathbf{b} + b_0 \mathbf{a} + \mathbf{a} \times \mathbf{b}),$$

$$q^* = (w, -v), \quad [0, \mathbf{p}_E] = q \otimes [0, \mathbf{p}_B] \otimes q^*.$$

All subsequent frame equations follow this convention directly.

The ordered event is

$$I_n = (\text{id}_n, t_n, \boldsymbol{\omega}_n, \mathbf{a}_n, \mathbf{m}_n, \text{validity}_n, \text{calibrationId}), \quad h_n = t_n - t_{n-1} > 0.$$

Time is in seconds on a declared monotonic scale. Gyro samples are calibrated rad/s; accelerometer values are  $\text{m/s}^2$  before direction normalization; magnetic components have one declared consistent unit. Accelerometer specific force must receive the declared sign convention before it is interpreted as the reference direction. A clock mapping to GPST is required for station/orbit composition.

Persistent data comprise the quaternion, previous event time/index, and optional gyro bias  $\mathbf{b}_\omega$  in rad/s. Parameters include  $\beta \geq 0$  in  $\text{s}^{-1}$ ,  $\zeta \geq 0$  in  $\text{s}^{-2}$ , fixed calibration, initialization, reference policy, clock/frame/mounting transforms and domain branches. A lagged magnetic variant additionally stores  $(b_x, b_z)$  and its initialization; the selected current-reference variant recomputes it and needs no independent persistent magnetic state. Fixed calibration bias and estimated drift are distinct terms.

Raw integer ADC values with exact scale ratios are rational literals. A finite IEEE value has an exact dyadic-rational meaning; an intended decimal ratio has a different literal type. Raw bit provenance can remain distinct from either mathematical value. Exact degree-to-radian conversion introduces symbolic  $\pi$ . Thus the purely algebraic profile assumes rational/algebraic radian inputs; symbolic  $\pi$  conversion belongs to an explicitly broader operator domain. An exact recorded value does not certify the corresponding physical measurement.

### 3.3 Normalization, rotation and residual operators

For a real vector  $v \neq 0$ , define the partial operator

$$\mathcal{N}(v) = \frac{v}{\sqrt{v^T v}},$$

where the square root is the positive root. Admitted observations give  $\hat{a} = \mathcal{N}(a)$  and  $\hat{m} = \mathcal{N}(m)$ . For unit  $q$ , the body-to-reference rotation is

$$R(q) = \begin{bmatrix} 1 - 2(y^2 + z^2) & 2(xy - wz) & 2(xz + wy) \\ 2(xy + wz) & 1 - 2(x^2 + z^2) & 2(yz - wx) \\ 2(xz - wy) & 2(yz + wx) & 1 - 2(x^2 + y^2) \end{bmatrix}.$$

Substitution into  $R(q)^T(0, 0, 1)^T - \hat{a}$  yields the accelerometer objective and its ambient Jacobian:

$$f_g(q) = \begin{bmatrix} 2(xz - wy) - \hat{a}_x \\ 2(wx + yz) - \hat{a}_y \\ 1 - 2(x^2 + y^2) - \hat{a}_z \end{bmatrix}, \quad (1)$$

$$J_g(q) = \begin{bmatrix} -2y & 2z & -2w & 2x \\ 2x & 2w & 2z & 2y \\ 0 & -4x & -4y & 0 \end{bmatrix}. \quad (2)$$

For a fixed reference  $b_E = (b_x, 0, b_z)^T$ , the magnetic residual is

$$f_b(q) = \begin{bmatrix} b_x(1 - 2(y^2 + z^2)) + 2b_z(xz - wy) - \hat{m}_x \\ 2b_x(xy - wz) + 2b_z(wx + yz) - \hat{m}_y \\ 2b_x(wy + xz) + b_z(1 - 2(x^2 + y^2)) - \hat{m}_z \end{bmatrix}. \quad (3)$$

Its Jacobian, with columns ordered  $(w, x, y, z)$ , is

$$J_b(q) = \begin{bmatrix} c_w & c_x & c_y & c_z \end{bmatrix},$$

$$c_w = \begin{bmatrix} -2b_z y \\ -2b_x z + 2b_z x \\ 2b_x y \end{bmatrix}, \quad c_x = \begin{bmatrix} 2b_z z \\ 2b_x y + 2b_z w \\ 2b_x z - 4b_z x \end{bmatrix}, \quad (4)$$

$$c_y = \begin{bmatrix} -4b_x y - 2b_z w \\ 2b_x x + 2b_z z \\ 2b_x w - 4b_z y \end{bmatrix}, \quad c_z = \begin{bmatrix} -4b_x z + 2b_z x \\ -2b_x w + 2b_z y \\ 2b_x x \end{bmatrix}.$$

Products such as  $b_x x$  mean  $b_x x$ , not an extra subscript. These expressions follow by direct polynomial differentiation: for example,  $\partial f_{b,1}/\partial y = -4b_x y - 2b_z w$  and  $\partial f_{g,3}/\partial w = 0$ .

For IMU, set  $f = f_g, J = J_g$ . For MARG, stack residuals and Jacobians vertically:

$$f = \begin{bmatrix} f_g \\ f_b \end{bmatrix}, \quad J = \begin{bmatrix} J_g \\ J_b \end{bmatrix}, \quad F(q) = \frac{1}{2}f(q)^T f(q), \quad s = \nabla F = J^T f.$$

The chain rule gives  $s = J_g^T f_g + J_b^T f_b$  in MARG mode. These are the source's unweighted residuals. Changing sensor weights defines a different recurrence. The derivatives refer to the displayed ambient polynomial extension evaluated at a unit quaternion. Agreement of two residual formulas on the unit sphere does not imply equality of their ambient gradients.

### 3.4 Magnetic reference and optional drift state

Before the current MARG residual is formed, compute

$$\mathbf{h}_n = R(q_{n-1})\widehat{\mathbf{m}}_n, \quad (5)$$

$$b_{x,n} = \sqrt{h_{x,n}^2 + h_{y,n}^2}, \quad b_{z,n} = h_{z,n}. \quad (6)$$

Hold these values fixed while evaluating  $J_b$ . Differentiating through the reference update would add chain-rule terms absent from the chosen source gradient. A dependency in the event graph therefore need not be a differentiation variable of every local objective.

**Proposition 3.1** (Reference norm). *If  $q_{n-1}$  and  $\widehat{\mathbf{m}}_n$  have unit norm, then  $b_{x,n} \geq 0$  and  $b_{x,n}^2 + b_{z,n}^2 = 1$ .*

*Proof.* Quaternion conjugation by a unit quaternion preserves vector norm. The square of  $b_x$  is  $h_x^2 + h_y^2$ , so adding  $b_z^2$  gives  $\mathbf{h}^T \mathbf{h} = 1$ . This proves a representation identity, not an undisturbed magnetic measurement.  $\square$

Define the correction direction by an explicit zero branch:

$$\mathbf{e} = \begin{cases} \mathbf{s} / \sqrt{\mathbf{s}^T \mathbf{s}}, & \mathbf{s}^T \mathbf{s} > 0, \\ 0, & \mathbf{s}^T \mathbf{s} = 0. \end{cases}$$

Zero gradient means no gradient correction, not proof that the attitude is true. For enabled MARG drift with both observations admitted and  $b_x > 0$ , use

$$\mathbf{d}_\omega = 2 \text{vec}(q_{n-1}^* \otimes \mathbf{e}), \quad (7)$$

$$\mathbf{b}_{\omega,n} = \mathbf{b}_{\omega,n-1} + \zeta h_n \mathbf{d}_\omega, \quad \boldsymbol{\omega}_c = \boldsymbol{\omega}_n - \mathbf{b}_{\omega,n}. \quad (8)$$

The updated bias is used in this sample. Otherwise the bias is held. For  $\mathbf{e} = (e_w, e_x, e_y, e_z)$ , the drift direction is explicitly

$$\mathbf{d}_\omega = 2 \begin{bmatrix} we_x - xe_w - ye_z + ze_y \\ we_y + xe_z - ye_w - ze_x \\ we_z - xe_y + ye_x - ze_w \end{bmatrix}.$$

The vector projection discards the scalar quaternion component. The ambient gradient need not be tangent to the unit sphere, so that scalar component must not silently be interpreted as a gyro axis. The selected IMU mode does not adapt all-axis gyro bias from an unobservable heading.

The source's gain parameterization can remain symbolic:

$$\beta = \frac{\sqrt{3}}{2} \tilde{\omega}_\beta, \quad \zeta = \frac{\sqrt{3}}{2} \tilde{\omega}_\zeta.$$

The error and drift-rate quantities are declared parameters; these identities do not establish their physical values or prove filter convergence.

### 3.5 The selected update and its unit-norm invariant

With corrected body rate  $\boldsymbol{\omega}_c$ , define

$$\dot{q}_\omega = \frac{1}{2} q_{n-1} \otimes [0, \boldsymbol{\omega}_c], \quad (9)$$

$$\mathbf{u}_n = q_{n-1} + h_n (\dot{q}_\omega - \beta \mathbf{e}), \quad (10)$$

$$q_n = \mathcal{N}(\mathbf{u}_n), \quad \mathbf{u}_n^T \mathbf{u}_n > 0. \quad (11)$$

All candidate state fields, including bias, are committed together only after the candidate quaternion satisfies the normalization domain.

**Proposition 3.2** (Unit norm after every successful update). *Starting with a unit quaternion, any finite sequence of successful updates (11) produces unit quaternions.*

*Proof.* For  $S = u_n^T u_n > 0$ , the positive-root semantics give  $\mathcal{N}(u_n)^T \mathcal{N}(u_n) = S/(\sqrt{S})^2 = 1$ . Induction over the accepted events proves the statement. Existence of a valid candidate and correct evaluation of the declared branches remain premises.  $\square$

To preserve the source’s other derivation without conflating algorithms, let

$$\mu = \alpha \|\dot{q}_\omega\| h, \quad \alpha > 1, \quad \gamma = \frac{\beta}{\mu/h + \beta}$$

when the denominator is positive. The finite-mixture candidate is

$$\begin{aligned} u_{\text{mix}} &= q + (1 - \gamma)h\dot{q}_\omega - \gamma\mu e \\ &= q + (1 - \gamma)h(\dot{q}_\omega - \beta e), \end{aligned}$$

because  $\gamma\mu = (1 - \gamma)\beta h$ . If both denominator terms vanish, the declared extension chooses  $\gamma = 0$ . At finite  $\gamma$ , this is generally a different effective step from (11). In particular,  $\dot{q}_\omega = 0$  and  $\beta > 0$  give  $\mu = 0$ ,  $\gamma = 1$ , and  $u_{\text{mix}} = q$ , even if  $e \neq 0$ . The large- $\alpha$  simplification is not an identity at every stationary input. An exact graph preserves the chosen recurrence; it does not remove this model choice.

### 3.6 Domain branches are part of the formal input

The following are explicit definitions resolving partial equations; they are not presented as pre-existing empirical rejection heuristics.

| Condition                                                      | Declared action                                                                              |
|----------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Zero initial quaternion                                        | No initialized state; require a declared nonzero initialization.                             |
| Untyped/nonfinite input, missing gyro or invalid clock mapping | No transition; record the input-domain failure.                                              |
| Duplicate or decreasing event time                             | No transition; a separate input policy must resolve ordering or duplicates.                  |
| Zero or missing acceleration                                   | Gyro-only update; no magnetic correction or drift adaptation.                                |
| Acceleration admitted; magnetic sample zero or missing         | IMU objective; hold all-axis drift estimate.                                                 |
| Both vectors admitted and $b_x = 0$                            | MARG residual remains defined; mark heading degeneracy and hold all-axis drift adaptation.   |
| Exactly zero gradient                                          | Set $e = 0$ ; no gradient-driven bias increment.                                             |
| $\beta = 0$                                                    | No attitude-gradient correction; independently enabled $\zeta$ remains a separate operation. |
| Exactly zero candidate $u_n$                                   | Hold the last complete state and report undefined normalization.                             |
| Observation-quality flag                                       | Use the declared input flag; do not invent an undeclared small-norm epsilon or outlier rule. |

An algebraic domain supports exact sign/equality decisions in principle; the cost is not bounded here. A broader transcendental domain must retain unresolved comparisons rather than silently treating a finite-precision threshold as exact.

### 3.7 Exact graph representation and conditional rewrites

The graph consists of typed literals, addition, subtraction, multiplication, division, positive square root, tuples/projections, dot/cross/Hamilton products and explicit branches. Roots bind the state to its operator profile, samples, clock and calibration. Division nodes carry nonzero-denominator obligations; square-root nodes carry nonnegative-operand obligations. With algebraic inputs, closure under these operators gives algebraic outputs for every defined finite event prefix. It does not give a finite machine-word bound on their representation.

**Proposition 3.3** (Positive-scale cancellation). *For  $c > 0$  and  $v \neq 0$ ,  $\mathcal{N}(cv) = \mathcal{N}(v)$ .*

*Proof.*  $\sqrt{c^2 v^T v} = c \sqrt{v^T v}$  by the positive-root convention; the factors cancel. For  $c < 0$  the normalized representative changes sign, and for  $c = 0$  it is undefined.  $\square$

**Proposition 3.4** (Rotation consumer without a normalization square root). *For quaternion  $v \neq 0$ ,  $S = v^T v$  and pure vector quaternion  $[0, p]$ ,*

$$R(\mathcal{N}(v))p = \frac{\text{vec}(v \otimes [0, p] \otimes v^*)}{S}.$$

*Proof.* Each quaternion factor contributes  $1/\sqrt{S}$ . Their product contributes  $1/S$ . This removes the square root for this rotation consumer only.  $\square$

**Proposition 3.5** (Pure-gyro projective step). *If the corrected rate is fixed for a step and there is no attitude-gradient correction, then, wherever defined,*

$$\mathcal{N}\left(\mathcal{N}(v) + \frac{h}{2}\mathcal{N}(v) \otimes [0, \omega_c]\right) = \mathcal{N}\left(v + \frac{h}{2}v \otimes [0, \omega_c]\right).$$

*Proof.* Factor the positive common factor  $1/\sqrt{v^T v}$  from the candidate and apply positive-scale cancellation. The rate must still be obtained under its declared attitude/bias convention, not recomputed using an unnormalized substitution. For real  $h$  and corrected rate, the candidate is automatically nonzero: Hamilton norm multiplicativity gives

$$\|v + \frac{h}{2}v \otimes [0, \omega_c]\|^2 = (v^T v) \left(1 + \frac{h^2}{4}\|\omega_c\|^2\right) > 0.$$

This establishes the normalization domain for this subcase only.  $\square$

The corrected homogeneous numerator is instead

$$v + \frac{h}{2}v \otimes [0, \omega_c] - h\beta\sqrt{v^T v}e(\mathcal{N}(v)).$$

Deleting its square root or substituting  $v$  into the unit-quaternion source polynomials changes the recurrence in general. Moreover, differentiation of the composed normalized objective gives

$$\nabla_v F(\mathcal{N}(v)) = \frac{(I - q q^T)s}{\|v\|}, \quad q = \mathcal{N}(v).$$

This is a projected gradient, not the source's ambient  $s$ . Therefore a projective optimization may apply the proved local identities above, but may not silently replace the descent direction or its effective magnitude.

**Proposition 3.6** (Quaternion sign cover). *The selected recurrence is sign-equivariant wherever defined: replacing its initial quaternion by  $-q$  changes every quaternion representative's sign while leaving its represented rotation and gyro-bias increments unchanged.*

*Proof.*  $R(-q) = R(q)$  because rotation is quadratic. The residuals are even in  $q$ , their displayed Jacobians are odd, and thus  $e(-q) = -e(q)$ , including the zero branch. The magnetic reference is unchanged. In  $q^* \otimes e$  the two signs cancel, so bias increments agree. The gyro derivative and candidate both change sign; normalization preserves that sign relation. Induction gives the result.  $\square$

This is equivalence of rotations under  $q \sim -q$ , not equality of stored tuples, raw IEEE bit patterns or hashes. No automatic sign canonicalization is assumed. Euler-angle display and shortest-arc comparison need their own explicit branch conventions and potentially transcendental operators.

### 3.8 Causal sample processing and graph cardinality

An event reads only the preceding state, its current sample and fixed parameters. Induction therefore proves that the state after event  $n$  depends on the seed and events  $1, \dots, n$ , never on an unobserved future sample. Checkpoint restart requires the complete selected state: attitude, optional bias, event time/index, and any lagged reference, calibration or scheduler state that the profile retains. Four or nine state roots do not imply four or nine fixed-size machine words.

For an input alphabet of  $K$  possible records there are  $K^n$  streams of length  $n$ . An injective  $B$ -bit representation of every such stream is impossible when  $K^n > 2^B$ . This concerns complete stream reconstruction; it does not claim that every distinct stream yields a distinct final quaternion. Graph sharing reduces duplicate expression trees but cannot repeal the counting bound or guarantee bounded algebraic degree, exact-predicate cost or evaluation memory.

A rounded checkpoint replaces exact lineage by a different numeric state. An exact checkpoint must retain a sufficient exact representation and every dependency not removed by a proved identity. Numerical evaluation may be deferred to an output request; a rounded display is not fed back as the next exact state.

JK/ASA may schedule observations or exports of an attitude already updated from all accepted sensor samples. Skipping samples, changing  $h$ , changing gains or reordering measurements changes the recurrence and must be a named profile or recorded control event. Two noncommuting angular increments already show why two ordered quaternion steps cannot generally be replaced by one summed-rate step. Observation cadence and sensor-integration cadence are separate contracts.

### 3.9 Station composition, observability and error meaning

Let  $L$  be station ENU,  $A$  the antenna/camera frame,  $C_{L \leftarrow E}$  the declared filter-to-ENU alignment, and  $C_{A \leftarrow B}$  the mounting rotation. For the simultaneous station-relative line of sight  $\ell_L(t)$ ,

$$\ell_A(t) = C_{A \leftarrow B} R(q(t))^T C_{L \leftarrow E}^T \ell_L(t).$$

The orbital kernel supplies satellite/station geometry; attitude changes its coordinates, not the satellite ephemeris. A moving station additionally requires its position and velocity at the same time. An attitude quaternion supplies neither. Between-sample interpolation, sensor latency, magnetic declination, reference-frame alignment and mounting updates are additional declared operators.

One measured reference direction leaves rotation about that direction unobservable. Two consistent nonparallel directions determine a rotation, with its two quaternion representatives;  $b_x = 0$  makes the reference directions collinear. Linear acceleration can violate the accelerometer reference model, and disturbed magnetism does not become true heading through exact arithmetic. A GNSS/INS position solution requires additional measurements and an estimator with position/velocity/bias state. This attitude formalization establishes no millimetre positioning or arcsecond pointing result.

Exact arithmetic of a discrete rule remains distinct from time discretization. For a sufficiently smooth ideal derivative  $F$  on a fixed branch,

$$q(t + h) = q(t) + hF(q(t), t) + O(h^2).$$

Evaluating the first two terms exactly does not remove the remaining terms. Normalization is smooth away from zero, but near zero candidates, zero gradients or mode switches, this local smoothness argument needs additional premises. The same separation applies to R10's RK4 rule, approximated forcing/frame coefficients, initial-state knowledge and physical force models. Sensor calibration, quantization and noise are further independent limitations.

**Logical strategy.** Retain exact typed literals, the chosen recurrence, sample provenance and state roots. Simplify only under proved domain and sign conditions. Defer finite-precision materialization to a declared output boundary. This supplies a formal route to removing intermediate arithmetic roundoff; it does not by itself bound expression growth, discretization error or physical error.

## 4 Exact domains for the retained numerical profiles

This section lifts the actual retained operators into mathematical domains. It defines their values, singularities and conditional invariants; it does not install an exact runtime or claim a new physical accuracy result. The retained orbital, navigation and coupled chapters remain dated implementation evidence. Their reported measurements were not repeated for this subversion.

### 4.1 Values, comparison and the meaning of a lift

An imported finite binary64 value denotes its exact dyadic rational. Its original bits are a separate serialization value: positive and negative zero are the same real number, but have different bit patterns. Exact decimal constants, an exact  $\pi$ , and the stored approximations to those constants are distinct atoms. A specification must choose which atoms it uses. In the ideal geometric formulas below,  $\pi$  and the elementary functions have their mathematical meanings; substituting them for rounded runtime constants is an explicit semantic change.

Integers, rationals and real algebraic numbers close the finite word, rational matrix and square-root constructions. Trigonometric and logarithmic coordinates require explicit real transcendental expressions. A finite recurrence is another finite expression, even when expensive to expand. An implicit operator ROOT or FLOW instead denotes a solution of stated equations, subject to existence and selection conditions. Naming an implicit operator supplies neither a closed-form solver nor a terminating evaluator.

Comparisons of exact rational or algebraic values are decidable. A general transcendental expression has a mathematical order, but this specification does not supply a universal decision procedure for it. An evaluator must retain an unresolved comparison obligation when its method cannot establish the answer. In particular,

$$n = \lfloor x \rfloor \iff n \in \mathbb{Z}, \quad n \leq x < n + 1. \quad (12)$$

No epsilon replaces equality at a cycle, quantization or event boundary. Piecewise expressions are lazy: an inactive force term does not require its unused denominator. These lifts remove floating-point rounding, overflow and underflow; consequently, they do not preserve every numerical branch or establish bit-equivalence to R10. Bit preservation belongs to the unchanged serialized payload and word codec, not to recomputation of its real-valued consequences.

## 4.2 Coefficient blocks, Earth frames and force evaluation

The source map is `orbit_dynamics.py`, `orbit_precision.py` and `orbit_kernel.hpp` in the pre-served R10 tree. The complete seed fields and physical scope are retained in Section 8. Let a coefficient block cover  $[a, b]$ , with  $b > a$ , and put  $u = (2t - a - b)/(b - a)$ . If several blocks contain  $t$ , the block with greatest stored index wins, including a shared endpoint. For each scalar row,

$$p(t) = \sum_{k=0}^d c_k T_k(u), \quad T_0 = 1, \quad T_1 = u, \quad T_{k+1} = 2uT_k - T_{k-1}, \quad (13)$$

$$b_{d+1} = b_{d+2} = 0, \quad b_k = c_k + 2ub_{k+1} - b_{k+2} \quad (k = d, \dots, 1), \quad (14)$$

$$p(t) = c_0 + ub_1 - b_2, \quad \dot{p}(t) = \frac{2}{b-a} \sum_{k=1}^d k c_k U_{k-1}(u). \quad (15)$$

For degree zero the value is  $c_0$  and the derivative is zero. The finite Clenshaw recurrence equals the Chebyshev sum by backward substitution; formal differentiation gives the derivative. No continuity between separately fitted blocks follows from this identity. A timestamp outside all blocks is undefined.

Write  $c_x = \cos x_p$ ,  $s_x = \sin x_p$ , and similarly for  $y_p$ . The actual passive frame convention is

$$P = \begin{pmatrix} c_x & 0 & s_x \\ s_y s_x & c_y & -s_y c_x \\ -c_y s_x & s_y & c_y c_x \end{pmatrix}, \quad R(a) = \begin{pmatrix} \cos a & \sin a & 0 \\ -\sin a & \cos a & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad M(t) = P(t)R(a(t))Q(t). \quad (16)$$

R1 uses constant polar motion and  $a = a_0 + \omega t$ . R2 uses stored EOP rows  $e_0, x_p, y_p$ , with  $a = a_0 + \omega t + e_0(t)$ . The nine entries of  $Q$  are stored polynomial rows, rather than a symbolic rotation parameterization. Position and velocity transform as

$$r_E = Mr, \quad v_E = Mv + \dot{M}r, \quad \dot{M} = \dot{P}RQ + P\dot{R}Q + PR\dot{Q}. \quad (17)$$

**Conditional invariant.** Exact trigonometry gives  $P^T P = R^T R = I$  and determinant +1. Thus  $M^T M = Q^T Q$ : length preservation and the identification of  $M^T$  with  $M^{-1}$  require a separate proof that  $Q^T Q = I$ . Independent polynomial fits do not supply that proof. Applying  $M^T$  to a force remains the specified algebraic operation even when this ideal rotation condition is absent; silently orthogonalizing  $Q$  would define a different model.

For the R2 external gravity field let  $p = Mr$ ,  $R = \|p\|$ , and let  $a_E > 0$  denote the reference radius. The declared profile requires  $R > a_E$ , in addition to nonzero denominators. Its finite Cartesian solid-harmonic construction is

$$X = a_E p_x / R^2, \quad Y = a_E p_y / R^2, \quad Z = a_E p_z / R^2, \quad D = a_E^2 / R^2, \quad (18)$$

$$V_{00} = a_E / R, \quad W_{00} = 0, \quad (19)$$

$$V_{mm} = (2m-1)(XV_{m-1,m-1} - YW_{m-1,m-1}), \quad (20)$$

$$W_{mm} = (2m-1)(XW_{m-1,m-1} + YV_{m-1,m-1}), \quad (21)$$

$$V_{nm} = \frac{2n-1}{n-m} ZV_{n-1,m} - \frac{n+m-1}{n-m} DV_{n-2,m} \quad (n > m), \quad (22)$$

$$U(p) = \frac{\mu}{a_E} \sum_{n=0}^d \sum_{m=0}^n (C_{nm} V_{nm} + S_{nm} W_{nm}), \quad a_g = M^T \nabla_p U. \quad (23)$$

The  $W_{nm}$  off-diagonal recurrence is identical with  $V$  replaced by  $W$ ; omit the second term when  $n = m + 1$ . The retained R2 degree is at most 12,  $C_{00} = 1$ ,  $S_{00} = S_{n0} = 0$ , and the convention has no

Condon–Shortley phase. Conversion from the documented fully normalized source is

$$C_{nm} = \sqrt{(2 - \delta_{m0})(2n + 1) \frac{(n - m)!}{(n + m)!}} \bar{C}_{nm}, \quad S_{nm} \text{ uses the same factor.} \quad (24)$$

The native forward derivative differentiates this finite recurrence by the sum, product and chain rules. In contrast, the Python construction/reference path uses  $\text{Im } U(p + i\epsilon e_j)/\epsilon$  with fixed  $\epsilon = 10^{-6}$  in its coordinate units. Exact arithmetic at a nonzero  $\epsilon$  does not turn that expression into  $\partial_j U$ . Its analytic limit, where the complex extension is regular, defines the derivative; the finite complex-step operator and the exact derivative must retain different names.

The older R1 gravity formula is also explicit. For  $s = p_z/R$  and unnormalized Legendre polynomials  $P_2 = (3s^2 - 1)/2$ ,  $P_3 = (5s^3 - 3s)/2$  and  $P_4 = (35s^4 - 30s^2 + 3)/8$ , put

$$a_{E,1} = -\mu p/R^3 + \sum_{n=2}^4 \frac{\mu J_n a_E^n}{R^{n+3}} (((n+1)P_n + sP'_n)p - RP'_n e_z) \quad (25)$$

$$+ \frac{\mu a_E^2}{R^5} \left( \nabla T - \frac{5T}{R^2} p \right), \quad T = 3C_{22}(p_x^2 - p_y^2) + 6S_{22}p_x p_y. \quad (26)$$

Its inertial gravity is  $M^T a_{E,1}$ . This is a separate finite field, not a claim that the R1 and R2 coefficient conventions are interchangeable.

Both orbital profiles add the declared active terms below; the relativistic term is an R2 option. Sun and Moon vectors  $r_b(t)$  come from their seed coefficient blocks. With  $r_* = \|r\|$ ,  $d = r - r_\odot$ , and  $\hat{s} = r_\odot/\|r_\odot\|$ ,

$$a_b = \mu_b \left( \frac{r_b - r}{\|r_b - r\|^3} - \frac{r_b}{\|r_b\|^3} \right), \quad (27)$$

$$a_{\text{SRP}} = \chi A \frac{\text{AU}^2}{\|d\|^3} d, \quad \chi = \begin{cases} 0, & \text{shadow enabled, } r \cdot \hat{s} < 0, \|r - (r \cdot \hat{s})\hat{s}\| < a_E, \\ 1, & \text{otherwise,} \end{cases} \quad (28)$$

$$\hat{R} = r/r_*, \quad \hat{N} = (r \times v)/\|r \times v\|, \quad a_{\text{RTN}} = a_R \hat{R} + a_T(\hat{N} \times \hat{R}) + a_N \hat{N}, \quad (29)$$

$$a_{\text{GR}} = \frac{\mu}{c^2 r_*^3} \left[ \left( \frac{4\mu}{r_*} - v \cdot v \right) r + 4(r \cdot v)v \right]. \quad (30)$$

Active body terms require  $r_b \neq 0$  and  $r \neq r_b$ ; SRP requires its Sun and displacement denominators, RTN requires  $r_* \neq 0$  and  $r \times v \neq 0$ , and GR requires  $r_* \neq 0$  and  $c > 0$ . An identically inactive RTN term does not impose its normal-vector requirement. Equality on either shadow boundary is lit under the retained strict inequalities. These formulas contain no unlisted maneuver, penumbra, drag or higher-degree force. Exact evaluation cannot supply one.

### 4.3 Exact discrete integration and a distinct continuous flow

Let  $y = (r, v)$  and  $F(t, y) = (v, a(t, r, v))$ . The exact lift of the implemented step is the finite operator  $\Psi_h$ :

$$k_1 = F(t, y), \quad k_2 = F(t + h/2, y + hk_1/2), \quad (31)$$

$$k_3 = F(t + h/2, y + hk_2/2), \quad k_4 = F(t + h, y + hk_3), \quad (32)$$

$$\Psi_h(t, y) = y + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4). \quad (33)$$

All four stage arguments must belong to the active domain, including the declared time interval and external-radius guard. For a positive lattice spacing  $h_0$  and elapsed time  $\tau$ , set  $n = \text{trunc}(\tau/h_0)$

toward zero, and  $\epsilon = \tau - nh_0$ . Compose signed steps from epoch to  $nh_0$ , then apply one step of size  $\epsilon$  if nonzero. Fractional query results are never used as canonical lattice states.

**Preservation proposition.** A cache returning only the states defined by this same canonical recurrence, under the same seed and branch choices, leaves every query unchanged. Proof is induction on the signed step index; the last fractional step is a pure function of the canonical state. This does not imply that arbitrary reblocking of RK4 steps, reversing a step, or changing the order of force additions preserves the floating-point result. Even over exact reals, RK4 generally has neither the continuous-flow semigroup law nor exact energy conservation or time reversibility.

Define the separate implicit operator by

$$\text{FLOW}_F(t, t_0, y_0) = y(t), \quad y(t) = y_0 + \int_{t_0}^t F(s, y(s)) ds. \quad (34)$$

On an open collision-free region with all active denominators bounded away from zero, fixed force branches and coefficient blocks,  $F$  is locally Lipschitz in  $y$  and continuous in  $t$ . The usual initial-value theorem then gives a unique maximal local solution. Piecewise continuous time coefficients can be concatenated across declared time boundaries under corresponding local existence conditions. No continuity of separately fitted coefficients is assumed. State-triggered shadow switches require a specified event solution concept and additional uniqueness conditions; isolated transversal crossings with a prescribed continuation and no accumulation of events are sufficient local hypotheses. The literal boundary convention alone does not prove these hypotheses for grazing, sliding or infinitely accumulating events.

When the unique flow exists on all intervening times, its composition law  $\text{FLOW}(t_2, t_0, y) = \text{FLOW}(t_2, t_1, \text{FLOW}(t_1, t_0, y))$  follows from uniqueness. It does not transfer to  $\Psi_h$ . A seed specifying  $F$  and  $y_0$  defines an initial-value problem; it is not a closed-form solution. Neither operator certifies agreement with a physical satellite or extends the seed's declared domain.

#### 4.4 Ground stations, angles, visibility and light time

The preserved `geodesy.py` and `orbit_query.py` specify a fixed ellipsoidal station. With semimajor axis  $a > 0$ , flattening  $0 < f < 1$ ,  $e^2 = f(2 - f)$ , latitude  $\varphi \in [-\pi/2, \pi/2]$ , longitude  $\lambda$ , and height  $h$ ,

$$N = a / \sqrt{1 - e^2 \sin^2 \varphi}, \quad (35)$$

$$r_g = ((N + h) \cos \varphi \cos \lambda, (N + h) \cos \varphi \sin \lambda, (N(1 - e^2) + h) \sin \varphi), \quad (36)$$

$$L = \begin{pmatrix} -\sin \lambda & \cos \lambda & 0 \\ -\sin \varphi \cos \lambda & -\sin \varphi \sin \lambda & \cos \varphi \\ \cos \varphi \cos \lambda & \cos \varphi \sin \lambda & \sin \varphi \end{pmatrix}, \quad z = L(r_E - r_g), \quad \dot{z} = L v_E. \quad (37)$$

Exact trigonometry proves  $LL^T = I$  and determinant +1; hence the station translation and ENU rotation preserve pairwise Euclidean distances and have inverse  $r_E = r_g + L^T z$ . The retained inverse geodesy implementation is a separate finite operator: outside its pole branch it starts with  $\varphi_0 = \text{atan2}(z, p(1 - e^2))$ ,  $p = \sqrt{x^2 + y^2}$ , and repeats  $\varphi_{j+1} = \text{atan2}(z + e^2 N(\varphi_j) \sin \varphi_j, p)$  at most 20 times, stopping at change below  $10^{-14}$ . Its pole threshold is  $p < 10^{-10}$ , with longitude zero; its height formula switches between  $p/\cos \varphi - N$  and  $z/\sin \varphi - N(1 - e^2)$  at  $|\cos \varphi| > 0.01$ . An implicit inverse of the forward ellipsoid chart needs its own uniqueness domain and longitude convention; it is not identical by definition to these 20 iterations.

For  $z = (E, N, U)$ , set  $H = \sqrt{E^2 + N^2}$  and  $D = \|z\| > 0$ . The simultaneous geometric observables

are

$$\alpha = \text{atan2}(E, N) \bmod 2\pi, \quad \varepsilon = \text{atan2}(U, H), \quad \dot{D} = z \cdot \dot{z}/D, \quad (38)$$

$$\dot{\alpha} = (N\dot{E} - E\dot{N})/H^2, \quad \dot{s} = \dot{U}/D - U\dot{D}/D^2, \quad s = U/D. \quad (39)$$

Azimuth and its rate are intrinsically undefined at  $H = 0$ . The retained query policy publishes them only for  $H > 10^{-6}$  metres; otherwise they are absent. Under that same policy, elevation rate is  $\dot{s}/\max(10^{-15}, \cos \varepsilon)$ , otherwise absent. The max and threshold are named implementation choices, distinct from the ideal derivative  $\dot{s}/\cos \varepsilon$  on its nonsingular domain. Visibility uses  $\varepsilon \geq \varepsilon_{\min}$ , so equality is visible. On a physical mask in  $[-\pi/2, \pi/2]$ , the event equation is  $s - \sin \varepsilon_{\min} = 0$ . An isolated bracket plus uniqueness or monotonicity conditions can specify one root; a sampled sign-change search is not a proof that every event or tangent contact was found.

This orbital station profile is geometric at one timestamp. It has no light-time or atmospheric-refraction correction. The separate live GPS profile below uses emission time and Earth rotation during flight. Their equations cannot be merged merely by calling both results a ground-station position.

## 5 Logical coverage and implementation boundary

The new chapters supply the expression types and bit-word representation, causal state semantics, exact arithmetic constructions, original attitude recurrences, and explicit lifts of the retained physical and spatial equations. The companion `formal/COVERAGE_AND_PROOFS.md` maps requirements F01–F14 to their definitions and conditional arguments. `formal/STRATEGY_REVISIONS.md` records the proposed changes, deductions and resulting strategy choices.

### 5.1 What the deductions establish

For a finite, well-formed seed, the declared grammar and inverse bit transpose preserve its exact record. Given defined operators, admitted inputs and valid rewrites, each retained state root denotes the exact selected recurrence. Positive-root normalization gives its stated invariant; homogeneous quaternion readout removes roots only where the identity applies. A shared graph can avoid duplicating a construction while retaining all its mathematical dependencies.

These conclusions have visible premises. The review compares definitions, source equations, signs, domains, timing and proof steps. It is human-readable mathematical reasoning, not execution evidence or a proof-assistant certificate. The release contains no new algorithm test, numerical simulation, benchmark or physical-accuracy experiment.

### 5.2 Corrections made by reasoning

The accepted strategy preserves source variants that initially appear similar: the original Madgwick report's magnetic-reference timing and its appendix code are distinct; finite-gain mixing and the simplified correction have different increments; an ambient quaternion gradient and the gradient through normalization are different derivatives. These differences determine the recurrence itself.

Likewise, nine fitted orbital-frame entries need not form an exact orthogonal matrix. Exact RK4 does not acquire the inverse/group property of a continuous flow. The native and Python receiver implementations have different range-domain policies. The formalization names these choices instead of merging them in an equivalence claim. Exact literal imports, lazy branch demand, root signs, input chronology and the winding discarded by modular keys remain explicit.

The adopted improvements are therefore structural: retain exact operators, reuse identical bound expressions, cancel roots under proved premises, separate readout from state, and record semantic

changes as new profiles. There is no claim of a measured execution-speed gain from this reasoning alone.

### 5.3 What is still implementation work

A future executable engine needs a conforming parser, exact integer/rational arithmetic, algebraic representations and decision procedures, scoped graph storage, a sound proof checker, and certified output evaluation. CPU and GPU execution must implement the same semantics and report resource exhaustion explicitly. Neither a finite graph nor one-bit word storage proves that a live workload fits a device's memory or timing budget.

Unrestricted transcendental and implicit-solution decisions do not have a promised general terminating oracle. Retaining a term, an unresolved decision or an unmet domain obligation is a specified outcome. Such an outcome is not a finite numerical prediction. A continuous ODE declaration needs justified existence and continuation, and is not an implemented exact orbit solver.

Physical conclusions additionally require supplied measurements, calibrated frames and clocks, an applicable force/sensor model, and a stated application tolerance. Exact internal arithmetic does not turn uncertain coefficients into physical truth or remove time discretization. The dated R10 measurements that follow remain evidence for their original numerical implementation and sample scope. They do not validate the newly specified exact expression engine.

**Delivered status.** Release 3.6.1.11 is a mathematical specification with logical strategy revisions and editable PDF sources. It is not a newly deployed CPU/CUDA kernel. The preserved 3.6.1.10 runtime remains the executable baseline; the new profile supplies the precise contract for subsequent implementation.

## Inherited 3.6.1.10 formalization and dated evidence

The following source chapters are preserved from the delivered R10 corpus. Their release-specific statements, timings, tests and measured errors refer to those earlier implementations. They are included for the actual equations and source lineage; they are not new evidence for the exact operator profile.

## 6 Seed execution optimization in 3.6.1.10

This release applies the usable word-organization ideas from the newly supplied 21-page design document to the complete orbital seed. The source audit in the next section distinguishes the source's equations from the corrected executable adaptations. The preserved parent is 3.6.1.9. The seed schema remains ORBIT-SEED-R1, schema version 3.6.1.9; release and schema versions are deliberately separate so existing seed bytes and content identities remain valid.

### 6.1 Exact six-stage word transpose

Let  $w_r$  be one of 64 numeric bit patterns and let  $a_{r,c} = ((w_r \gg c) \& 1)$  for  $0 \leq r, c < 64$ . The storage operation is the bit-matrix transpose

$$p_c = \sum_{r=0}^{63} a_{r,c} 2^r.$$

Both the old and optimized implementations produce exactly these words. The new implementation places the block in an exact multiword temporary,  $V = \sum_r w_r 2^{64r}$ . For each  $s = 32, 16, 8, 4, 2, 1$ , define

$$M_s = \sum_{\substack{r \& s = 0 \\ c \& s \neq 0}} 2^{64r+c}, \quad d_s = 63s,$$

and perform the simultaneous exchange

$$T = (V \oplus (V \gg d_s)) \& M_s, \quad V \leftarrow V \oplus T \oplus (T \ll d_s).$$

Each stage exchanges one row-index bit with its corresponding column-index bit. The six exchanges transpose the complete block. Applying the transpose twice is the identity. This is a binary permutation, not a Hadamard transform of the satellite coordinates. Python's integer is an exact multiword temporary; the serialized representation still uses little-endian uint64 words.

The codec retains plane-major order, numeric type tags, zero tail padding, canonical byte encoding and SHA-256 content identity. Negative zero, subnormal binary64 values and signed/unsigned integer limits retain every bit. Both legacy canonical and bit-plane codecs remain readable. The public encoder and decoder now share the 300,000-word count limit.

### 6.2 Measured packing improvement

An independent mapping check covers all 4,096 basis bits. Because every operation is linear over GF(2), the basis check establishes the specified mapping for every block. Random tests also cover 137 lengths, including partial blocks and transitions between groups. All eight January/February example seeds repack to exactly their original bytes and retain their physical-model hashes.

| Operation            | Parent median | New median | Ratio |
|----------------------|---------------|------------|-------|
| Words to planes      | 5.858 ms      | 0.228 ms   | 25.72 |
| Planes to words      | 6.167 ms      | 0.145 ms   | 42.55 |
| Complete seed pack   | 8.783 ms      | 3.162 ms   | 2.78  |
| Complete seed unpack | 17.103 ms     | 5.528 ms   | 3.09  |

These are local host measurements for the C03 seed, with nine measurements of five calls after warmup. Full ranges, Python version, source hashes and seed identities are recorded in `results/seed_word_optimization/benchmark_final.json`. The ratios describe packing and loading; they are not claims about orbital integration throughput. Seed sizes and physical prediction errors do not change as a result of this permutation.

### 6.3 Prepared queries with consistent state identity

The GCRS-to-ECEF query evaluates the frame and derivative together,  $\mathcal{F}(t) = (M(t), \dot{M}(t))$ , sharing the same precession, Earth rotation, polar motion and polynomial evaluations. The resulting operations retain the parent’s arithmetic ordering. Station ECEF origins and ENU matrices use a bounded cache keyed by the exact binary64 latitude, longitude and height bits. Parsed Boolean inputs use a bounded cache keyed by canonical feedback content. Signed zero and edited values cannot alias a different cached input.

A worker holds the physical model snapshot used to create its numeric transport. A session holds one complete seed snapshot and checks its physical-model identity against the worker. Public seed/model views are read-only. To edit, use `session.copy_seed()` or `worker.copy_model()` and construct a new session or worker. A station or feedback edit can reuse the same worker when the physical model is unchanged. This repairs the prior possibility of using an edited Python frame with an older native model or a stale trace identity.

With one fixed parent binary/adaptor to isolate Python preparation, five-repeat median measurements give 1.15–1.29 times faster 256-query workloads and 1.20–1.38 times faster first-day schedules across the four objects. Complete mathematical outputs match the parent exactly. This comparison does not claim a whole-release cold integration speedup. See `results/query_overhead_comparison_final.json` and `results/query_snapshot_independent_check.json`.

### 6.4 Native domain, surface and repeated-query behavior

ORBIT-SEED-NATIVE-3 carries the dynamics kind and exact declared time interval before the existing numeric model fields. Versions 1 and 2 remain readable; their domain is the intersection of the required forcing/frame series. The new worker rejects queries outside the declared model interval, inconsistent epochs and physically undefined propagation at or below the reference Earth surface. Python already rejected Earth-penetrating acceleration; the old native path could return it as a valid state. These checks now agree.

For the canonical integration lattice  $t_n = nh$ , each RK4 call receives its exact indexed endpoint  $t_{n+1}$ , or the exact requested fractional endpoint. This avoids rejecting a valid closed boundary when floating-point evaluation of  $nh + h$  differs from  $(n + 1)h$ . Midpoint labels stay inside that endpoint interval. The existing 30 s and 7.5 s frozen model grids retain their numerical states.

Within a batch, timestamps are keyed by their exact binary64 representation, including the distinction between positive and negative zero. First occurrences are propagated; later identical entries reuse the result and preserve input order. For 64 copies of the 300 s test timestamp, actual RK4 work decreases from 640 steps to 10. Reused rows report zero additional steps and an explicit reuse flag. CUDA still pays launch/upload overhead; this work reduction is not a claim of a

64-fold wall-clock improvement.

## 6.5 Fresh execution and compatibility evidence

The CPU replay reconstructs all 15,553 archived January/February forecast positions exactly, at the original reference timestamps. All eight first-day reference comparisons remain below the adopted 10 m engineering budget. Packing itself adds zero numerical error. The eight complete seed files remain 8,716–9,147 bytes across both dates and retain their original hashes.

The native audit compares 1,040 actual CUDA trajectories and 256 complete word transitions, plus 108 cache replay queries. The largest sampled CPU/CUDA position difference remains  $4.31961135 \times 10^{-5}$  m, and velocity difference remains  $8.07049 \times 10^{-9}$  m/s. Compute Sanitizer reports zero errors for its 129-query trajectory/word workload. Both CPU and CUDA builds pass all four CTests. The broader archived CUDA replay checks 15,553 epochs: all 8,064 January predictions match the parent CUDA positions exactly. February's 7,489 comparisons use the CPU baseline and reach 0.220015 mm maximum difference, within the 10 mm CPU/CUDA contract. This broader sample set is distinct from the 1,040-query audit above. The fresh packed-only replay denies reads of the original release, exercises both ASA/NA sets and a counterfactual JK equation, and refines seven-day events for all four satellites at both stations. Independent retained SATNAV verification checks 256 epochs and 4,828 fields.

The final machine-readable index `validation_results.json` binds the new runtime files, binaries and individual reports. It distinguishes inherited physical-reference evidence from these new execution checks. Detailed old investigation folders retain their original status and version; they do not supersede the final index.

## 6.6 Validation scope

The following physical equations and January/February accuracy measurements were established in 3.6.1.9. They remain baseline evidence for the same frozen models. The new release checks serialization, native execution, replay and query results separately. An exact pack does not remove floating-point integration error or unknown future manoeuvres. The previously reported longer-horizon failures and missing reference intervals remain part of the measured system limits.

## 6.7 Supplied phi source: exact transfers and corrections

The useful transfer is an explicit finite-state, bitfield architecture. R10 applies an exact bit permutation to its existing lossless bitplane format. The supplied text does not establish replacement orbital dynamics or a more accurate geometric bearing. Its claimed properties must follow from its equations, separately from the names given to them.

**Provenance.** The supplied 21-page PDF is an exported Google Search AI conversation, with pseudocode and mathematical claims. References here use its visible physical page numbers, checked against rendered pages. The source is `C:/TOMONETMALFORMED/3.6-phi-2pi-fibonacci-36-10---07---1990-gregorian-calander-Tom-Klootwijk.pdf`. Its SHA-256 is `0fddb55ba01bad0c94b435ddf89c4b778e8a94776946f6cc60cfbed860c5116`. The full decision record is `docs/PHI_SOURCE_AUDIT.md`. No future target observations or changed force parameters enter this transfer.

**Metric and information counterexamples.** For source page 5, let  $\phi = (1 + \sqrt{5})/2$ ,  $H_4 = H_2 \otimes H_2$  and  $A = \text{diag}(\phi, \phi^{-1}, 1, -1)H_4$ . Then

$$H_4^T H_4 = I, \quad |\det A| = 1, \quad \|A(1, 0, 0, 0)^T\|_2^2 = \frac{\phi^2 + \phi^{-2} + 2}{4} = \frac{5}{4}.$$

Table 1: Source-to-kernel decisions. Page numbers refer to the supplied PDF.

| Pages    | Construction                     | Decision and usable scope                                                                                     |
|----------|----------------------------------|---------------------------------------------------------------------------------------------------------------|
| 7–12     | Word fields and Boolean pipeline | Accept explicit widths, masks, signedness and update order; retain all physical numeric bits.                 |
| 4, 8     | Phase sweep                      | Correct to a declared periodic phase plus cycle count. A sawtooth is not itself a Fourier transform.          |
| 5, 9     | Hadamard and pinion scaling      | The normalized Hadamard is orthogonal. The complete scaled map is not; masked integer code loses information. |
| 5, 11    | Golden-ratio rotation            | A declared sampling sequence is valid; dyadic shifts are not exact multiplication by $\phi$ .                 |
| 3–5, 15  | One-bit shape and LUT            | Occupancy predicates, not signed distances. Require a domain, geometry and complete LUT.                      |
| 6–7, 10  | Branch culling                   | Valid only if the bound contains all transformed descendants; the printed proof does not establish this.      |
| 6, 12–14 | Loop and Klein claim             | Finite-state feedback is usable. A parity flip does not prove a Klein-bottle surface or non-orientability.    |
| 13–14    | Log radius, binary angle         | Valid chart and modular arithmetic after domain and mask corrections; quantization is explicit.               |
| 16–17    | Source OTAN2                     | Radius and occupancy cannot recover geometric direction. Preserve the actual station-relative vector.         |
| 17, 20   | Infinite nesting in a word       | A 64-bit state has at most $2^{64}$ distinct values. External state must be accounted for.                    |

Thus  $A$  preserves four-dimensional volume magnitude, not Euclidean lengths or all areas. Page 9’s integer implementation also maps both  $(0, 0, 0, 0)$  and  $(-1024, -1024, 0, 0)$  to zero after its masks: the second input has sums  $(-2048, 0, -2048, 0)$ . Both inputs fit the declared fields. This arithmetic cannot serve as a lossless physical-state encoding.

**Executable corrections.** Page 8’s conditional mask is always zero because a right shift by 63 yields only 0 or 1, while its mask has bit zero clear. Page 9’s bit complement represents  $-x - 1$ , not  $-x$ . Page 12’s addition of 4 is not a bit-2 toggle:  $4 + 4 = 8$ , whereas XOR toggles exactly the desired bit. On page 14, a half-turn in the 11-bit angle field [51 : 41] requires XOR of word bit 51; the printed mask flips ten bits. Finally, declaring left-to-right notation does not alter C/C++ precedence. For page 16’s expression, log-radius 2 and occupancy 0 produce 2 under C rules but 0 under strict left-to-right evaluation. Explicit parentheses or a specified parser are necessary.

**Ground-station direction and chart limits.** The points  $(1, 0)$  and  $(0, 1)$  have equal radius and can have equal occupancy, yet their bearings differ by  $\pi/2$ . The source’s radius/occupancy OTAN2 therefore cannot replace geometric azimuth; it is also distinct from the existing UGTS OTAN2 profile. Keep the physical relative position, ENU axes, range, and defined cases for azimuth/elevation. An 11-bit full-turn angle has a half-bin rounding error of  $0.087890625^\circ$ . At an illustrative  $R = 42,164,000$  m,

$$e_{\text{chord}} = 2R \sin(\pi/4096) = 64,678.76 \text{ m.}$$

Even a single-angle 10 m quantization budget at this radius requires at least 24 angular bits; this allocates nothing to other error sources. A station calculation uses actual slant range. Log radius requires  $\rho = \log_2(r/r_\star)$  with  $r, r_\star > 0$ ; the origin has no finite value. The page-4 log of a trigonometric product is undefined at zeros and unbounded below, so it cannot establish page 7’s assumed nonnegative monotonic motion. A parent-local interval  $[1, 2]$  misses the slit  $x = 0$ , while its child translated by  $-1$  intersects it. Culling needs an envelope of every descendant. Similarly,  $\det \text{diag}(-1, 1, -1) = +1$ : two axis flips do not establish non-orientability.

**Exact time and derived keys.** For integer ticks  $n$ , reference tick  $n_0$  and positive integer period  $P$ ,

$$n - n_0 = qP + r, \quad q = \left\lfloor \frac{n - n_0}{P} \right\rfloor, \quad 0 \leq r < P.$$

Keeping both  $q$  and  $r$  preserves time exactly, including negative elapsed ticks; phase alone aliases times separated by a period. Orbital GPST/UTC/TT/TDB conversion and the full epoch remain authoritative. The calendar and constants in the filename derive no orbital initial condition or force constant. Log-polar values, occupancy and phase can supply derived display addresses or keys with declared origin, domain, quantization and winding. They do not replace metre-valued geometry or establish cryptographic unpredictability.

**The accepted R10 packing construction.** For a  $64 \times 64$  bit array, the transpose is

$$B'_{c,r} = B_{r,c}, \quad T^2 = I.$$

Six stages exchange corresponding row-index and column-index bits. Each stage is a permutation, so no numeric information is lost. R10 uses this exact transpose for its existing bitplane format, retaining binary64 bit patterns, types, ordering, padding and canonical digest. This is an engineering construction inspired by the source's binary-word organization; the source does not itself specify or prove this algorithm. It imposes no Hadamard arithmetic or golden-ratio quantization on physical coordinates. Release round-trip checks and measured benchmarks establish correctness and speed; this source audit makes no timing claim. Representational exactness does not remove the independently measured physical-model discrepancy.

## 7 Gregorian calendar continuity and exact phase carry

One January-to-December sequence does not cover the Gregorian leap rules. The calendar proceeds into the next year; a wrapped address phase does not restart the calendar. The new verification enumerates every date from 1 January 2000 through 31 December 2399 and includes the following 1 January 2400 boundary. This is a calendar and address check, separate from satellite forecast accuracy.

### 7.1 Independent date arithmetic

For year  $Y$ , month  $M$  and day  $D$ , the arithmetic oracle is

$$\begin{aligned} a &= \lfloor (14 - M)/12 \rfloor, & y &= Y + 4800 - a, & m &= M + 12a - 3, \\ J &= D + \lfloor (153m + 2)/5 \rfloor + 365y \\ &\quad + \lfloor y/4 \rfloor - \lfloor y/100 \rfloor + \lfloor y/400 \rfloor - 32045, \\ T &= 86400 [J - J(1980, 1, 6)]. \end{aligned}$$

Here  $J$  is the astronomical-noon Julian day number associated with a date; integer date differences produce the corresponding midnight tick  $T$ . Expected dates use explicit month enumeration, checked against both this March-based formula and an independent January-based ordinal sum. The expected value oracle does not use the runtime's `datetime` implementation.

A year is leap when divisible by four, except centuries not divisible by 400. The complete cycle therefore has

$$97 = 100 - 4 + 1 \quad \text{leap days}, \quad 400 \cdot 365 + 97 = 146097 = 7 \cdot 20871 \quad \text{days}.$$

The month/leap pattern and weekdays repeat after 400 years. The year number and absolute time continue. For each enumerated date, the arithmetic difference to the same date 400 years later is also checked; this does not add another 400 years of formatter or physical-orbit coverage.

## 7.2 The phase wraps; absolute time continues

For integer reference  $T_0$  and period  $P = 2^{14}$ , the literal address helper uses

$$(w, r) = \text{divmod}(T - T_0, P), \quad T = T_0 + Pw + r, \quad 0 \leq r < P, \quad f = r/P.$$

Arithmetic right shift and a 14-bit mask form the independent test oracle. The fraction  $f$  is exactly representable in binary64. All phase values are tested at three signed winding offsets, with additional signed tick/reference cases reaching  $10^{100} + 16385$ . This arbitrary-size claim concerns the integer helper, not binary64 timestamp input.

The contiguous and Morton key layouts store the remainder  $r$  in their time field. A key may repeat after  $P$  ticks while winding  $w$  increases. Tests check both layouts against independently assembled bit strings and retain absolute tick, winding and remainder in every daily chart record. At a one-second tick period, the phase repeats after 16,384 seconds; it is not a calendar month or year.

## 7.3 Recorded result and limits

| Check coverage                                 | Count               |
|------------------------------------------------|---------------------|
| Dates in the complete cycle                    | 146,097             |
| Daily midnights including the next boundary    | 146,098             |
| Month starts / year starts                     | 4,800 / 400         |
| Leap days                                      | 97                  |
| Daily GPST formatter comparisons               | 438,293             |
| Additional century/epoch formatter comparisons | 22                  |
| Phase wraps / adjacent tested ticks            | 770,433 / 2,311,299 |
| Complete phase cases at three signed offsets   | 49,152              |
| Large signed tick/reference cases              | 210                 |
| Complete daily chart/key records               | 146,098             |
| Exact comparisons / failures                   | 9,519,315 / 0       |

Seven focused regression tests also pass, including checks that deliberately broken calendar-reset and lost-winding substitutes are detected. Explicit century cases cover common years 1900, 2100, 2200 and 2300, and leap years 2000 and 2400. The 2400 leap-day cases are separate checks outside the enumerated cycle. Daily checks exercise midnight, an exact 0.125-second fraction, and the preceding date's last whole second.

The machine-readable result is `results/calendar_phase_36110/summary.json`; it records the checked source hashes and all eight unchanged seed hashes. Reproduce it with `python tools/validate_calendar_phase.py`. No runtime equations, floating-point propagation or force-model values changed.

The formatter labels uniform 86,400-second days as GPST. This audit does not validate a UTC leap-second table, predict future UTC offsets, certify sub-millisecond display accuracy, or expand any seed's physical timestamp domain. The calendar check supplies no 400-year orbit forecast or additional satellite-position accuracy guarantee.

## 8 ORBIT-SEED-R1: compact physical seeds and timestamp queries

This profile predicts the position of a satellite relative to a specified ground station. It includes real MEO, GEO, inclined geosynchronous and high elliptical examples. A small, complete model seed is the input; a requested timestamp is the query. The model regenerates position and velocity with explicit dynamics. The two ordered ASA/NA sets and the synchronous JK recurrence select subsequent query times in ordered replay. Direct scrubbing obtains the same physical state without replaying the preceding display interactions.

### Complete binary seeds; sizes are measured in the execution table.

Each seed includes the initial state, force constants, compact Sun/Moon and frame coefficients, provenance, both demonstration stations, time/query settings, two ASA/NA mask sets and editable Boolean equations. No future target trajectory table is stored. The native executable and its declared runtime are shared dependencies, accounted for separately from per-seed bytes.

The four fitted states use data at or before 2 January 2025 00:00 GPST. The following seven days are excluded from fitting. Their earlier R1 results were inspected before R2 development, so this interval is a reused benchmark. These are archived examples, not a statement of current satellite positions. An arbitrary future manoeuvre cannot be recovered from a prior seed that does not contain that manoeuvre. Compactness is a consequence of a specified dynamical model and initial conditions; it does not provide lossless compression of every possible orbit.

### 8.1 The complete state seed

Let the decoded seed be

$$S = (I, t_0, \mathbf{r}_0, \mathbf{v}_0, \mathcal{F}, C, M, N, \mathcal{G}, Q, \mathcal{W}, \mathcal{P}), \quad (40)$$

where  $I$  identifies the object and model version,  $\mathcal{F}$  contains all force constants and fitted accelerations,  $C$  contains compact planetary forcing,  $M$  defines the frame/time conversion,  $N$  defines numerical integration,  $\mathcal{G}$  defines the stations,  $Q$  defines query/chart parameters,  $\mathcal{W}$  defines the literal word equations and masks, and  $\mathcal{P}$  records construction provenance. Position and velocity are six IEEE 754 binary64 SI numbers, not quantized UGTS addresses.

The upgraded physical model is ORBIT-DYNAMICS-R2; the decoder also retains R1 support. Its state is geocentric and aligned with GCRS/ICRF axes. Full GPST seconds since 6 January 1980 are retained. For an elapsed query  $\tau$ ,  $t = t_0 + \tau$  and

$$TT = GPST + 51.184 \text{ s}, \quad JD_{TT} = 2444244.5 + (t + 51.184)/86400. \quad (41)$$

The Chandra input is labelled TDB and is explicitly converted during construction and review. GPST, UTC, TT and TDB calendar labels are not interchanged. At this example epoch GPST is 18 seconds ahead of UTC.

## 8.2 Explicit Earth frame and compact forcing

Define the passive Earth rotation matrix and polar-motion matrix as

$$R_z(a) = \begin{pmatrix} \cos a & \sin a & 0 \\ -\sin a & \cos a & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad (42)$$

$$P = \begin{pmatrix} c_x & 0 & s_x \\ s_y s_x & c_y & -s_y c_x \\ -c_y s_x & s_y & c_y c_x \end{pmatrix}, \quad c_x = \cos x_p, \quad s_x = \sin x_p, \quad c_y = \cos y_p, \quad s_y = \sin y_p. \quad (43)$$

With the embedded slowly varying celestial matrix  $Q(\tau)$ ,

$$a(\tau) = a_0 + \dot{a}\tau + \delta a(\tau), \quad M(\tau) = P(\tau)R_z(a(\tau))Q(\tau), \quad (44)$$

$$\mathbf{r}_E = M\mathbf{r}, \quad \mathbf{v}_E = M\mathbf{v} + \dot{M}\mathbf{r}. \quad (45)$$

Thus Earth-fixed velocity includes the time derivative of the frame. The construction uses IAU 2006/2000A and the IERS Bulletin A published on 26 December 2024, before the forecast epoch. Published daily  $x_p, y_p$  and DUT1 values are linearly interpolated. The ERA correction is  $\delta a(\tau) = \dot{a}[\text{DUT1}(\tau) - \text{DUT1}(0)]$ , with  $\dot{a} = 2\pi(1.00273781191135448)/86400$ . The velocity derivative includes  $\dot{P}$ ,  $\dot{\delta a}$  and  $\dot{Q}$ . TIO locator  $s'$  is zero. Published EOP predictions carry their own uncertainty; their interpolation is not a future observed-EOP measurement.

Each slowly varying scalar coefficient block uses a finite Chebyshev expansion:

$$u(\tau) = \frac{2\tau - \tau_a - \tau_b}{\tau_b - \tau_a}, \quad f(\tau) = \sum_{k=0}^d c_k T_k(u), \quad T_0 = 1, \quad T_1 = u, \quad T_{k+1} = 2uT_k - T_{k-1}. \quad (46)$$

The seed stores the interval, degree and every coefficient.  $Q$  has nine degree-eighteen rows; Sun and Moon forcing have three rows of degree sixteen and thirty respectively over the fifteen-day construction domain. Daily EOP blocks have three degree-one rows:  $\delta a, x_p, y_p$ . Rapid Earth rotation remains a direct trigonometric rotation. At shared segment boundaries the later segment applies. Queries outside the embedded domain return an error rather than silently extrapolating the coefficient blocks.

The Sun/Moon coefficients approximate the DE440s planetary model published in 2020. They describe known forcing from other bodies, not withheld positions of the target satellite. The measured off-node forcing discrepancies are about 1.99 m for the Sun and 0.060 m for the Moon. These quantities are not satellite orbit accuracy. Original DE440 files are unnecessary during packed replay.

## 8.3 Actual orbital differential equations

The physical state evolves as

$$\frac{d\mathbf{r}}{d\tau} = \mathbf{v}, \quad \frac{d\mathbf{v}}{d\tau} = M^T \nabla_{\mathbf{p}} U(\mathbf{p}) + \mathbf{a}_{\odot} + \mathbf{a}_{\text{Moon}} + \mathbf{a}_{\text{SRP}} + \mathbf{a}_{\text{RTN}} + \mathbf{a}_{\text{GR}}, \quad \mathbf{p} = M\mathbf{r}. \quad (47)$$

Here the Moon acceleration symbol denotes the declared lunar third-body term. For  $R = \|\mathbf{p}\|$ , latitude sine  $s = p_z/R$ , longitude  $\lambda$ , Earth radius  $a_E$  and parameter  $\mu$ , the full degree/order twelve field is

$$U(\mathbf{p}) = \frac{\mu}{R} \sum_{n=0}^{12} \left(\frac{a_E}{R}\right)^n \sum_{m=0}^n P_{nm}(s) [C_{nm} \cos(m\lambda) + S_{nm} \sin(m\lambda)]. \quad (48)$$

$P_{nm}$  has no Condon–Shortley phase. The seed stores unnormalized  $C_{nm}$ ,  $S_{nm}$ , converted from the EGM96 fully normalized coefficients by

$$C_{nm} = \sqrt{(2 - \delta_{m0})(2n + 1) \frac{(n - m)!}{(n + m)!}} \bar{C}_{nm}, \quad (49)$$

and the same factor for  $S$ . It includes  $C_{00} = 1$  and  $S_{00} = 0$ . The positive-potential convention uses  $+\nabla U$ .

Native evaluation avoids longitude and polar division. Define  $X = a_E p_x / R^2$ ,  $Y = a_E p_y / R^2$ ,  $Z = a_E p_z / R^2$ ,  $D = a_E^2 / R^2$ ,  $V_{00} = a_E / R$  and  $W_{00} = 0$ . The Cartesian Cunningham recurrence is

$$V_{mm} = (2m - 1)(XV_{m-1,m-1} - YW_{m-1,m-1}), \quad (50)$$

$$W_{mm} = (2m - 1)(XW_{m-1,m-1} + YV_{m-1,m-1}), \quad (51)$$

$$V_{nm} = \frac{2n - 1}{n - m} ZV_{n-1,m} - \frac{n + m - 1}{n - m} DV_{n-2,m}, \quad (52)$$

$$W_{nm} = \frac{2n - 1}{n - m} ZW_{n-1,m} - \frac{n + m - 1}{n - m} DW_{n-2,m}, \quad (53)$$

$$U = \frac{\mu}{a_E} \sum_{n=0}^{12} \sum_{m=0}^n (C_{nm} V_{nm} + S_{nm} W_{nm}). \quad (54)$$

The diagonal recurrence uses  $m = 1, \dots, 12$ ; the off-diagonal recurrence uses  $m = 0, \dots, 12$  and  $n = m + 1, \dots, 12$ . For  $n = m + 1$  the second term is absent. Each native recurrence value carries three analytic Cartesian derivatives; the product rule produces  $\nabla U$ . Independent Python validation differentiates the scalar potential with complex steps, and native tests compare a separately coded spherical-gradient oracle. The constants are  $\mu = 3.986004415 \times 10^{14}$  SI and  $a_E = 6378136.3$  m. This is the tide-free EGM96 field truncated at degree/order twelve.

For each third body  $b$  with Earth-centred position  $\mathbf{r}_b$ ,

$$\mathbf{a}_b = \mu_b \left[ \frac{\mathbf{r}_b - \mathbf{r}}{\|\mathbf{r}_b - \mathbf{r}\|^3} - \frac{\mathbf{r}_b}{\|\mathbf{r}_b\|^3} \right]. \quad (55)$$

The indirect term removes the acceleration of the Earth. With  $\mathbf{d} = \mathbf{r} - \mathbf{r}_\odot$ , the fitted radiation-pressure term is

$$\mathbf{a}_{SRP} = \chi A \left( \frac{\text{AU}}{\|\mathbf{d}\|} \right)^2 \frac{\mathbf{d}}{\|\mathbf{d}\|}. \quad (56)$$

$\chi = 0$  in the declared cylindrical Earth shadow when the seed's shadow flag is enabled, and 1 otherwise. With shadow disabled,  $\chi = 1$  everywhere. Explicitly, if  $\hat{\mathbf{s}} = \mathbf{r}_\odot / \|\mathbf{r}_\odot\|$ , shadow requires  $\mathbf{r} \cdot \hat{\mathbf{s}} < 0$  and  $\|\mathbf{r} - (\mathbf{r} \cdot \hat{\mathbf{s}})\hat{\mathbf{s}}\| < a_E$ . There is no hidden penumbra, attitude or manoeuvre model. Finally,

$$\hat{\mathbf{R}} = \mathbf{r} / \|\mathbf{r}\|, \quad \hat{\mathbf{N}} = (\mathbf{r} \times \mathbf{v}) / \|\mathbf{r} \times \mathbf{v}\|, \quad \hat{\mathbf{T}} = \hat{\mathbf{N}} \times \hat{\mathbf{R}}, \quad (57)$$

$$\mathbf{a}_{RTN} = a_R \hat{\mathbf{R}} + a_T \hat{\mathbf{T}} + a_N \hat{\mathbf{N}}. \quad (58)$$

The Schwarzschild correction is explicit, with  $r = \|\mathbf{r}\|$ :

$$\mathbf{a}_{GR} = \frac{\mu}{c^2 r^3} \left[ \left( \frac{4\mu}{r} - \mathbf{v} \cdot \mathbf{v} \right) \mathbf{r} + 4(\mathbf{r} \cdot \mathbf{v})\mathbf{v} \right], \quad c = 299792458 \text{ m/s}. \quad (59)$$

The three signed empirical accelerations and  $A$  are fitted only on the declared training arc. Their continuation is a physical approximation, not knowledge of future spacecraft operations. The same ten-parameter fit is fixed for all four objects: state six, SRP one and RTN three. Earlier training-window candidates of one, three and seven days are compared against the earlier January

1 interval; the lowest maximum error selects the window. The final fit uses only data at or before the January 2 cutoff. This past-only selection is distinct from the already inspected January 2–9 comparison benchmark. Harmonics above degree twelve, tides, detailed radiation pressure, attitude changes and manoeuvres remain possible sources of forecast disagreement.

#### 8.4 Bottom-up timestamp reconstruction and scrubbing

Native CPU/CUDA integrate these equations with classical RK4. For a six-vector  $\mathbf{y} = (\mathbf{r}, \mathbf{v})$  and derivative  $F$ ,

$$\mathbf{k}_1 = F(t, \mathbf{y}), \quad \mathbf{k}_2 = F(t + h/2, \mathbf{y} + h\mathbf{k}_1/2), \quad (60)$$

$$\mathbf{k}_3 = F(t + h/2, \mathbf{y} + h\mathbf{k}_2/2), \quad \mathbf{k}_4 = F(t + h, \mathbf{y} + h\mathbf{k}_3), \quad (61)$$

$$\mathbf{y}^+ = \mathbf{y} + \frac{h}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4). \quad (62)$$

MEO/GEO/IGSO use  $h = 30$  seconds; Chandra uses 7.5 seconds, selected through an independent numerical-convergence check. Its fitted initial state and forces were not refitted against future positions. Independent Python propagation uses DOP853 rather than the native RK4 implementation.

For any elapsed time  $\tau$ , define  $n = \text{trunc}(\tau/h)$  and  $\epsilon = \tau - nh$ . Canonical lattice states are obtained by repeated signed steps from the packed epoch. The exact requested state is one final fractional step of length  $\epsilon$  from the canonical state at  $nh$ . Fractional results never become canonical checkpoints. This makes the physical query independent of the order in which the timeline was scrubbed.

The CPU worker retains at most the declared number of sparse checkpoints (256 in the examples, at one-hour spacing). Missing checkpoints regenerate from an available earlier canonical state or the epoch. Eviction changes work, not the mathematical trajectory. Cold-query work is proportional to elapsed integration steps; compact storage does not imply constant-time propagation. CUDA batches assign independent timestamp queries to device work and integrate from the epoch. They currently do not share the CPU checkpoint cache. CPU scrubbing and parallel CUDA batch execution are distinct measured paths.

#### 8.5 Two vacuum/set stages and synchronous JK

The user’s “double vacuum, double set” is concretely defined here as exactly two independently stored ASA/NA/boundary mask triples,  $(A_1, N_1, B_1)$  and  $(A_2, N_2, B_2)$ , applied in order. No unspecified operation is inferred from the phrase. All operators below are 32-bit word operators, and  $P$  is the present mask. For stage  $i$  and input  $w$ ,

$$a_i(w) = w \& A_i \& P, \quad n_i(w) = a_i(w) \& N_i, \quad (63)$$

$$h_i(w) = \text{popcount}(n_i(w) \& B_i), \quad V_i(w) = \begin{cases} 0, & h_i(w) > 0, \\ n_i(w), & h_i(w) = 0. \end{cases} \quad (64)$$

One selected boundary hit therefore absorbs the entire output word. It does not merely clear the intersecting bit. At ordered query  $j$ ,

$$\mathbf{s}_j = \Phi(\mathcal{S}, \tau_j), \quad g_j = G(\mathbf{s}_j, \mathcal{G}, \tau_j), \quad (65)$$

$$d_j = E_D(g_j, \tau_j), \quad x_j = E_X(q_j, d_j), \quad (66)$$

$$y_j = V_2(V_1(x_j)), \quad J_j = E_J(q_j, y_j, d_j), \quad (67)$$

$$K_j = E_K(q_j, y_j, d_j), \quad q_{j+1} = [(J_j \& \neg q_j) \mid (\neg K_j \& q_j)] \& P, \quad (68)$$

$$\tau_{j+1} = \min(\tau_{\text{end}}, \tau_j + \Delta(q_{j+1})). \quad (69)$$

Both JK right-hand sides read the same old  $q_j$ . Boundary absorption does not implicitly freeze JK. The editable X expression accepts  $q, d$ ; J and K accept  $q, y, d$ . They support word constants zero/ones and bitwise AND, OR, XOR and complement with parentheses. They compile to literal Boolean truth tables; an independent AST interpreter checks the complete trace.

Default predicate bits are, in ascending order: above the station elevation mask, within three degrees of that mask, absolute seed age  $|\tau|$  at least 24 hours, positive elevation rate, and approaching range. The actual default expressions are

$$x = q \mid d, \quad J = y \& d, \quad K = \neg d. \quad (70)$$

With both unabsorbed identity mask sets these reduce to  $q^+ = d$ . The state still selects the next physical query timestamp:  $\Delta = 30$  seconds when  $(q^+ \& 6) \neq 0$ , and 300 seconds otherwise. Different mask/equation inputs change that recurrence, as the counterfactual and absorption tests demonstrate. Physical acceleration never reads  $q$ ; the CGK spring equations are retained separately rather than renamed orbital gravity.

## 8.6 Station-relative geometry and both horizons

For WGS84 station position  $\mathbf{r}_g$  and its standard East/North/Up rotation  $L(\varphi, \lambda)$ ,

$$\mathbf{z} = (E, N, U)^T = L(\mathbf{r}_E - \mathbf{r}_g), \quad \dot{\mathbf{z}} = L\mathbf{v}_E. \quad (71)$$

Set  $H = \sqrt{E^2 + N^2}$  and  $D = \|\mathbf{z}\|$ . The returned quantities are

$$\text{az} = \text{atan2}(E, N) \bmod 2\pi, \quad \text{el} = \text{atan2}(U, H), \quad (72)$$

$$\dot{D} = \mathbf{z} \cdot \dot{\mathbf{z}}/D, \quad \frac{d \sin \text{el}}{dt} = \dot{U}/D - U\dot{D}/D^2. \quad (73)$$

Azimuth is clockwise from North. At the horizontal origin azimuth and its derivative are explicitly undefined. Angles are simultaneous geometric values: there is no atmospheric refraction or retarded light-time correction here.

Visibility events solve

$$f(t) = \sin \text{el}(t) - \sin \text{el}_{\text{mask}} = 0. \quad (74)$$

The search refines sign brackets and derivative-bracketed extrema, so it can recover a short pass whose sampled endpoints are both below the mask. It reports grazing candidates and numerical root tolerances. The validation compares 120-second and 30-second searches over seven days for both stations and every object. Without a global interval bound this is not a proof that arbitrarily many roots cannot occur between samples. Always-visible/hidden labels mean visible/hidden over the searched interval under the stated search policy.

The second horizon is an error-budget horizon. Given withheld reference samples  $\mathbf{r}_k^*$ ,  $e_k = \|\mathbf{r}(t_k) - \mathbf{r}_k^*\|$ . For budget  $b$  the report records the first sample with  $e_k > b$  and its predecessor. The interval brackets a sampled exceedance; it does not certify continuous error below  $b$  before it. An epoch can be inside the model's coefficient domain while outside a useful accuracy horizon. Maximum error across the whole interval is reported alongside endpoint error because an orbit can diverge and later approach the reference.

## 8.7 Original UGTS keys and complete time

The orbital chart uses  $\rho = \log(H/r_0)$ ,  $\theta = \text{atan2}(N, E)$  and the explicit separate hoop angle  $\phi$ . The source chart is used only when  $-20 \leq \rho \leq 0$ ; out-of-range coordinates are not wrapped into a valid key. The full tick is  $k = \lfloor t/\delta t \rfloor$ , with integer winding and remainder from  $k = 16384w + k_{\text{phase}}$ . The normalized fraction is separately labelled  $k_{\text{phase}}/16384$ . Source quantization produces the

original (20, 18, 14, 12)-bit tuple. Both the contiguous and original Morton layouts are retained. Full position, velocity, Up, absolute time and provenance remain in the result. Source OTAN2 uses  $\Delta\theta = \text{wrap}(\theta_j - \theta_{j-1})$  and is still  $\arctan(\Delta\theta/\Delta\rho)$ , with its explicit undefined-ratio cases, and is not replaced by azimuth.

## 8.8 Binary packing, integrity and executable entry points

The 52-byte little-endian header contains magic ATOMORB1, format/codec versions, compressed and raw lengths, and the SHA-256 of the domain-prefixed canonical payload. The payload uses tagged null/Boolean/signed and unsigned 64-bit integer/binary64/string/list/map values. Map keys are sorted, unique UTF-8 strings. Codec 1 compresses those canonical bytes directly. The default codec 2 packs the numeric bit patterns into one-bit planes before Zlib compression. It preserves all 64 bits per numeric value. For numeric word  $w_j$ , define

$$P_{b,g} = \sum_{\ell=0}^{63} [(w_{64g+\ell} \text{ shr } b) \& 1] 2^\ell, \quad b = 0, \dots, 63. \quad (75)$$

Absent final lanes are zero. The inverse restores bit  $b$  of word  $j$  from lane  $j \bmod 64$  of  $P_{b,[j/64]}$ . The plane-major uint64 words follow a structural template with original numeric type tags, and a 16-byte BITPLN64 header with template length and word count. Signed integers, unsigned integers and binary64 preserve their original bits; the structural template retains Boolean tokens. Both codecs share the same canonical content digest. Plane packing adds zero numerical error, but does not change floating-point arithmetic in propagation. The measured example packs are slightly larger with this representation than with codec 1. The decoder checks exact lengths, bounded expansion and nesting, complete stream termination, canonical re-encoding, digest, model schema and time domains. There is no executable payload or pickle.

The digest domain is atomOS:ORBIT-SEED-R1:canonical-binary followed by a zero byte. Physical-model and complete-seed hashes differ by scope. Runtime timings and cache occupancy are excluded from the deterministic trace digest; all mathematical query fields and both mask/JK stages remain covered. Independent verification reconstructs the native orbit, checks the full station geometry, keys/time, OTAN2, the AST recurrence, next query time and final chain.

```
python tools/orbital_seed.py inspect --seed examples/orbit/seeds/C03.orbseed
python tools/orbital_seed.py query --seed examples/orbit/seeds/C03.orbseed --worker bin
    /cpu/orbit_worker.exe --station singapore_demo --time-s 12345.678 --out NEW_QUERY.
    json
python tools/orbital_seed.py predict --seed examples/orbit/seeds/G05.orbseed --worker
    bin/cpu/orbit_worker.exe --end-s 86400 --out NEW_RUN
python tools/verify_orbit_run.py --seed examples/orbit/seeds/G05.orbseed --worker bin/
    cpu/orbit_worker.exe --run NEW_RUN --out NEW_VERIFY.json
python tools/orbit_scrub.py --worker bin/cpu/orbit_worker.exe
```

The local timeline is served at <http://127.0.0.1:3619>. It accepts exact fractional-second queries, a slider, object/station selection, ordered feedback playback and visibility events. The JSON companions are editable; packing them rebuilds the binary .orbseed files with a new content identity. Accuracy reports are associated only when their canonical physical-model hash matches the loaded seed.

## 8.9 Packed execution and regression evidence

The following numbers are derived from the final decoded seeds and verified native binaries. The isolated replay child receives only the packed seed, shared runtime source and native executable; an audit hook denies every read under the original release directory. The reconstructed fractional, past and seven-day timestamp states match the normal native run bit for bit.

| Object  | Seed bytes | Default queries | Altered JK | Warm max (ms) |
|---------|------------|-----------------|------------|---------------|
| G05     | 9,066      | 397             | 2,881      | 0.816         |
| C03     | 9,074      | 289             | 2,881      | 0.956         |
| C06     | 9,073      | 417             | 2,881      | 0.834         |
| CHANDRA | 8,716      | 520             | 2,881      | 2.075         |

Query counts cover one day. The counterfactual sets  $J = 1, K = 0$  and changes the actual timestamps selected by the native recurrence. Physical states at shared timestamps remain identical. Warm timing is end-to-end Python/native geometry latency on this host for the recorded fractional seek sequence, not a universal performance bound. Cold seeks may require thousands of integration steps.

Across eight seven-day station/object searches, 83 rise/set events were found. The largest event-time change under refinement from 120-second to 30-second sampling was 0.002508 seconds. The event algorithm's grazing and no-global-root-exclusion qualifications still apply.

A five-minute, seven-day table containing time and six binary64 state components would occupy 112,952 bytes before its metadata. This compares a generated state table to its specified model, not arbitrary-trajectory compression. The demonstration archive also contains large reference datasets and audit traces, which are not per-seed replay inputs.

The table is 12.45 times the size of the largest complete packed seed. The shared CPU executable is 77,824 bytes, and the isolated shared Python source is 89,342 bytes at validation. Installed libraries and the OS/C++ runtime are additional shared dependencies. The default codec packs 64 one-bit planes into uint64 words without discarding any numeric bit. The canonical codec remains readable; bit-plane packing is slightly larger for these examples.

The new native orbital checks compare 1,040 CPU/CUDA positions across the four models, with maximum disagreement 0.00004320 m. All 256 complete CPU/CUDA two-stage word traces match. Bounded-cache stress tests remain bit-exact through repeated eviction. Compute Sanitizer reports zero errors for 129 orbital queries and the word operation. The full Python suite passes 139 tests, including bit-plane reconstruction, rehashed-false-geometry, OTAN2 rejection and missing-reference-interval acceptance.

Fresh retained-component regression passes four CTests on each build, 256 SATNAV epochs per backend with independent reconstruction, 4,225 SRK transitions and 3,308 CGK transitions per backend, and 6,861 persistent live-worker comparisons per backend. The required GPU review tool also records zero errors under both texture and global-memory Compute Sanitizer runs on the actual RTX 5070 Ti Laptop.

The first orbital RK4 test exposed an omitted final state assignment; that defect was fixed before these final results. Earlier failed and superseded investigation folders are retained as such and are not used as final validation. Numerical agreement is separate from the physical future-reference errors below.

## 8.10 R2 physical-position results and separate confirmation

The R2 models stay below the adopted **10 m three-dimensional reference-discrepancy budget at every tested sample through 24 hours** for all four targets, on both the January benchmark and the separate February confirmation date. This statement concerns sampled positions of these seeds; it is not a universal or continuous-time orbit guarantee. The January window was previously inspected during R1 development and is explicitly a **reused benchmark**. The February selection procedure was frozen before its future errors were evaluated; no model was retuned afterward.

The forecast quantity is

$$e_i = \|\hat{\mathbf{r}}(t_i) - \mathbf{r}_{\text{reference}}(t_i)\|_2, \quad t_i > t_0.$$

The January origin is 2 January 2025 00:00 GPST,  $t_0 = 1419811200$ ; February uses 2 February 2025 00:00 GPST,  $t_0 = 1422489600$ . All cutoff and earlier target rows are excluded from the forecast curves. R2 selects among 1-, 3- and 7-day training windows using an internal validation day entirely before the forecast origin. The selected January windows are 3/3/3/1 days for G05/C03/C06/Chandra; the unchanged rule selects 1/7/3/3 days for February. External EGM96 gravity and planetary forcing are distinct from future target observations.

Table 2: Maximum discrepancy over all first-day samples (metres). Each R2 column has 288 samples per target through 24 h; Chandra’s final sample is approximately 51.185 s before the exact endpoint.

| Target        | R1 January | R2 January | R2 February |
|---------------|------------|------------|-------------|
| G05 / MEO     | 285.330    | 1.057      | 3.072       |
| C03 / GEO     | 20.600     | 5.304      | 4.654       |
| C06 / IGSO    | 44.684     | 1.172      | 2.108       |
| Chandra / HEO | 76.506     | 8.093      | 0.290       |

Table 3: R2 January discrepancy in metres at the nearest actual sample to each requested horizon. Individual samples are not worst-case interval errors.

| Target        | 15 min | 1 h   | 6 h   | 24 h  | 72 h   | 7 d      |
|---------------|--------|-------|-------|-------|--------|----------|
| G05 / MEO     | 0.474  | 0.655 | 0.272 | 0.713 | 2.228  | 12.531   |
| C03 / GEO     | 1.368  | 1.702 | 4.350 | 4.172 | 17.939 | 73.481   |
| C06 / IGSO    | 0.547  | 0.609 | 0.615 | 1.172 | 2.335  | 2.360    |
| Chandra / HEO | 0.003  | 0.008 | 0.074 | 2.052 | 28.338 | 3680.247 |

**Time grid.** GNSS is on the exact requested GPST grid. Chandra is supplied on a 300 s TDB grid, converted using the periodic TDB–TT correction and  $\text{GPST} = \text{TT} - 51.184 \text{ s}$ . The nearest 15 min sample is 848.816057 s after the January origin and 848.815204 s after the February origin. The final samples are respectively 604748.815856 s and 604748.815036 s. Exact offsets remain in the CSV/JSON, including all six requested horizons for both dates.

Table 4: R2 whole-window statistics (metres). January has 2,016 samples per target. February has 2,016 each except C03, which has 1,441 and a 48 h gap.

| Target        | Jan. RMS | Jan. maximum | Feb. RMS  | Feb. maximum |
|---------------|----------|--------------|-----------|--------------|
| G05 / MEO     | 5.294    | 12.998       | 33.113    | 71.800       |
| C03 / GEO     | 37.434   | 78.717       | 39382.862 | 100604.174   |
| C06 / IGSO    | 2.943    | 8.084        | 3.494     | 7.629        |
| Chandra / HEO | 5412.235 | 25635.951    | 2055.685  | 5663.182     |

The seven-day 10 m budget is **not met for all targets**. C06 stays below it at every seven-day sample on both dates; G05 exceeds it later on both dates. C03 and Chandra have additional reference qualifications below. The original R1 seven-day maxima were 7402.960/532.582/1237.242/19399.307 m in the same object order. The R2 first-day improvement must not be extended into a claim that every seven-day maximum improved or passed. RMS and percentiles describe correlated samples, not independent-sample confidence intervals. Reports additionally retain interval RMS/maxima and sample percentiles.

## 8.11 Sampled thresholds and incomplete reference coverage

Table 5: Preceding sample and first sampled exceedance, in hours after the origin, rounded to 0.001 h. These pairs are not certified continuous crossing brackets. “None” means no exceedance at the available samples.

| Date     | Target        | 10 m               | 100 m             | 1 km              |
|----------|---------------|--------------------|-------------------|-------------------|
| January  | G05 / MEO     | (145.083, 145.167] | None at samples   | None at samples   |
| January  | C03 / GEO     | (29.000, 29.083]   | None at samples   | None at samples   |
| January  | C06 / IGSO    | None at samples    | None at samples   | None at samples   |
| January  | Chandra / HEO | (50.402, 50.486]   | (82.569, 82.652]  | (83.986, 84.069]  |
| February | G05 / MEO     | (50.833, 50.917]   | None at samples   | None at samples   |
| February | C03 / GEO     | (72.000, 120.000]  | (72.000, 120.000] | (72.000, 120.000] |
| February | C06 / IGSO    | None at samples    | None at samples   | None at samples   |
| February | Chandra / HEO | (83.986, 84.069]   | (83.986, 84.069]  | (83.986, 84.069]  |

For 10 km, January Chandra first exceeds the level at 302648.815956 s, following the sample at 302348.815956 s; all January GNSS targets remain below 10 km at their samples. February C03 first exceeds 10 km at the first sample after its gap, 432000 s; the previous sample is at 259200 s. Other February targets remain below 10 km at the sampled times.

**C03 February coverage.** GFZ contains no C03 records on February 5 and 6. Adjacent daily endpoints leave a 48-hour reference gap and 575 missing five-minute samples. Wuhan also lacks C03 on February 6 and cannot certify the missing interval. GFZ remains the predeclared primary source. The first post-gap discrepancy is 29256.547 m and the later maximum is 100604.174 m; these observed exceedances are retained. Full-window coverage is **indeterminate**, and no continuous tolerance passage through the gap is inferred. No maneuver or other cause is assigned from missing records alone. The figures leave the missing interval blank.

## 8.12 Numerical error and reference discontinuities

Table 6: Separate numerical and frame comparisons, in metres. January’s independent physical audit uses 11 times across the past/future domain; February compares nine future checkpoints with adaptive DOP853. The frame column is the larger of the two complete sampled frame-approximation maxima.

| Target        | Jan. RK4/DOP853 | Feb. RK4/DOP853 | Frame approximation |
|---------------|-----------------|-----------------|---------------------|
| G05 / MEO     | 0.031716        | 0.026461        | 0.002007            |
| C03 / GEO     | 0.001232        | 0.001023        | 0.003260            |
| C06 / IGSO    | 0.001252        | 0.001026        | 0.003169            |
| Chandra / HEO | 0.036582        | 0.147044        | 0.009135            |

Native RK4 uses 30 s for GNSS and 7.5 s for Chandra. Same-seed integration agreement tests arithmetic, not orbit truth. The January forecast report’s own DOP853 check covers only its first sample; the wider January evidence above comes from the separate, target-independent physical audit. Native CPU/CUDA checks and exact word/packing checks remain separate evidence. The frame test compares the embedded daily-EOP/Chebyshev frame with direct SOFA IAU 2006/2000A evaluation of the same published prediction. It excludes actual future EOP error, station uncertainty and a complete relativistic ICRF/GCRS coordinate error budget; it is not subtracted as an independent scalar variance.

**Chandra source discontinuities.** The January error rises from 123.204 m to 23839.033 m between two five-minute samples. A separate one-second Horizons query shows that at January 5 TDB 12:01:09–12:01:10 the source position displacement differs from the integral of its own reported velocity by **24317.077 m**; outside the neighboring jump intervals the same local check is at most 0.0654 m. Native RK4 agrees with independently tightened DOP853 to about 1.5 mm at the two forecast endpoints surrounding that jump. The source header identifies CFA’s merged trajectory, but no local join or maneuver cause. The large discontinuity therefore cannot be attributed solely to smooth force error or native integration.

The February reference has a corresponding kinematic discontinuity at February 5 TDB 12:01:09–12:01:10: **2487.123 m** displacement-minus-velocity integral mismatch, versus at most 2.26  $\mu\text{m}$  away from its neighboring intervals. The forecast’s next five-minute error rises from 1.053 m to 2492.098 m. These diagnostics use only the source’s positions and velocities; no model is fitted to them. **All original discrepancy samples and whole-window maxima are retained.** The cause of the provider discontinuities is not established, and no corrected “true orbit” accuracy is invented. Original queries, hashes and calculations are in `source/orbit_data/diagnostics/`.

**Reference uncertainty.** January GFZ/Wuhan comparisons give 3D RMS 0.0263 m for G05, 2.074 m for C03 and 0.1692 m for C06; C03’s maximum provider disagreement is 3.575 m. These are product differences, not certified absolute bounds. Providers share some underlying observations. Chandra supplies no state covariance or absolute position guarantee. A millimetre or sub-metre match to a selected reference is therefore not a matching physical-accuracy claim.

### 8.13 Chronology, source conventions and reproducible evidence

The R2 physical model uses the documented degree/order-12 EGM96 field, cutoff-available daily Earth orientation, and external Sun/Moon forcing. EGM96’s original mu, reference radius, tide-free convention and fully normalized coefficients are preserved with explicit conversion in `source/orbit_data/gravity_reference/`. The format’s normalization default is documented by ICGEM: <https://icgem.gfz.de/docs/ICGEM-Format-2023.pdf>.

The EOP sources were published before their cutoffs: IERS Bulletin A of 26 December 2024 and 30 January 2025, respectively: <https://datacenter.iers.org/data/6/bulletina-xxxvii-052.txt> and <https://datacenter.iers.org/data/6/bulletina-xxxviii-005.txt>. Both permit public release and unlimited distribution. The February GFZ reference changes from IGS20 to IGB20. IGSMAIL-8543, published 9 December 2024, specifies zero datum transformation parameters because origin, scale and orientation remain aligned; individual reference-station coordinates are updated. Preserve each source label and apply no fitted alignment: <https://lists.igs.org/pipermail/igsmail/2024/008539.html>.

Precise target products were retrieved retrospectively; their earlier estimates can incorporate later observations within provider processing. Neither date proves that identical seeds were available live in 2025. January is a reused comparison, while February applies the already-frozen selection rule to a separate date. The confirmation freeze manifest binds the rule and model-file hashes. Raw source acquisition and coverage inspection preceded any February prediction-error evaluation. No later error was used to refit a seed.

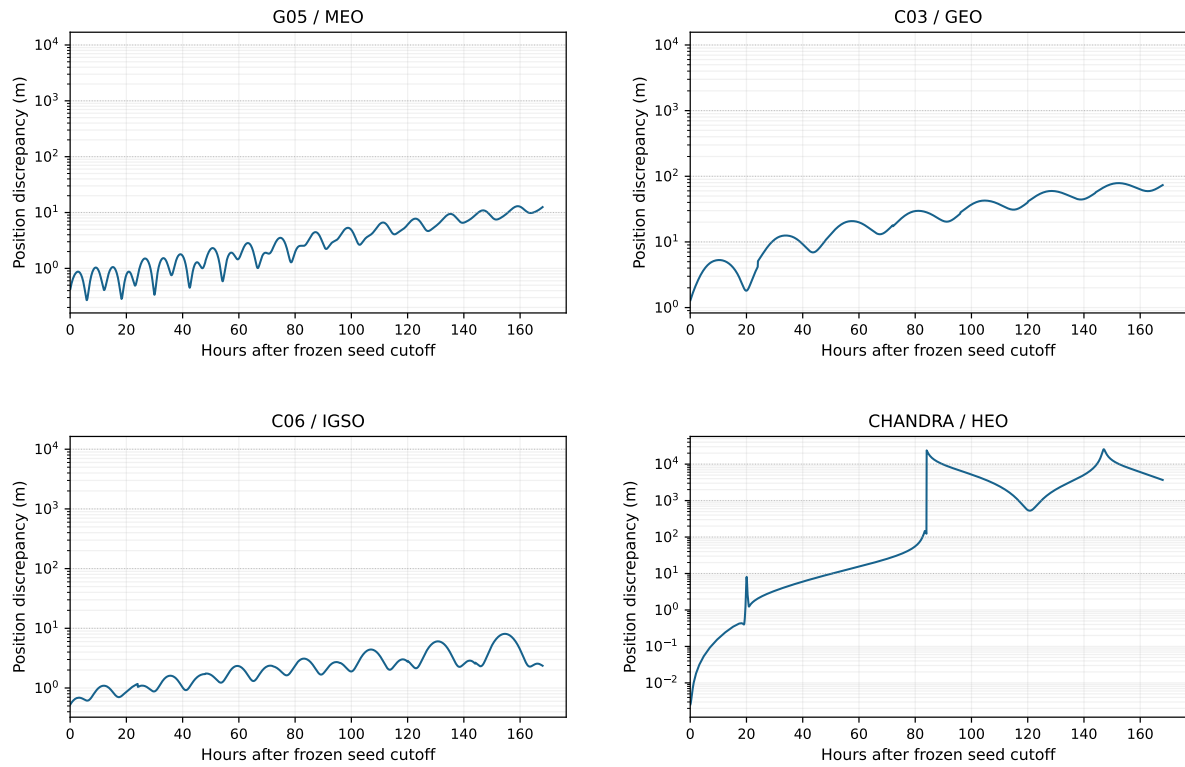
- **GFZ and IGS.** Retain GFZ/IGS provider and network attribution. The cited GFZ product-series DOI is <https://doi.org/10.5880/GFZ.1.1.2016.003>; its DataCite record declares **CC BY-NC 4.0**. The DOI names an ultra-rapid series whereas these GBM files are rapid products without a separate per-file grant. The fixture bundle conservatively retains that noncommercial qualification and is not claimed as unrestricted commercial data. IGS terms: <https://www.igs.org/wp-content/uploads/2020/09/IGS-Data-and-Product-Disclaimer-and-Terms-of-Use-200805.pdf>.

- **CODE and Wuhan.** CODE's reference <https://doi.org/10.48350/197028> lists open access/BORIS terms, not a blanket MIT grant. Wuhan final products retain provider/IGS attribution. Original archive URLs and per-file hashes remain in the source manifests.
- **NASA/JPL and CFA.** Acknowledge Giorgini and the JPL Solar System Dynamics Group, NASA/JPL Horizons, and CFA for Chandra: <https://ssd.jpl.nasa.gov/horizons/>. Retrieval date is 15 September 2026. Geometric ICRF/TDB settings and original responses are retained. Sun/Moon Horizons responses identify DE441; the construction oracle is DE440s, a distinct planetary ephemeris version.

**Authoritative reports.** Use `results/orbit_accuracy_r2_cpu_verified/accuracy_summary.json`, its CUDA counterpart, and `results/orbit_accuracy_confirmation_cpu_verified/accuracy_summary.json`. R1 evidence remains in `results/orbit_accuracy_cpu_final/` for comparison. The pre-fix failed trial and superseded intermediate models are not final accuracy evidence. CSVs retain every measured error, exact horizon offset, source and frame. Each report distinguishes `model_file_sha256` from `physical_model_sha256=orbit_seed.digest(model)`; legacy `model_sha256` is explicitly the file-hash alias. Accuracy displayed for a custom packed seed applies only when the physical-model digest matches. Reference fixtures, publications and the DE440s construction binary are separate from the complete replay seed byte count.

## 8.14 R2 January benchmark curves

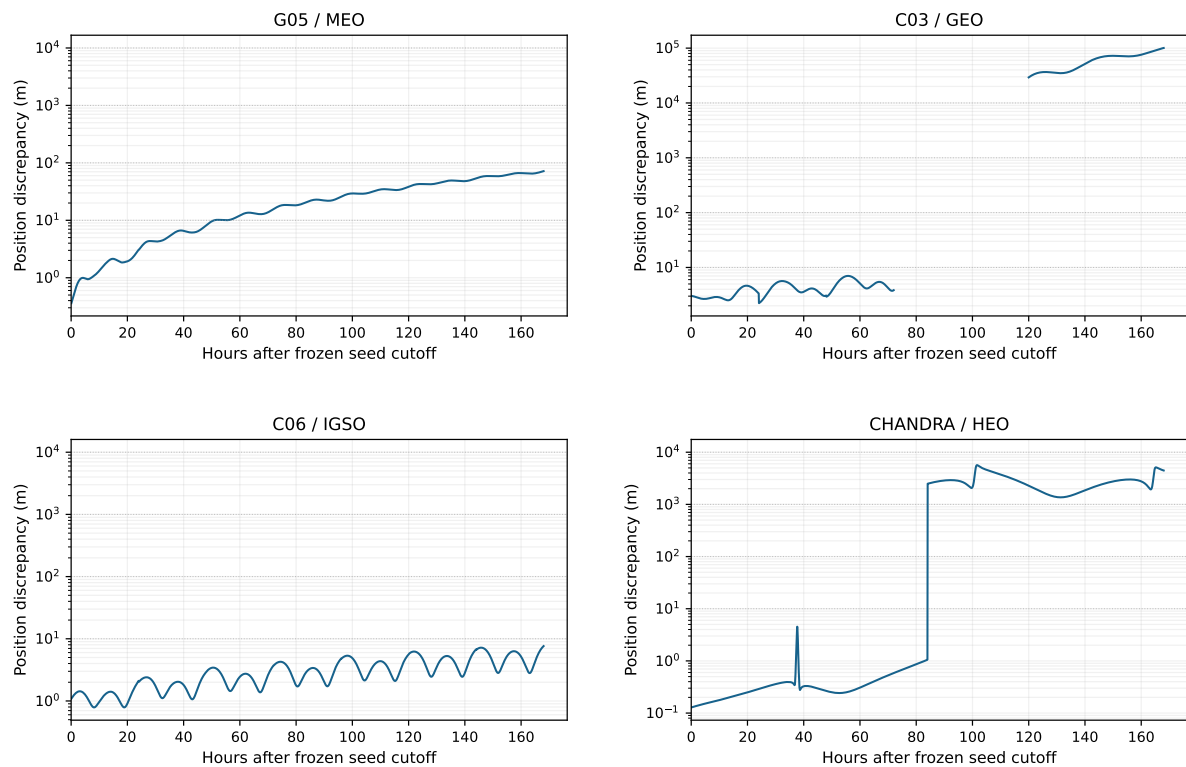
Frozen-seed forecast versus later external trajectory samples



This is the reused January benchmark. Dotted lines are reporting thresholds, not fitted constraints. The Chandra source discontinuity remains visible and its errors remain counted. First-day detail is also supplied as `forecast_error_first_day.pdf` in the same result directory.

## 8.15 Separate February confirmation curves

Frozen-seed forecast versus later external trajectory samples



The selection rule was frozen before this date's errors were evaluated. The C03 reference gap is blank; connecting its endpoints would imply unavailable measurements. Chandra's source discontinuity is retained. Lines join only available adjacent samples and do not certify error between samples or beyond the declared window.

## Inherited receiver, geometry and source contracts

The component versions and measurements below keep their original historical scope. The R11 domain-lift chapter states what exact expression semantics would change and what source equations remain unchanged.

## 9 Live GPS receiver-to-position pipeline

**LIVE-GPS-L1-R1** supplies the actual GPS equations and transport inputs missing from the prepared-observation kernel. The implemented chain is

$$\text{RTCM bytes} \longrightarrow (P_i, e_i, t_r) \longrightarrow (x_{s,i}^{rx}, \delta t_{s,i}, I_i, T_i) \longrightarrow (x_r, y_r, z_r, b_r). \quad (76)$$

The raw code  $P_i$  and satellite ephemeris  $e_i$  come from a real remote receiver; the physical corrections are calculated locally. The orbit and atmospheric model is Python; the position/clock solution is the existing FP64 native CPU/CUDA kernel, reached through a persistent process. This closes the reception-to-position chain without assuming an upstream program has already solved the geometry.

### Meaning of the live demonstration

The user has no local GNSS hardware. The Internet supplies observations from the LIENSS reference receiver in the public Centipede network. Every computed coordinate in this experiment locates that remote receiver. Its transmitted antenna reference point (ARP) is held out of the estimator and used only for subsequent comparison. No host-computer location is inferred from it.

### 9.1 What crosses the network boundary

The bounded client performs a read-only NTRIP/HTTP GET, or reads a TCP byte stream. It supports HTTP chunk framing, optional ICY headers, timeouts and bounded reconnection. Centipede permits one NTRIP client per public IP; the recorded CPU and GPU sessions run sequentially. The public service is documented at [Centipede: connecting to the caster](#). The exact source URL, bytes, UTC reception records, connection boundaries and SHA-256 fingerprints accompany each capture. No local coordinate/GGA is sent.

RTCM3 uses preamble D3, a ten-bit payload length and CRC24Q with polynomial 0x1864CFB. CRC is transport-error detection, not authentication. The implemented receiver extracts GPS LNAV message 1019; station metadata 1005/1006; legacy GPS code 1004; and GPS MSM4/5/6/7 (1074–1077). Other constellation MSM messages are consumed so their terminal flags correctly close the shared observation epoch. Their measurements do not enter the GPS-only solve.

An epoch is keyed by station ID and full GPST time. The multiple-message flag keeps fragments together until a terminal fragment arrives. CRC loss, reconnect, changed epoch before termination and end-of-input produce explicit incomplete records. RTCM has no fragment count or sequence-start flag: a newly connected client cannot prove that it saw the first fragment. This limitation is retained in the event metadata; missing ranges are not reconstructed from station geometry.

The one selected signal per satellite is C1C (L1 C/A), mapped to channel  $i = \text{PRN} - 1$ . MSM7's finer positive pseudorange takes precedence over legacy 1004 for the same signal. A valid legacy value may fill an unavailable MSM value. All original signal fields are retained; zero, absent or invalid code does not become a solver row. Carrier phase is decoded/preserved where present but no ambiguity state is estimated by this profile.

## 9.2 Time, provenance and reproducible execution

Times are unwrapped GPS seconds from 1980-01-06. RTCM week ambiguity is resolved against the recorded reception date, with an explicit GPS-minus-UTC offset of 18 seconds for these sessions. This value is stated by [USNO Earth-orientation values](#); it is a configurable input, not an automatically maintained leap-second service. RINEX epochs already tagged GPST receive no additional UTC offset.

No RTCM 1019 message contains the Klobuchar coefficients. The demonstrated sessions use only the GPSA/GPSB header of the same-day BKG broadcast file `BRDC00WRD_R_20262570000_01D_MN.rnx`:

$$\alpha = (1.2107 \times 10^{-8}, 1.4901 \times 10^{-8}, -5.9605 \times 10^{-8}, -1.1921 \times 10^{-7}), \quad (77)$$

$$\beta = (96256, 81920, -196610, -393220). \quad (78)$$

Its orbital records are *not* loaded during RTCM operation. Every usable satellite orbit/clock arrives in message 1019 from the stream itself. Download URLs, UTC acquisition time and hashes are in the source-data manifests. With no ionosphere header, the model explicitly labels its 5 ns baseline approximation.

Run the included Windows CPU worker from the release directory:

```
python tools/live_satnav.py
--binary bin/cpu/satnav_stream.exe
--url ntrip://caster.centipede.fr:2101/LIENSS
--iono-nav source/live_data/
BRDC00WRD_R_20262570000_01D_MN.rnx
--seconds 180 --out results/my_live_run
```

The display wraps one command for readability; join it onto one line and keep the navigation-file path continuous. Select the CUDA binary and add `-backend cuda` for the GPU path. Use a new output directory for each run. The dated ionosphere file reproduces this experiment; future-day operation should supply a current header or explicitly select the documented fallback.

The same final received bytes can be replayed with an explicit full-time reference:

```
python tools/live_satnav.py
--binary bin/cpu/satnav_stream.exe
--rtcm source/live_data/
centipede_lienss_20260914/capture.rtc3
--reference-gpst 1473457283
--iono-nav source/live_data/
BRDC00WRD_R_20262570000_01D_MN.rnx
--out results/my_recorded_run
```

RINEX input uses `-rinex-obs` and `-nav`; it is explicitly recorded as file replay. The input contract, all optional arguments, live-to-CGK bridge and independent oracle commands are supplied in `docs/LIVE_INPUT.md`, `docs/LIVE_HANDOFF.md` and the source-data README.

### 9.3 LIVE-R1 time contract and broadcast orbit

A raw observation is  $(t_r^{\text{tag}}, P_i, \text{PRN}_i, \sigma_{0i})$ . Time is full, unwrapped GPS seconds from 1980-01-06;  $P_i$  is the original GPS L1 C/A pseudorange in metres. Receiver time tags and code share one clock convention. GPS-labelled RINEX calendar fields receive no UTC conversion. The final physical reception-time estimate is

$$\hat{t}_r = t_r^{\text{tag}} - \hat{b}_r/c, \quad c = 299\,792\,458 \text{ m s}^{-1}. \quad (79)$$

Here  $b_r$  is the estimated receiver clock bias in metres. The raw tag remains in the trace. RTCM week rollover is resolved against an explicit reception-date reference; age tests use full time, never a modulo-week age.

Each GPS ephemeris provides  $\sqrt{A}, e, i_0, \Omega_0, \omega, M_0, \Delta n, \dot{i}, \dot{\Omega}$ , six harmonic coefficients,  $t_{oe}, t_{oc}$ , three clock coefficients,  $T_{GD}$ , health and issue identifiers. The selected record has matching PRN, healthy status and IODC mod 256 = IODE. Both clock and orbit age are bounded by 7200 s; a shorter declared half-fit interval further limits orbit age. A nearer unhealthy issue is not replaced by an older healthy issue.

For signal emission time  $t_s$ , define

$$A = (\sqrt{A})^2, \quad t_k = t_s - t_{oe}, \quad (80)$$

$$n = \sqrt{\mu/A^3} + \Delta n, \quad M = M_0 + nt_k, \quad \mu = 3.986005 \times 10^{14} \text{ m}^3 \text{ s}^{-2}. \quad (81)$$

Newton's method solves  $E - e \sin E = M$ :

$$E_{j+1} = E_j - \frac{E_j - e \sin E_j - M}{1 - e \cos E_j}. \quad (82)$$

The implementation reduces  $M$  modulo  $2\pi$ , starts at  $E_0 = M$ , and requires an increment below  $10^{-14}$  rad within 30 iterations. For  $0 \leq e < 1$ ,

$$\nu = \text{atan2}\left(\sqrt{1-e^2} \sin E, \cos E - e\right), \quad \phi = \nu + \omega, \quad (83)$$

$$u = \phi + C_{us} \sin 2\phi + C_{uc} \cos 2\phi, \quad (84)$$

$$r = A(1 - e \cos E) + C_{rs} \sin 2\phi + C_{rc} \cos 2\phi, \quad (85)$$

$$i = i_0 + \dot{i} t_k + C_{is} \sin 2\phi + C_{ic} \cos 2\phi, \quad (86)$$

$$\Omega = \Omega_0 + (\dot{\Omega} - \omega_e)t_k - \omega_e(t_{oe} \bmod 604800), \quad (87)$$

$$\omega_e = 7.2921151467 \times 10^{-5} \text{ rad s}^{-1}. \quad (88)$$

All angles are radians. Radius harmonics are metres; the other harmonics are angular corrections. These definitions implement the GPS LNAV user algorithm in [IS-GPS-200N, section 20.3.3.4.3](#); they are new GPS-specific additions to this kernel package.

## 9.4 Satellite clock, emission time and reception axes

Let  $x' = r \cos u$  and  $y' = r \sin u$ . The satellite position in the Earth-fixed axes at emission is

$$\tilde{\mathbf{x}}_s = \begin{pmatrix} x' \cos \Omega - y' \cos i \sin \Omega \\ x' \sin \Omega + y' \cos i \cos \Omega \\ y' \sin i \end{pmatrix}. \quad (89)$$

With  $\Delta t_c = t_s - t_{oc}$ , the broadcast clock polynomial, eccentric-orbit relativity correction, and L1 code clock are

$$\delta t_{\text{poly}} = a_{f0} + a_{f1} \Delta t_c + a_{f2} \Delta t_c^2, \quad (90)$$

$$\delta t_{\text{rel}} = F e \sqrt{A} \sin E, \quad F = -4.442807633 \times 10^{-10} \text{ s m}^{-1/2}, \quad (91)$$

$$\delta t_{s,L1} = \delta t_{\text{poly}} + \delta t_{\text{rel}} - T_{GD}. \quad (92)$$

The trace stores all three contributions separately. The  $-T_{GD}$  sign is specific to this L1 profile. An external differential C1–P1 bias is not invented. Alternate code types are rejected until their signal-bias model is explicit. See [IS-GPS-200N, sections 20.3.3.3.1–20.3.3.3.2](#).

The measured code connects the receiver and satellite clock tags:

$$t_s^{\text{tag}} = t_r^{\text{tag}} - P/c, \quad t_s^{(j+1)} = t_s^{\text{tag}} - \delta t_{s,L1}(t_s^{(j)}). \quad (93)$$

The adapter uses three clock iterations starting at  $t_s^{\text{tag}}$ . The receiver clock cancels in  $t_r^{\text{tag}} - P/c$ , because it occurs in both raw quantities. Subtracting the receiver clock again here would double-correct the emission time. This construction follows ESA's [pseudorange-based emission-time algorithm](#).

Earth rotation over the travel time is applied once, before the native solve:

$$R_3(a) = \begin{pmatrix} \cos a & \sin a & 0 \\ -\sin a & \cos a & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad (94)$$

$$\mathbf{x}_s^{rx} = R_3(\omega_e \tau) \tilde{\mathbf{x}}_s, \quad \tau = \|\mathbf{x}_s^{rx} - \mathbf{x}_r\|/c. \quad (95)$$

Four range iterations, initially  $\tau_0 = P/c$ , determine the reception-frame satellite position for the current receiver estimate. The native model never rotates it a second time. The rotation sign and frame convention follow [ESA's reception-frame coordinate construction](#).

The first GSI epoch supplies an independent component check: all eight emission positions match the native RTKLIB trace within 0.000578 m and its clock values within 0.000438 ns. The RTKLIB trace rounds coordinates to millimetres and clocks to 0.001 ns, and excludes  $T_{GD}$  from its reported satellite clock. The comparison therefore adds  $T_{GD}$  back to Equation (92) for that check only.

## 9.5 Complete Klobuchar L1 ionosphere

WGS84 ECEF-to-geodetic conversion supplies receiver latitude  $\varphi_r$ , longitude  $\lambda_r$  and ellipsoidal height  $h$ . Rotating the line of sight into local ENU gives azimuth  $A_z$  clockwise from north and elevation  $\mathcal{E}$ . A default  $10^\circ$  elevation mask is applied after the receiver geometry has been initialized.

Klobuchar latitude, longitude and elevation arguments are in semicircles:  $\varphi_u = \varphi_r/\pi$ ,  $\lambda_u = \lambda_r/\pi$  and  $\epsilon = \mathcal{E}/\pi$ . Azimuth remains in radians. The complete pierce-point and geomagnetic construction is

$$\psi = \frac{0.0137}{\epsilon + 0.11} - 0.022, \quad (96)$$

$$\varphi_I = \min(0.416, \max(-0.416, \varphi_u + \psi \cos A_z)), \quad (97)$$

$$\lambda_I = \lambda_u + \frac{\psi \sin A_z}{\cos(\pi \varphi_I)}, \quad (98)$$

$$\varphi_m = \varphi_I + 0.064 \cos(\pi(\lambda_I - 1.617)), \quad (99)$$

$$t_{\text{local}} = (43200\lambda_I + t_r^{\text{tag}}) \bmod 86400. \quad (100)$$

With the four broadcast alpha and four beta coefficients,

$$A_I = \max\left(0, \sum_{j=0}^3 \alpha_j \varphi_m^j\right), \quad (101)$$

$$P_I = \max\left(72000, \sum_{j=0}^3 \beta_j \varphi_m^j\right), \quad (102)$$

$$X = \frac{2\pi(t_{\text{local}} - 50400)}{P_I}, \quad (103)$$

$$F_I = 1 + 16(0.53 - \epsilon)^3. \quad (104)$$

Amplitude  $A_I$  is seconds and period  $P_I$  is seconds. The slant L1 delay, converted to metres, is

$$I = cF_I \begin{cases} 5 \times 10^{-9} + A_I (1 - X^2/2 + X^4/24), & |X| < 1.57, \\ 5 \times 10^{-9}, & \text{otherwise.} \end{cases} \quad (105)$$

Every trigonometric operation on a semicircle value explicitly multiplies by  $\pi$ . The limits on pierce latitude, amplitude and period are part of the published model. See [ESA's Klobuchar equations](#), and GPS LNAV ionosphere parameters in [IS-GPS-200N, section 20.3.3.5.2.5](#).

When broadcast coefficients are unavailable, the adapter explicitly selects  $\alpha_j = \beta_j = 0$  and records `klobuchar_zero_coefficients_5ns_baseline`. This still produces  $cF_I 5 \times 10^{-9}$  metres of delay; it is not zero ionosphere. For the internet capture, an independently downloaded BKG broadcast navigation header provides the coefficients, with its source hash retained. Receiver ephemerides can still come from the actual RTCM 1019 stream.

## 9.6 Troposphere, observation variance and native solve

LIVE-R1 uses a specified standard atmosphere with relative humidity  $H = 0.7$ . For  $-100 \leq h \leq 10000$  m and  $\mathcal{E} > 0$ , define

$$h_0 = \max(h, 0), \quad (106)$$

$$p = 1013.25(1 - 2.2557 \times 10^{-5} h_0)^{5.2568} \quad [\text{hPa}], \quad (107)$$

$$T = 288.16 - 0.0065 h_0 \quad [\text{K}], \quad (108)$$

$$e_w = 6.108 H \exp\left(\frac{17.15T - 4684}{T - 38.45}\right) \quad [\text{hPa}]. \quad (109)$$

The hydrostatic and wet Saastamoinen delays are

$$T_h = \frac{0.0022768p}{(1 - 0.00266 \cos 2\varphi_r - 0.00028 h_0/1000) \sin \mathcal{E}}, \quad (110)$$

$$T_w = \frac{0.002277(1255/T + 0.05)e_w}{\sin \mathcal{E}}, \quad T_d = T_h + T_w. \quad (111)$$

Negative terrestrial heights use sea-level weather and carry a specific status. Outside the height domain, atmosphere is omitted with an explicit diagnostic; no calibrated weather is inferred. These constants and equations match the independent [RTKLIB tropmodel source](#).

The default nominal code uncertainty is  $\sigma_{0i} = 3$  m. The executed native profile weights observations by

$$\sigma_i = \frac{\sigma_{0i}}{\max(\sin \mathcal{E}_i, 0.1)}, \quad W_{ii} = \sigma_i^{-2}. \quad (112)$$

CN0 is retained when supplied, without an invented CN0-to-variance calibration. These are nominal independent variances, not measured position errors or protection levels.

The full measurement equation and its single additive correction are

$$P_i = \|\mathbf{x}_r - \mathbf{x}_{s,i}^{rx}\| + b_r - c\delta t_{s,i,L1} + I_i + T_{d,i} + \epsilon_i, \quad (113)$$

$$a_i = c\delta t_{s,i,L1} - I_i - T_{d,i}, \quad P_{i,\text{corr}} = P_i + a_i. \quad (114)$$

The native CPU/CUDA solver estimates  $\mathbf{s} = (x_r, y_r, z_r, b_r)^\top$  by weighted least squares. For  $\rho_i = \|\mathbf{x}_r - \mathbf{x}_{s,i}^{rx}\|$ ,

$$r_i = P_{i,\text{corr}} - \rho_i - b_r, \quad H_i = \left( (\mathbf{x}_r - \mathbf{x}_{s,i}^{rx})^\top / \rho_i \quad 1 \right). \quad (115)$$

An online Givens QR solve of the whitened system gives  $\Delta \mathbf{s}$ ;  $\mathbf{s} \leftarrow \mathbf{s} + \Delta \mathbf{s}$ . Rank, iteration, finite-value and residual-fit results remain explicit. Literal ASA/NA selection is unchanged, including whole-word absorption on a selected boundary hit.

## 9.7 Persistent receiver execution and correction feedback

The persistent native worker receives complete epochs. The preparation layer and native solve form the outer state-feedback map

$$O_n^{(k)} = \text{Prepare}(P_n, t_n^{\text{tag}}, \mathcal{E}_n, s_n^{(k)}), \quad (116)$$

$$s_n^{(k+1)} = \text{NativeSolve}(O_n^{(k)}, s_n^{(k)}), \quad (117)$$

where  $\mathcal{E}_n$  denotes the available ephemeris set. Preparation recomputes reception-frame geometry, elevation selection, ionosphere, troposphere and weights from the new position. The default outer limit is eight passes. A valid position requires initialized terrestrial geometry, a converged native solve, and

$$\max(\|x_n^{(k+1)} - x_n^{(k)}\|, |b_n^{(k+1)} - b_n^{(k)}|) < 10^{-3} \text{ m}. \quad (118)$$

At the first receiver epoch, the complete state is zero. Geometry initialization uses all available healthy GPS codes with orbit and clock corrections; the trace explicitly marks atmosphere and elevation as uninitialized. After the first coarse Earth position is solved, the next pass activates the physical models. Reference-station coordinates are never read as seeds. Subsequent epochs start from this worker's preceding valid solution:

$$s_{n+1}^{(0)} = \hat{s}_n. \quad (119)$$

A failed epoch does not overwrite that state. A detected receiver/station change resets the seed. Duplicate or out-of-order epochs do not produce a new fix. This is per-epoch code positioning with a warm start; it does not silently add a motion prior or carrier-phase ambiguities.

The geometric mechanics profile remains a downstream consumer of position results. Its JK state is not injected into satellite orbit/clock corrections or used as a fictitious GNSS observation. The live feedback in Equation (117) closes the formerly missing receiver preparation loop around the existing position kernel.

**Separate observable evidence.** The journal stores raw measurements, every preparation pass, every native solution and final availability. Correction components retain signs and units. A recorded-file replay, a local timed replay, and a current internet receiver stream are distinct source modes. A remote reference receiver's position is its antenna position; it does not measure the position of the user's computer.

The GSI 120-epoch run starts at ECEF zero, converges on all epochs, and agrees with independent NumPy solves of all 263 converged passes within  $1.17 \times 10^{-8}$  m per state component. The executed weighting differs from RTKLIB's broadcast-variance weighting: their final positions differ by a mean 0.529 m and a maximum 1.589 m. A controlled NumPy re-solve with the same physical corrections and RTKLIB weights reduces those differences to a mean 0.000174 m and a maximum 0.000346 m. This experiment isolates the weighting choice; it is recorded separately from the executed native profile. The full report is `results/real_0759_cpu/independent_comparison.json`.

## 9.8 Persistent execution and exact records

`satnav_stream` starts once and accepts successive complete requests. Each request contains the epoch ID/time, four-component seed, unchanged ASA/NA/boundary words and up to 32 observation rows. Replies contain status, fit classification, ECEF/clock, formal variance, residuals, iteration count and kernel timing. Both CPU and CUDA dispatch the same SATNAV equations. A

rejected request produces an error record; a Python response timeout closes the worker so a late response cannot be assigned to the next epoch.

Each completed decoded epoch triggers correction preparation and native solves while the socket session is still open. There is no whole-file staging step in the live path. Warm starts carry only the last complete position/clock estimate; they do not introduce velocity filtering or modify raw ranges. Native failure, missing ephemeris and exhausted outer correction iterations publish a null position with an explicit status. Diagnostics still retain the partial iterate. Fit classification remains separate from numerical convergence.

| Artifact                             | Recorded content                                                                               |
|--------------------------------------|------------------------------------------------------------------------------------------------|
| <code>capture.rtc3 + sidecars</code> | Raw received body, SHA-256, host times and connection boundaries.                              |
| <code>events.jsonl</code>            | Decoded original fields, orbital issues, station metadata and incomplete/error records.        |
| <code>trace.jsonl</code>             | Raw epoch; every preparation and native pass; final position/clock/time and validity.          |
| <code>prepared/</code>               | Exact final-pass CSV geometry/corrections for independent Householder and native batch replay. |
| <code>summary.json</code>            | Source identity, counts, worker binary identity, timing scope and capture outcome.             |

Only one synchronous callback operates on received bytes. If decoding, disk I/O or positioning becomes slower than delivery, TCP backpressure accumulates upstream. The local demonstration measures a 1 Hz station and is bounded in duration/bytes; it establishes neither continuous service availability nor a hard real-time bound. Socket timeouts, CRC/epoch-loss tests and recorded continuity expose the tested failure semantics. Clock-tag freshness is reported separately from local compute time: this host's UTC clock was about two seconds behind the incoming GPST tags, so a negative tag-to-host interval is not interpreted as negative network latency.

## 9.9 Current live execution and independent validation

Both final sessions ran on 14 September 2026 UTC (the local Netherlands date crossed into 15 September during the work). Each captured 180 complete one-second epochs using one public caster connection. CPU and CUDA ran sequential sessions: CPU: 21:55:20–21:58:20 UTC; CUDA: 21:58:54–22:01:54 UTC. The source is `caster.centipede.fr:2101/LIENSS`.

| Measured quantity                             | CPU    | CUDA   |
|-----------------------------------------------|--------|--------|
| Complete received epochs                      | 180    | 180    |
| Published native positions                    | 172    | 175    |
| Initial unavailable epochs (ephemeris warmup) | 8      | 5      |
| Initial ECEF/clock seed                       | Zero   | Zero   |
| Mean 3D separation from advertised ARP (m)    | 1.644  | 1.854  |
| Maximum ARP separation (m)                    | 2.232  | 2.267  |
| Mean 3D difference from RTKLIB (m)            | 0.649  | 0.622  |
| Maximum RTKLIB difference (m)                 | 0.969  | 0.931  |
| Median complete-input-to-position (ms)        | 2.205  | 5.947  |
| Maximum complete-input-to-position (ms)       | 11.611 | 14.346 |
| All positions emitted during socket capture   | Yes    | Yes    |

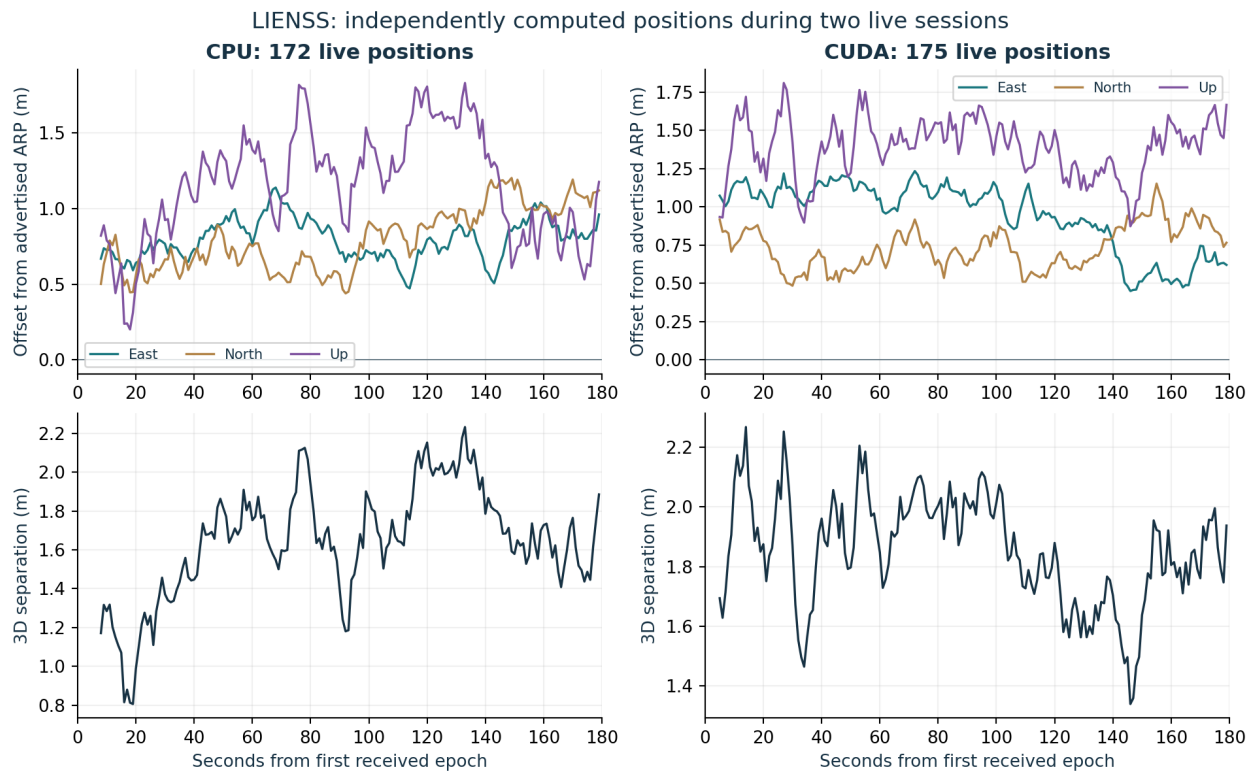
The native outputs and their timestamps were checked against the exact raw-frame hashes and receive-chunk sidecars. The latency above runs from the host receiving the final required input fragment to publication of a valid position. It includes local decoding and solving; it is not a calibrated radio/network transit time. It excludes unavailable warmup epochs. CUDA's first unavailable live epoch took 2.59 seconds inside processing, including initial device work. A separate persistent-worker regression measured about 1.05 seconds for its first request. CPU is faster for this small 1 Hz workload; these results establish CUDA correctness, not a speedup. The CUDA device was the RTX 5070 Ti Laptop, compute capability 12.0, built with CUDA 12.8 and driver 591.59.

The comparison coordinate is the stream's advertised antenna reference point:

$$(4426043.0455, -89429.1998, 4576296.6447) \text{ m.} \quad (120)$$

It was neither a supplied initial estimate nor an observation. It is not an independently surveyed truth coordinate in this experiment. Consequently the table reports separation from advertised ARP, not certified absolute accuracy. RTKLIB also performs single-point code positioning here; its output quality code is 5, not an RTK fixed-ambiguity solution.

## 9.10 Measured position series and independent checks



**Figure.** East/North/Up and three-dimensional separation from the advertised ARP in the two actual live sessions. The blank beginning is ephemeris warmup; no position is fabricated for it. These are different time windows, so their physical differences are not a CPU-versus-GPU numerical comparison.

For that numerical comparison, the same final raw capture was replayed through both backends with the same preparation model and receive order. Both produced 175 positions from 180 epochs; their maximum ECEF/clock component difference was  $1.21 \times 10^{-8}$  m. Live CUDA and its archived-byte replay also passed the same epoch/status/position comparison. `results/live_cross_backend.json` records it.

The independent published RTKLIB `convbin` decoded each capture to RINEX. The final CPU and CUDA decoded satellite/epoch sets matched exactly; 1980 and 2094 common C1C ranges agreed to 0.279 mm, within RINEX decimal rounding. Respectively 280 and 588 ephemeris numeric fields agreed. A separate `pyrtcm` field audit on the five-minute source capture passed 96,880 comparisons. The orbit/clock oracle matched eight satellite positions to 0.578 mm and clocks to 0.000438 ns, limited by its printed output precision. Source hashes and commands are retained.

## 9.11 Regression, real-model checks and retained state connection

The older GSI recording solved all 120 real epochs from ECEF zero. Its raw physical model and weight-isolation experiment are described with the equations above. Prepared final live CPU observations passed 3,324 independent Householder checks across all 180 epochs, with maximum state-component difference below  $8.4 \times 10^{-9}$  m. All partial/warmup statuses were retained. The persistent CUDA worker processed those same real final-pass inputs under Compute Sanitizer: zero memory errors, 180 responses and 1,048 independent checks.

Current native builds passed all three CTest suites on CPU and CUDA. Python passed 118 test methods. The persistent-interface regression passed 6,861 comparisons per backend, including rejected requests followed by successful recovery. The existing texture/global CUDA paths also passed independent comparison and Compute Sanitizer through `validate_gpu.py -sanitizer`. Analytic/parser tests cover wrong CRC, fragments, missing data, stream truncation, HTTP chunking, reconnection, callback failure, stale epochs, native failure and cold-start rules.

Independent NumPy re-solves verified all 698 converged outer passes across both final live sessions to  $1.40 \times 10^{-8}$  m per component. The physical audit checked 6,284 prepared rows, including original code, correction signs and proof that every chosen 1019 issue had already arrived before its epoch was solved. All 347 published fixes preserve the corrected reception-time equation.

The same live position records also cross the retained CGK-R1 handoff after capture. Only completed positions enter that downstream replay; omitted warmup epochs remain listed in its manifest. ENU is anchored at the first computed position, never at the advertised ARP. Original IDs, ECEF, receiver clock and full time remain attached. Both key layouts are emitted wherever the chart is defined. The stationary demonstration does not require the word state to switch, and its modeled mechanics are not interpreted as a measured receiver velocity. The final CPU bridge retained 172 fixes and listed eight warmup omissions; its native batch position/clock replay exactly matched the stream. CGK replay passed 16,684 field comparisons, and a separate binding audit passed 1,042 checks. The word remained  $q = 0$  in this static session. Both keys were emitted for 153 positions; 19 remained inside the chart core. These facts do not assert dynamic switching.

**Evidence index.** `results/validation_status.json` links current reports and their hashes. `results/network_probes/` contains the independent conversion, raw-field and RTKLIB comparisons. `source/live_data/README.md` records data provenance and third-party attribution. Parent 3.6.1.7 totals in the following chapters are explicitly historical and are not added to these live-session results. The original sibling release remains unchanged.

## 9.12 Scope of the resulting navigation kernel

This release now contains real raw-data ingestion, satellite ephemeris propagation, explicit physical code corrections, continuous native position/clock computation, and independent real-observation evidence. It is a working single-point GPS L1 receiver backend. The validation uses a static public base station and an older real receiver recording; it does not establish moving-receiver field performance.

The retained CGK-R1 equations provide an explicit geometry/word/mechanics model downstream. Their modeled motion must remain distinguishable from receiver measurements; ASA/NA and JK feedback alone do not add independent observables or cancel unknown GNSS errors. Full ECEF, Up, receiver clock, linear time and both UGTS key layouts remain represented under their existing contracts.

Additional applications can consume the position/time records: maps, logging, spatial queries, digital twins or a later motion estimator. A moving local satnav still needs its own receiver observation source; higher precision needs additional models such as differential corrections or carrier ambiguities. RTK/PPP, Doppler velocity, multi-constellation clock states, map routing and calibrated antenna/code bias estimation are separate implementations. The delivered static-session statistics and formal covariance do not stand in for those missing capabilities.

| Use case                | Available output or mechanism                                           | Additional work for deployment                                                        |
|-------------------------|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Remote position logging | Live ECEF, clock, full time, raw observations and reproducible lineage. | Long-duration operation, service monitoring and application storage.                  |
| Local moving satnav     | Executable receiver-to-position backend.                                | Local receiver observations, motion validation and the desired map/routing interface. |
| Spatial indexing        | ENU/Up, chart state, full time and both exact UGTS key layouts.         | Application database, spatial extent and query policy.                                |
| Digital simulation      | Editable literal equations and coupled geometry/mechanics.              | Calibrated model parameters and validation against the intended system.               |
| Precise positioning     | Raw code and available carrier metadata retained.                       | Differential/precise corrections, bias models and ambiguity estimation.               |

## 10 The completed geometry–state feedback connection

Version 3.6.1.7 defines and implements **CGK-R1**, a coupled continuous and discrete state kernel. Prepared GNSS observations are solved by SATNAV-R1. Their unquantized position estimates become inputs to a persistent modeled mechanical state. Geometry of that state is encoded into the literal ASA/NA and J–K word operations. Their result changes the mechanical operator, and the advanced position supplies geometry for the following step.

### The complete executed dependency

Corrected-code observations  $\rightarrow$  SATNAV ECEF estimate  $\rightarrow$  local target  $z_n \rightarrow$  geometry of the carried state  $p_{n-1} \rightarrow$  predicate words  $\rightarrow$  editable drive/X/J/K equations  $\rightarrow q_n \rightarrow$  stiffness and force  $\rightarrow (p_n, v_n) \rightarrow$  next geometry.

The actual UGTS handoff publishes the carried J/K state and both the original measurement estimate and the modeled physical state.

The source did not supply complete M2 circle-plus, predicate-encoder or physical eigenmatrix definitions. This release supplies explicit new definitions for all three roles. Their equations, input parameters, units and execution order are given here and in docs/COUPLED\_CONTRACT.md. They are new formalizations using the retained primitives, not claims of reconstructing unavailable text.

### 10.1 State, observations and time identity

For each independently seeded trajectory, retain

$$S = (p, v, q, \phi, t, \rho_{\text{prev}}, \theta_{\text{prev}}, \chi_{\text{prev}}), \quad (121)$$

where  $p \in \mathbb{R}^3$  is ENU position in metres,  $v \in \mathbb{R}^3$  is velocity in m/s,  $q$  is a 32-bit word,  $\phi$  is the separate hoop phase,  $t$  is full unwrapped time and  $\chi_{\text{prev}}$  records whether the previous chart exists. Satellite geometry, clock bias, formal covariance and original observation identity remain available in the measurement record outside this state and its compact keys.

Sample  $n$  supplies time  $t_n$ , target  $z_n$ , observation validity  $o_n \in \{0, 1\}$ , and external force  $f_n$ . Define  $\Delta t = t_n - t_{n-1} > 0$ . Predicates read the old state together with the endpoint observation. The observed target and force are the piecewise constant inputs of the declared discrete observer interval; the resulting mechanical state is labeled  $t_n$ . The trace records both interval endpoints, and input rows are never silently reordered.

For a solved ECEF position  $\hat{r}_n$ , the fixed ENU transform gives

$$z_n = R_{\text{ENU}}(\hat{r}_n - r_0^{\text{ECEF}}). \quad (122)$$

The modeled position  $p_n$  is a separate quantity. It is not substituted into the original receiver solution or represented as another independent GNSS measurement.

## 11 Literal chart motion and an explicit circle-plus

For the old modeled position  $p = (E, N, U)$ , set  $r = \sqrt{E^2 + N^2}$ . For  $r \geq r_{\text{core}} > 0$ , use the retained chart

$$\rho = \log(r/r_0), \quad \theta = \text{atan2}(N, E). \quad (123)$$

The preceding chart belongs to the preceding step's *before* state. Thus the next step compares the newly advanced physical geometry with the geometry that produced that advance. The first step has no chart increment.

For two defined charts, set

$$\Delta\theta = \text{wrap}(\theta - \theta_{\text{prev}}), \quad \Delta\rho = \rho - \rho_{\text{prev}}, \quad \alpha = \arctan(\Delta\theta/\Delta\rho) - a_0. \quad (124)$$

This is literal source OTAN2. A zero pair, zero denominator, missing chart or floating-point range failure retains its explicit status. It is never replaced by quadrant-resolving  $\text{atan2}$  while keeping the source name.

### 11.1 CGK-R1 circle-plus definition

Let  $\text{wrap} : \mathbb{R} \rightarrow [-\pi, \pi)$ , with the  $\pi$  tie represented by  $-\pi$ . For  $0 \leq w \leq 1$ , define

$$a \oplus_w b := \text{wrap}(a + w \text{wrap}(b - a)). \quad (125)$$

This is interpolation along the selected shortest angular arc. It is not Boolean XOR, and associativity is not assumed. At  $w = 0$  it selects  $\text{wrap}(a)$ ; at  $w = 1$  it selects  $\text{wrap}(b)$ . Common full-turn changes in either input do not change the selected phase, subject to the declared tie convention.

With  $e = z_n - p$ , the target bearing  $\beta = \text{atan2}(e_N, e_E)$  exists when  $o_n = 1$  and the horizontal error is at least  $r_{\text{core}}$ . The actual blended phase used by the encoder is

$$\psi = \text{wrap}(\beta \oplus_w \alpha + \phi). \quad (126)$$

Both angular inputs must be defined. If they are not, only the alignment predicate is false. Limit, fringe, support, editable Boolean equations and the physical step still evaluate according to their definitions.

### 11.2 Independent hoop evolution

The hoop has its own supplied angular rate  $\omega_h$  and evolves as

$$\phi_n = \text{wrap}(\phi_{n-1} + \Delta t \omega_h). \quad (127)$$

Its role stays distinct from geodetic latitude, modeled chart angle and full time winding. It enters the blend explicitly and is preserved in the key tuple.

## 12 The geometry-to-bit encoders

Each present channel  $i$  supplies a unit mechanical direction  $u_i$ , angle center  $\beta_i$ , alignment tolerance  $\epsilon_i$ , projected-error limit  $L_i$ , support geometry and fringe width  $\eta_i$ . Parameters and their units are input data; changing them changes the executed model.

For a sphere with center  $c_i$  and radius  $R_i$ ,

$$s_i(p) = \|p - c_i\| - R_i. \quad (128)$$

A channel may instead select the inherited finite-cone signed-distance function: reduce the point to axial/radial coordinates and take the signed distance to its closed triangular meridian. Slant length, half-angle, apex and unit axis are supplied explicitly. The cone definition is the same one given later in this volume; it is evaluated on the evolving modeled position.

The four Boolean features are exactly

$$A_i^{\text{geom}} = [\psi \text{ defined}] \wedge [|\text{wrap}(\psi - \beta_i)| \leq \epsilon_i], \quad (129)$$

$$L_i^{\text{geom}} = [o_n = 1] \wedge [|u_i^T e| > L_i], \quad (130)$$

$$F_i^{\text{geom}} = [|s_i(p)| \leq \eta_i], \quad (131)$$

$$S_i^{\text{geom}} = [s_i(p) \leq 0]. \quad (132)$$

Channel  $i$  occupies bit  $i$ ; absent channels contribute zero. The resulting words are named *align*, *limit*, *fringe* and *support*. The separate valid word is all ones for a valid sample and zero otherwise.

### 12.1 Four actual input equations

The editable JSON supplies drive, input, J and K expressions. The default handoff profile uses

```
"equations": {
  "d": "limit | fringe",
  "x": "(q | d) & support",
  "j": "y & (limit | fringe)",
  "k": "align & ~limit & ~fringe"
}
```

The vocabulary is the exact Boolean algebra  $\&$ ,  $|$ ,  $\wedge$ ,  $\sim$ , parentheses and word constants  $0/1$ . All expressions read the old  $q$ . The drive sees six inputs, X sees those plus drive, and J/K additionally see the ASA output  $y$ . The compiler enumerates their exact 64-, 128- and 256-bit truth tables. The direct Python reference evaluates the original expression trees rather than these tables.

The source ASA masks and whole-word boundary operation then produce

$$a = x \wedge A \wedge V, \quad n = a \wedge N, \quad h = \text{popcount}(n \wedge B), \quad y = \begin{cases} 0 & h > 0, \\ n & h = 0, \end{cases} \quad (133)$$

followed by the synchronous literal update

$$q_n = [(J \wedge \neg q_{n-1}) \vee (\neg K \wedge q_{n-1})] \wedge V. \quad (134)$$

Absorption determines  $y$ ; it does not override the supplied J/K equations.

### 13 The physical operator supplied by the feedback state

The modeled dynamics have diagonal positive mass  $M$ , diagonal damping  $C$ , and a supplied symmetric base stiffness  $K_0$ . For each present channel,

$$\kappa_i(q_i) = \kappa_i^{\text{off}} + (\kappa_i^{\text{on}} - \kappa_i^{\text{off}})q_i, \quad (135)$$

$$g_i(q_i) = g_i^{\text{off}} + (g_i^{\text{on}} - g_i^{\text{off}})q_i, \quad (136)$$

$$K(q) = K_0 + \sum_{i \in V} \kappa_i(q_i) u_i u_i^T, \quad (137)$$

$$F(q) = f_n + \sum_{i \in V} g_i(q_i) u_i. \quad (138)$$

Mass entries are kilograms, damping entries kg/s, stiffness entries N/m and force entries newtons. Directions are dimensionless. Signed stiffness and damping are accepted; they can represent an active or unstable model and are reported without silently clipping them to a passive case.

The declared equation of motion over this sample interval is

$$M\ddot{p} + C\dot{p} + o_n K(q_n)(p - z_n) = F(q_n). \quad (139)$$

The newly committed word therefore changes a physical operator, not just an annotation on an independently calculated trajectory. On an unavailable observation,  $o_n = 0$  removes the target spring term. The discrete equations, external force, state-dependent force and velocity evolution continue.

#### 13.1 A complete discrete advance

Backward Euler gives the actual linear system solved at every step:

$$G = M + \Delta t C + \Delta t^2 o_n K(q_n), \quad (140)$$

$$b = Mv_{n-1} + \Delta t [F(q_n) + o_n K(q_n)(z_n - p_{n-1})], \quad (141)$$

$$v_n = G^{-1}b, \quad p_n = p_{n-1} + \Delta t v_n. \quad (142)$$

The implementation solves the three-dimensional system with pivoting; it does not form an inverse as its numerical algorithm. The independent reference uses a separate numerical linear solver. Every matrix entry, right-hand side, before state and after state is retained in the trace.

For signed coefficients  $G$  can become singular. This is an explicit numerical failure, not a valid state advance. The computed  $q_n$  remains recorded; the mechanical state and its valid time stay unchanged. Later rows of that trajectory carry a previous-failure status, while other trajectories continue. Nonfinite diagnostic components are represented as null only on failure records.

## 14 The physical eigenmatrix and mechanical checks

Because  $M$  is positive diagonal and  $K$  symmetric, define

$$D(q) = M^{-1/2}K(q)M^{-1/2}. \quad (143)$$

This is the explicit CGK-R1 physical eigenmatrix. It is distinct from the SATNAV observation Jacobian. The native code computes its sorted real eigenvalues and orthonormal eigenvectors, retaining  $D$ ,  $K$ , eigenvectors and a residual. Upper-triangle construction is mirrored to maintain exact floating-point symmetry. The native Jacobi solver scales by the largest matrix entry and uses a pair-relative  $2 \times 10^{-15}$  off-diagonal criterion with at most 64 rotations, preserving small independent modes in matrices with widely separated stiffness scales.

If  $Dw_j = \lambda_j w_j$  and  $\varphi_j = M^{-1/2}w_j$ , then

$$K\varphi_j = \lambda_j M\varphi_j, \quad w_i^T w_j = \delta_{ij}. \quad (144)$$

For  $\lambda_j > 0$  the undamped natural angular frequency is  $\sqrt{\lambda_j}$  rad/s. For  $\lambda_j < 0$ ,  $\sqrt{-\lambda_j}$  is the undamped modal growth rate; a negative value is not reported as a real oscillation frequency. Zero eigenvalues represent neutral stiffness modes. Damping can change the actual transient growth or decay; these are stiffness-mode quantities.

The recorded eigensystem residual is

$$r_\lambda = \max_{a,j} |(DW - W \text{diag } \lambda)_{aj}|. \quad (145)$$

Verification also checks orthonormality and the independently computed spectrum. Signs of eigenvectors and the choice of basis inside repeated eigenspaces are not used as correctness criteria.

### 14.1 What stability can actually be proved

For a fixed target, zero force, fixed  $q$ ,  $o_n = 1$ ,  $C \geq 0$  and  $K \geq 0$ , let  $E = \frac{1}{2}v^T M v + \frac{1}{2}(p - z)^T K(p - z)$ . The implemented backward-Euler step satisfies

$$\begin{aligned} E_n - E_{n-1} = & -\Delta t v_n^T C v_n - \frac{1}{2}(v_n - v_{n-1})^T M(v_n - v_{n-1}) \\ & - \frac{1}{2}(p_n - p_{n-1})^T K(p_n - p_{n-1}) \leq 0. \end{aligned} \quad (146)$$

This follows by substituting the discrete equation and the quadratic difference identity. A changed word changes  $K$  and can change stored potential energy; signed coefficients, external forces and moving observations are also outside this fixed-operator inequality. No general convergence claim is substituted for those actual dynamics.

## 15 Actual SATNAV/UGTS handoff and editable model input

The updated `tools/ugts_handoff.py` invokes the coupled profile by default. It consumes the actual solution and profile files, creates the coupled model input, executes the native backend and independently replays its complete trace. The earlier hold-only behavior is available explicitly as `-legacy-hold`.

Each published record retains the original solver result and a separately labeled modeled state. J/K before, input values and after now come from the executed recurrence. The modeled ENU state is transformed back to ECEF for publication, and its local chart feeds both original 64-bit codecs. Full time, winding, Up, clock bias and original formal covariance remain outside the compact addresses. Failed and unavailable observation rows retain their identity and lineage.

### 15.1 Inputs are inspectable and editable

`examples/coupled/model.json` declares version 3.6.1.7, profile CGK-R1, the four Boolean equations and a list of trajectories. Each trajectory defines initial time/position/velocity/word/phase, mass, damping, base stiffness, chart and hoop parameters, masks, channel geometries and timed samples. Sphere and cone supports, on/off force and stiffness, directional encoders and the observation sequence are all supplied as data. No geometric-to-bit encoder or physical coefficient is silently inferred from a name.

The native compiler writes four exact CSV schemas: trajectory manifest, channels, samples and equation truth tables. Their order and field meanings are specified in `COUPLED_CONTRACT.md`. The resulting trace has one row per sample, including full geometry, predicate words, ASA tuple, J/K operands, physical matrices, eigensystem and state advance. Hashes bind the supplied model and trace.

### 15.2 Run the complete connection

From the package directory, using the delivered Windows binaries:

```
bin/cpu/satnav.exe --input examples/demo \  
  --out results/my_satnav --backend cpu --verify  
python tools/ugts_handoff.py --input examples/demo \  
  --run results/my_satnav --backend cpu
```

The coupled tool also runs a directly supplied model:

```
python tools/coupled.py --input examples/coupled/model.json \  
  --out results/my_coupled \  
  --binary bin/cuda/coupled_kernel.exe --backend cuda
```

Output directories are new for each run. A CUDA thread owns one complete trajectory; it advances that trajectory in time order. Parallelism spans independent trajectories and does not break the carried dependence within a trajectory.

## 16 Preserved 3.6.1.7 coupled implementation validation

These results were executed for parent version 3.6.1.7 on the local Windows machine.

source/baseline\_3\_6\_1\_7\_validation.json preserves its index. The unchanged sibling 3.6.1.7 release contains its full run files. These totals are historical evidence, not new 3.6.1.8 live-receiver tests.

| Executed check                         | Result                                 |
|----------------------------------------|----------------------------------------|
| CPU and CUDA builds / CTest            | Passed; three suites per build         |
| Coupled C++ assertions                 | 202 passed                             |
| Retained Boolean / SATNAV assertions   | 5,060,172 / 105,559 passed             |
| Python test methods                    | 70 passed                              |
| Retained command-line scenarios        | 25 Boolean + 11 SATNAV passed          |
| Editable coupled example, each backend | 72 steps; 6,984 comparisons            |
| Full coupled suite, each backend       | 272 trajectories; 3,308 steps          |
| Independent full-suite comparisons     | 320,876 per backend                    |
| Native analytic / causal invariants    | 20 passed per backend                  |
| Deliberate trace corruptions rejected  | Eight per backend                      |
| Coupled CUDA Compute Sanitizer         | Zero errors; full replay passed        |
| Actual SATNAV-to-coupled handoff       | 256 steps per backend                  |
| Independent actual handoff comparisons | 24,832 per backend                     |
| Separate publication/key/lineage audit | Passed on CPU and CUDA                 |
| Retained SATNAV texture/global CUDA    | Independent replay and memcheck passed |

The comprehensive suite includes 257 trajectories with varying sample counts, so the CUDA run uses three 128-thread blocks with a partial final block. It produces 3,301 advanced states, three deliberately induced singular/overflow failures and four correctly frozen subsequent records. Expected failure records are checked against the independent reference; they are not counted as successful advances.

### 16.1 Evidence that both directions actually execute

Counterfactual native trajectories share observations while changing the initial word: position and velocity change, and later geometric drive changes. A second pair shares its initial state while changing target geometry: drive and J/K state change, then the physical trajectory changes. Edited Boolean equations also change both state and later predicates. These are executed causal comparisons, beyond agreement between two implementations.

Known free-motion, constant-force and diagonal-spring updates are checked against closed forms of the stated discrete method. Off-diagonal generalized eigenmodes, signed coefficients, missing observations, whole-word absorption, chart-domain cases and numerical failure semantics are exercised. CPU regression checks also retain small independent eigenmodes beside a stiffness of  $10^{308}$ .

## 17 Coupled handoff evidence and reproducibility

The actual 256-row SATNAV stream produces 253 observation-driven advances and three prediction advances. There are 51 J/K changes across nine distinct word states. At unavailable-observation epochs 250, 251 and 252, modeled motion continues by approximately 2.10713 m, 1.61596 m and 1.24055 m respectively. Residual-alert epoch 253 remains an input observation with its diagnostic intact.

A separate publication audit checks every complete native trace binding, original solver/epoch/residual/clock/variance record, both key encodings, full time, winding, Up and hoop, all 255 state-carry links, all 256 state-dependent stiffness sums and every hash-chain record. Its implementation reads the output independently from the publisher. Original solver records remain distinct from modeled positions.

### 17.1 Reproduce the full native fixture suite

```
python tools/validate_coupled.py \  
  --binary bin/cpu/coupled_kernel.exe \  
  --backend cpu --out results/my_cpu_validation  
python tools/validate_coupled.py \  
  --binary bin/cuda/coupled_kernel.exe \  
  --backend cuda --out results/my_cuda_validation  
compute-sanitizer --tool memcheck --error-exitcode 99 \  
  bin/cuda/coupled_kernel.exe \  
  --input results/my_cuda_validation/compiled \  
  --out results/my_memcheck --backend cuda  
python tools/coupled.py \  
  --input results/my_cuda_validation/model.json \  
  --out results/my_memcheck_replay --verify results/my_memcheck
```

The replay-only command labels execution as not run because it reads an existing trace; the preceding native command and sanitizer log establish that execution. The native report retains actual backend/device identity and binary/model hashes.

### 17.2 Toolchain and limits of these results

The builds use MSVC 19.44.35221.0 and NVIDIA CUDA compiler 12.8.61. CUDA executes on an NVIDIA GeForce RTX 5070 Ti Laptop GPU, compute capability 12.0, with driver 591.59. Short build directories avoid this workspace's long Windows scratch paths. Python replay uses NumPy 2.1.2. CPU Address/UndefinedBehavior Sanitizer was not run in this Windows release; its earlier Linux record remains archived.

These checks establish the declared coupled implementation on the supplied finite fixtures. Physical coefficients are editable model inputs; this release has not calibrated them against a measured mechanical system or validated live GNSS data. No general switched-system convergence or measured speedup is inferred. The new blend, encoders and mechanical matrix are fully specified CGK-R1 definitions. They do not purport to recover unavailable original M2 equations.

## 18 Literal self-reference: the completed recurrence

Version 3.6.1.6 adds the executable profile **SRK-R1**. Its state is a 32-bit word  $q_t$  that occurs in its own next-state equation. The old release provided literal ASA/NA and J-K primitives but its observation adapter used a constant hold event. This subversion supplies an explicit feedback composition, editable equation input, CPU/CUDA execution and independent equation replay.

Let  $W = \{0, \dots, 2^{32} - 1\}$ , and let  $V, A, N, B, d \in W$  denote the present, ASA, NA, boundary and external drive words. All complements below are 32-bit complements. The initial state satisfies  $q_0 \wedge \neg V = 0$ . A trajectory supplies three Boolean expressions  $E_X, E_J, E_K$  and a requested number of steps  $T > 0$ . The complete transition, in evaluation order, is

$$x_t = E_X(q_t, d), \quad (147)$$

$$a_t = x_t \wedge A \wedge V, \quad (148)$$

$$n_t = a_t \wedge N, \quad (149)$$

$$h_t = \text{popcount}(n_t \wedge B), \quad (150)$$

$$y_t = \begin{cases} 0, & h_t > 0, \\ n_t, & h_t = 0, \end{cases} \quad (151)$$

$$J_t = E_J(q_t, y_t, d), \quad K_t = E_K(q_t, y_t, d), \quad (152)$$

$$q_{t+1} = [(J_t \wedge \neg q_t) \vee (\neg K_t \wedge q_t)] \wedge V. \quad (153)$$

The ASA tuple  $(a_t, n_t, h_t, y_t)$  is the retained whole-word rule. The bracketed part of (153) is the retained literal J-K equation. The supplied expression bindings and final projection to  $V$  are explicit SRK-R1 definitions. For a single active bit they give hold, set, reset and toggle for  $(J, K)$  equal to  $(0, 0), (1, 0), (0, 1), (1, 1)$  respectively.

### Self-reference is an executable dependency

$q_t \longrightarrow x_t \longrightarrow (a_t, n_t, h_t, y_t) \longrightarrow (J_t, K_t) \longrightarrow q_{t+1}$ , with  $q_t$  also read directly by both J/K expressions and by the final update. Every right-hand side reads the same old state. Only the final line commits the next state.

Boundary absorption sets the whole output word  $y_t$  to zero. It does not silently replace the user's J/K expressions with hold: those expressions still evaluate with that zero word. Thus the equations determine the transition even in an absorbed step. Numerical GNSS residual classifications do not gate SRK-R1.

## 19 Actual equations are input data

The editable input is `examples/self_reference/equations.json`. The root declares version 3.6.1.6, profile SRK-R1, optional default equations, and a `trajectories` array. Each trajectory supplies an ID, initial state, drive, four masks and step count; it may override the equations. For example:

```
{
  "version": "3.6.1.6", "profile": "SRK-R1",
  "equations": {"x": "q | d", "j": "y", "k": "y"},
  "trajectories": [{
    "id": 1, "q0": 0, "drive": 15, "present": 15,
    "asa_mask": 15, "na_mask": 15, "boundary_mask": 0,
    "steps": 6
  }]
}
```

The expression vocabulary is `&`, `|`, `^`, `~`, parentheses, the names `q`, `d`, and, for `J/K`, `y`. The literal `0` means the zero word; `1` means the all-ones word. Decimal state and mask fields remain ordinary unsigned integers. Operator precedence is complement, AND, XOR, OR; parentheses can specify every grouping explicitly.

The expression trees define an exact finite Boolean algebra. They are parsed and evaluated as that algebra; they are not passed to a general-purpose evaluator. Each expression is applied independently to each bit. Whole-word absorption couples the bits through  $h_t$ . Arbitrary cross-bit arithmetic, shift operators and continuous-state formulae would require a separately defined extension.

### 19.1 Exact truth-table compilation

For bits  $q_i, y_i, d_i \in \{0, 1\}$ , the compiler creates

$$L_X = \sum_{u,v \in \{0,1\}} E_X(u, v) 2^{2u+v}, \quad (154)$$

$$L_J = \sum_{u,v,w \in \{0,1\}} E_J(u, v, w) 2^{4u+2v+w}, \quad (155)$$

$$L_K = \sum_{u,v,w \in \{0,1\}} E_K(u, v, w) 2^{4u+2v+w}. \quad (156)$$

The CPU/CUDA kernel reads the corresponding table bit. For the example above,  $(L_X, L_J, L_K) = (14, 204, 204)$ . For the latch with  $K = 0$ , the tuple is  $(14, 204, 0)$ . Every allowed expression has a truth table on these inputs, and every Boolean function on these inputs is expressible in this vocabulary. The compilation is exact: each possible input assignment is evaluated, and the native table read retrieves that assignment's value.

## 20 Fixed points, cycles and complete finite execution

During one trajectory, equations, drive and masks are constant. They define a deterministic function  $F : Q_V \rightarrow Q_V$ , where

$$Q_V = \{q \in W : q \wedge \neg V = 0\}, \quad |Q_V| = 2^{\text{popcount}(V)}, \quad q_{t+1} = F(q_t). \quad (157)$$

**Proposition 20.1** (Closure and eventual periodicity). *Every computed state belongs to  $Q_V$ . Every infinite orbit of this fixed-input transition is eventually periodic. A fixed point is a cycle of length one.*

*Proof.* The input requires  $q_0 \in Q_V$  and the final AND with  $V$  places every successor in  $Q_V$ . Among  $|Q_V| + 1$  consecutive states at least two are equal. If  $q_i = q_j$  for  $i < j$ , determinism gives  $q_{i+k} = q_{j+k}$  for every  $k \geq 0$ . This proves periodicity after the first repeated state; it does not assert that the period is one.  $\square$

The implementation emits every requested step, including those after a repeat. Independent replay records a fixed point, a cycle, or no repeat observed within the requested budget. The last case is finite evidence only; it does not imply divergence or the absence of a later repeat. With 32 active bits the general state bound is  $2^{32}$ , so this theorem alone gives no short runtime bound.

### 20.1 Worked exact trajectories

Set  $V = A = N = d = 15$ ,  $B = 0$ ,  $q_0 = 0$ , and  $E_X = q \vee d$ ,  $E_J = y$ . For the latch  $E_K = 0$ ,  $x = a = n = y = 15$  and

$$q_{t+1} = y \vee q_t = 15, \quad (q_0, q_1, q_2, \dots) = (0, 15, 15, \dots). \quad (158)$$

For toggle  $E_K = y$ ,  $J = K = 15$  and

$$q_{t+1} = q_t \oplus 15, \quad (q_0, q_1, q_2, q_3, \dots) = (0, 15, 0, 15, \dots). \quad (159)$$

With  $B = 8$  and  $E_J = E_K = y$ ,  $h = 1$  absorbs all of  $y$  and the state holds. With the same absorbed word but  $E_J = d$ ,  $E_K = 0$ , the state becomes 15. These outcomes follow directly from the supplied equations and expose the difference between output absorption and state reset.

Synchronous evaluation matters. For one active bit with  $E_J = \neg q$  and  $E_K = q$ , the result is toggle. Both expressions use  $q_t$ , never a partially updated state. State projection preserves inactive bits as zero, including when an input expression uses the all-ones constant.

## 21 Native execution and independent equation replay

The shared `include/self_reference.hpp` implements the truth-table and literal transition operations. The CPU walks each trajectory in step order. CUDA assigns one thread to each independently seeded trajectory, then executes its recurrence sequentially within that thread. No ordering between independent trajectories is needed; different trajectories never share a mutable state.

The compilation interface is a decimal CSV with columns

```
trajectory_id,q0,drive,present,asa_mask,na_mask,boundary_mask,
x_lut,j_lut,k_lut,steps
```

The header is physically one line. Every trace row retains

```
trajectory_id,step,before,drive,x,asa,na,hits,output,j,k,after
```

The native run also records backend, device, timing and trace lengths. The Python reference evaluates the original expression trees on whole integer words; it does not use the compiled truth tables as its transition oracle. It compares every trace field, row count and trajectory ordering, and records equation/input provenance. This cross-check can detect table compilation, transition and output corruption; it is finite execution evidence rather than a machine-checked proof.

### 21.1 Reproduce the equation-input run

From the package directory, build CPU and run:

```
cmake -S . -B build_cpu
cmake --build build_cpu --config Release --parallel 2
python tools/self_reference.py \
  --input examples/self_reference/equations.json \
  --out results/selfref_local \
  --binary build_cpu/Release/self_reference.exe --backend cpu
```

For single-configuration systems the binary is `build_cpu/self_reference`. For CUDA, enable `SATNAV_ENABLE_CUDA=ON`, choose the GPU binary and use `-backend cuda`. Each run uses a new output directory. The JSON expressions are editable without recompiling C++ or CUDA; the compiler regenerates their tables.

### 21.2 Relation to the existing SATNAV and UGTS equations

SATNAV-R1 retains its corrected-code equations, QR solve, OTAN2 source ratio, both 64-bit codecs and full coordinate/time records. The earlier observation adapter's hold event is retained by the explicit legacy option. SRK-R1 is the retained formal word state-feedback computation and runs without a GNSS residual admission condition. The new bindings are identified as definitions of this release. CGK-R1 now provides explicit new circle-plus, encoder and physical-matrix definitions and connects the observation path, as detailed in Section 10. The unavailable earlier M2 definitions are not claimed to have been reconstructed.

## 22 SRK-R1 baseline validation, version 3.6.1.6

All entries here concern the executable 3.6.1.6 source on the current Windows machine. The 3.6.1.5 preparation record is preserved separately under `results_3_6_1_5/`. Finite comparisons check the requested trajectories; the closure and eventual-periodicity argument is given separately above.

| Executed check                        | Result                                                        |
|---------------------------------------|---------------------------------------------------------------|
| CPU CMake build and CTest             | Passed, 2 test programs                                       |
| Retained SATNAV C++ checks            | 105,559 passed                                                |
| New literal feedback C++ checks       | 5,060,172 passed                                              |
| Python test methods                   | 31 passed                                                     |
| Native command-line scenarios         | 25 feedback + 11 SATNAV passed                                |
| Editable six-trajectory example       | 34 transitions, 408 exact field comparisons per backend       |
| 257-trajectory stress input           | 4,225 transitions, 50,700 exact field comparisons per backend |
| CPU and CUDA equation replay          | Both passed sample and stress inputs                          |
| Feedback GPU Compute Sanitizer        | Zero memory errors; sanitized trace replay passed             |
| Retained SATNAV independent reference | 256 epochs, 4,828 comparisons passed                          |
| SATNAV CUDA texture/global paths      | Both passed independent comparisons and memcheck              |
| CPU sanitizers in this Windows run    | Not run; earlier Linux evidence archived                      |

### 22.1 Recorded toolchain and device

MSVC 19.44.35221.0, Visual Studio 2022 BuildTools; NVIDIA CUDA compiler 12.8.61; NVIDIA GeForce RTX 5070 Ti Laptop GPU, compute capability 12.0, driver 591.59. The source targets `sm_120` and `compute_120`. Device, runtime, allocation sizes and kernel-only times are retained in each native run record. The short external build directories used on this machine avoid Windows compiler scratch-path limits caused by the long workspace path.

The stress input uses seed 3616, varying 1–32 steps per trajectory, full 32-bit operands and 257 trajectories so that the 128-thread CUDA blocks include a partial final block. Direct AST replay checks all 12 trace columns. The Python suite also rejects deliberate mutations of every trace column, missing/extra rows and reordered rows. The C++ checks include independent bit oracles and exhaustive small domains.

### 22.2 Evidence and delivered implementation

`source/baseline_3_6_1_6_validation.json` indexes the archived baseline evidence. The exact equation JSON, compiled CSV, complete native/reference traces and their hashes are retained. The baseline delivered Windows CPU and CUDA executables under `bin/windows/`; CUDA execution uses the installed NVIDIA runtime and driver. No measured speedup is inferred from these small fixtures. The PDF is built with Tectonic 0.17.0 and visually checked in that release. Source and delivered artifacts were listed in the regenerated SHA-256 manifest.

## 23 The imported UGTS and aTOMos contracts

The supplied prototype defines a finite, query-first state-and-relation package. It includes cone/-sphere relations, a bounded translational sweep, log-polar metric and kinematics, explicit time and winding, separate topology profiles, finite grammar, 64-bit key codecs and audited uncertainty/memory claims. Its authoritative handoff is

support → compatibility → guard → verified event → transition → lineage. (160)

This order is retained in the downstream satnav event adapter. [U, pp. 1, 9–12]

### 23.1 What U does not define

U contains no satellite ephemeris propagation, receiver clock observation equation, code-delay correction model, geodetic datum or GNSS position estimator. These are new SATNAV-R1 definitions, grounded in primary GNSS references rather than attributed to the earlier prototype. [N01]

The base aTOMos contracts retained here are the four-field ASA word rule, explicit JK bit semantics, source-ratio OTAN2, typed unavailable values, immutable input snapshots and ordered result/provenance records. The new matrix in the receiver solve is the explicit *GNSS design matrix*. It is not a silently completed physical beamforming/eigenmatrix from an earlier profile. [A, retained M2 project contract]

### 23.2 Source figures and key table

U's Figure 4 on printed p. 5 keeps the half-turn state bundle distinct from the orientation-reversing Klein quotient. Its Table 1 and Figure 6 on printed p. 8 keep contiguous packing distinct from MSB-round-robin Morton interleaving. Figure 8 on printed p. 10 returns the result to the canonical event sequence. All three distinctions survive the specialization.

The complete earlier 50-operator catalog remains a source catalog, not a claim that every operator is necessary for positioning. The delivered helpers implement the subset selected below. A source mechanism omitted from the default satnav path is identified rather than repurposed without a declared meaning.

## 24 Receiver state and source-symbol separation

The authoritative positioning state is

$$z_e = (x_e, y_e, z_e, b_e) \in \mathbb{R}^4, \quad b_e = c \delta t_e, \quad (161)$$

where the first three components are ECEF metres and the fourth is receiver clock bias in metres. Clock seconds are  $b_e/c$  with  $c = 299792458$  m/s in this profile. This is the standard code-positioning unknown set. [G1]

| Quantity                    | Role                     | Representation                                                      |
|-----------------------------|--------------------------|---------------------------------------------------------------------|
| $r$                         | Receiver location        | Three unquantized ECEF doubles.                                     |
| $b$                         | Receiver clock bias      | Metres, not a branch bit.                                           |
| $P_i$                       | Code observation         | Metres, after the explicitly recorded correction convention.        |
| $\rho$                      | Local log-radius         | UGTS chart quantity, not the code range.                            |
| $\theta$                    | Local angular coordinate | $\text{atan2}(N, E)$ in a fixed ENU chart.                          |
| $\phi_h$                    | UGTS hoop phase          | Separate supplied periodic field; not silently geographic latitude. |
| $\varphi_g, \lambda_g, h_g$ | Geodetic coordinates     | Latitude, longitude and ellipsoidal height.                         |
| $X, w$                      | Tick and winding         | Full integer time identity retained beside its modular key field.   |
| $K_C, K_M$                  | Quantized addresses      | Two distinct 64-bit codecs, not complete navigation states.         |

The source's lower-case hoop  $\phi$  is retained as  $\phi_h$  to avoid colliding with geodetic latitude. The earlier aTOMos phase alias is resolved through a named adapter: OTAN2 receives increments of local  $\rho, \theta$ , while the key retains its separate UGTS hoop field. No 3D coordinate is silently removed. [U, p. 2][N02]

The authoritative records also include epoch identity, time scale, satellite/channel mapping, input/product provenance, statuses and uncertainty assumptions. All fit results preserve the full ECEF estimate and clock value outside the compact key.

## 25 The corrected-code observation model

A decoded code measurement can be modelled as a geometric range, receiver/satellite clock terms and selected propagation/instrumental delays. SATNAV-R1 accepts an upstream additive correction  $a_i$  and defines

$$\tilde{P}_i = P_i + a_i = \|s_i - r\| + b + \varepsilon_i. \quad (162)$$

This is the reduced standard code-positioning model after the declared upstream terms are accounted for. [G1, G4]

For the convention

$$P_i = R_i + b - c\delta t_i^{\text{sat}} + T_i + I_i + B_i + \varepsilon_i, \quad (163)$$

the matching additive correction is

$$a_i = c\delta t_i^{\text{sat}} - T_i - I_i - B_i. \quad (164)$$

The adapter must define the exact code, satellite clock convention, relativistic clock terms and hardware/group biases it uses. The kernel adds the supplied term *once*. It does not repeat an already applied correction or invent a missing ionosphere/troposphere model.

### 25.1 Satellite coordinates and Earth rotation

$s_i$  is the satellite position at signal emission expressed in the Earth-fixed axes associated with reception. The upstream stage must perform the necessary transmit-time and frame handling consistently. A satellite coordinate computed in emission-time ECEF axes cannot simply be treated as already in the reception axes. [G2]

The solver keeps these supplied positions and corrections fixed during its four-state iteration. The profile is therefore an adapter boundary for prepared code data, not a complete broadcast-navigation decoder. Raw receiver or RINEX values require an upstream modelling adapter before using the named CSV contract. [N01]

### 25.2 One clock system

There is exactly one receiver-clock unknown. Unaligned measurements from multiple GNSS time systems generally need additional bias treatment; this release does not estimate a hidden inter-system offset. Use compatible code/clock data, or supply and document an upstream alignment under a separate profile.

## 26 Input epochs, channels and aTOMos selection

Each independent epoch binds its receive-time label, approximate initial state and 32-bit channel masks. Each observation row points to that epoch and names a unique channel in  $0, \dots, 31$ . Channel IDs, not row order, define mask bits.

Let  $V$  contain present channels and  $X$  contain channels marked ready. The retained word selection is

$$a = X \wedge M_{\text{ASA}} \wedge V, \quad (165)$$

$$n = a \wedge M_{\text{NA}}, \quad (166)$$

$$h = \text{popcount}(n \wedge M_{\partial}), \quad (167)$$

$$y = \begin{cases} 0, & h > 0, \\ n, & h = 0. \end{cases} \quad (168)$$

The solver uses precisely the channels selected by  $y$ . These four fields are emitted unchanged in the result. The default ASA/NA masks are all ones and the default boundary is zero. [A, retained M2 project contract] [N04]

### 26.1 Whole-word behavior remains literal

A boundary bit intersecting the selected word absorbs the entire epoch group. It does not mean “drop only this satellite.” To explicitly omit an individual observation, use a selector bit, a not-ready flag or an upstream policy with its own provenance. Replacing the boundary behavior with per-bit clearing would change the base engine.

A remaining support of fewer than four channels returns T00\_FEW. Four channels must still provide a numerically independent design matrix. A full bit count cannot prove observability by itself. The new GNSS solver checks the resulting matrix rather than treating POPCNT as a complete position solution. [G1][N03]

### 26.2 Timeline and schema consistency

An epoch identifier binds rows from one receiver event. The time column is full unwrapped GPST seconds under the supplied upstream convention; modular UGTS time is added later. Duplicate epoch IDs, duplicate channels or duplicate satellite IDs within an epoch are errors. Correcting a false upstream time label is not something a CSV parser can infer from equal strings.

The current CSV reader uses finite bounded numeric fields and explicit nonnegative IDs. Physical correction provenance belongs to the input profile; metadata consistency is not a physical synchronization measurement.

## 27 The weighted positioning step

For current estimate  $z^{(k)} = (r^{(k)}, b^{(k)})$ , define

$$R_i^{(k)} = \|r^{(k)} - s_i\|, \quad (169)$$

$$v_i^{(k)} = \tilde{P}_i - R_i^{(k)} - b^{(k)}, \quad (170)$$

$$H_i^{(k)} = \left[ \frac{x^{(k)} - s_{ix}}{R_i^{(k)}}, \frac{y^{(k)} - s_{iy}}{R_i^{(k)}}, \frac{z^{(k)} - s_{iz}}{R_i^{(k)}}, 1 \right]. \quad (171)$$

Linearization gives  $H\Delta z \simeq v$ . With declared independent code standard deviations  $\sigma_i > 0$ , whiten the rows as

$$A_i = H_i/\sigma_i, \quad d_i = v_i/\sigma_i, \quad \Delta\hat{z} = \arg \min_{\Delta} \|A\Delta - d\|_2^2. \quad (172)$$

Then update  $z^{(k+1)} = z^{(k)} + \Delta\hat{z}$ . The nonlinear model and iterative linearization follow the code-positioning references; the QR implementation is the explicit numerical choice of this specialization. [G1, G4][N03]

### 27.1 No normal-equation update

The C++/CUDA implementation uses streaming Givens QR on the whitened rows. It does not form and invert  $A^T A$  for the update. This avoids squaring the design matrix's condition number as part of the solution method. The independent Python reference uses a different Householder QR implementation on the full row matrix.

All four unknowns use metres, including the clock bias. A floating-point conversion to seconds is applied only to a displayed/consumer clock offset. This keeps a factor of the speed of light out of the fourth Jacobian column.

The default iteration stops when both position-increment norm and absolute clock increment are at most  $10^{-4}$  m, with at most 12 steps. These are numerical convergence controls, not a statement that real measurements yield 0.1 mm accuracy. The convergence domain depends on the input model and initial approximation.

## 28 Streaming QR, rank and formal covariance

The kernel maintains an upper-triangular  $4 \times 4$  matrix  $R$  and a transformed right-hand side  $z$ . Each whitened row is appended by rotations that eliminate its components against the current triangular diagonal.

For diagonal  $R_{jj}$  and current row component  $a_j$ , let

$$\gamma = \text{hypot}(R_{jj}, a_j), \quad c_j = R_{jj}/\gamma, \quad s_j = a_j/\gamma. \quad (173)$$

The same rotation is applied to the remaining row and right-hand-side entries. A zero pair needs no rotation. The result is solved by back substitution.

### 28.1 The implemented rank screen

The retained check is

$$|R_{jj}| > \tau \max_k |R_{kk}| \quad \text{for all } j, \quad \tau = 10^{-10}. \quad (174)$$

This is a declared unpivoted triangular-diagonal screen. It detects the tested rank-deficient cases, but is not an SVD condition-number certificate or a proof of well-conditioned geometry for all possible inputs. A failed screen produces `RANK_DEFICIENT`, not a coordinate filled with zeros. [N03]

### 28.2 Covariance is tied to the assumed sigmas

After convergence the Jacobian is rebuilt at the final estimate. From  $R^{-1}$ , compute the diagonal of

$$P = (H^T W H)^{-1} = R^{-1} R^{-T}, \quad W = \text{diag}(\sigma_i^{-2}). \quad (175)$$

The implementation obtains this from triangular solves, not a dense matrix inverse of  $H^T W H$ . It returns coordinate/clock variances in squared metres. This is formal uncertainty under the declared local linear and noise model. [G5]

It is not measured error against truth, and not a guaranteed bound on faulty or mis-modelled measurements. The independent fixture comparisons check both estimates and these formal diagonal values; true-error metrics are separately computed only because the artificial fixture has a known generating position.

## 29 Solver status and residual diagnostics

Convergence, residual fit and the later UGTS event are separate fields.

| Solver status   | Meaning                                                               |
|-----------------|-----------------------------------------------------------------------|
| CONVERGED       | Numerical increment thresholds reached with valid final calculations. |
| MASK_ABSORBED   | Selected word intersected the retained whole-word boundary.           |
| TOO_FEW         | Fewer than four selected observations.                                |
| RANK_DEFICIENT  | Triangular rank/covariance check failed.                              |
| ITERATION_LIMIT | The selected number of iterations did not satisfy convergence.        |
| NUMERIC_FAILURE | Invalid range, nonfinite result or unusable numeric intermediate.     |

For a converged solution, preserve every selected post-fit residual  $r_i = \tilde{P}_i - \|s_i - \hat{r}\| - \hat{b}$ , and report

$$\text{RMS} = \sqrt{\frac{1}{m} \sum_i r_i^2}, \quad S = \sum_i (r_i/\sigma_i)^2, \quad Z_{\max} = \max_i |r_i/\sigma_i|. \quad (176)$$

The default comparison  $Z_{\max} \leq 6$  is a named residual budget. It is not a leverage-normalized test or a claimed false-alarm probability. The solver does not automatically discard a channel and retry until this flag passes. [N09]

Exactly four observations give NO\_REDUNDANCY, even if their residuals are small. For more observations, the flag is WITHIN\_RESIDUAL\_BUDGET or RESIDUAL\_ALERT. A single gross-outlier fixture converges numerically but produces the latter alert; the data record retains its large actual position error.

RAIM/ARAIM, HPL/VPL, continuity risk, certified spoof detection and navigation-action authority are not supplied by these flags. A valid solution record can be useful for replay and analysis without representing those additional guarantees.

### 30 Local ENU and the UGTS log chart

The receiver solution remains three-dimensional. For a declared geodetic anchor  $(\varphi_0, \lambda_0, h_0)$ , convert ECEF differences to East/North/Up using

$$R_{ENU} = \begin{bmatrix} -\sin \lambda_0 & \cos \lambda_0 & 0 \\ -\sin \varphi_0 \cos \lambda_0 & -\sin \varphi_0 \sin \lambda_0 & \cos \varphi_0 \\ \cos \varphi_0 \cos \lambda_0 & \cos \varphi_0 \sin \lambda_0 & \sin \varphi_0 \end{bmatrix}. \quad (177)$$

Then  $(E, N, U)^T = R_{ENU}(\mathbf{r} - \mathbf{r}_0)$ . The transform is an explicit GNSS/geodesy addition, separate from the source's abstract local chart. [G3]

With horizontal radius  $r_h = \sqrt{E^2 + N^2}$ , use

$$\rho = \ln(r_h/r_0), \quad \theta = \text{atan2}(N, E). \quad (178)$$

Up remains a separate value. A declared core radius returns `origin_core` instead of evaluating a singular logarithm. Out-of-range  $\rho$  is not automatically wrapped into a different geographic position. [U, pp. 3–4][N02]

#### 30.1 Metric and finite steps carried from U

For the planar chart,

$$ds^2 = r_h^2(d\rho^2 + d\theta^2), \quad \Delta r_h = r_h(e^{\Delta\rho} - 1), \quad (179)$$

$$J = r_h \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}. \quad (180)$$

These identities explain the local representation; the new satellite-range solve still uses full ECEF coordinates. Quantizing this chart does not improve the original measurement precision or create a third coordinate. [U, p. 4]

The geodetic utility uses the declared WGS84 ellipsoid constants  $a = 6378137$  m and  $1/f = 298.257223563$ . Its height is ellipsoidal height, not an unprovided geoid-corrected height above mean sea level. The reference tests its conversions and inverse relation over a finite set of points. [G6]

## 31 OTAN2 in satnav: preserve the observable

The inherited literal operator remains

$$\text{otan2}_s(\Delta\theta, \Delta\rho) = \arctan(\Delta\theta/\Delta\rho), \quad \Delta\rho \neq 0, \quad (181)$$

with a separately supplied reference-axis subtraction when selected. This is a *log-chart motion slope*. It is not used to solve pseudoranges, and is not relabeled as the receiver's geographic heading or satellite azimuth. [A, retained M2 project contract]

The passive adapter compares successive defined local chart records in the same frame. It uses an explicitly named principal-difference policy for  $\Delta\theta$ ; that policy does not reconstruct missed revolutions. Full linear time and the chosen phase convention are stored with the observation.

### 31.1 Missing direction remains a status

A first observation has no previous chart increment. A zero pair has no direction. A pure angular increment has an undefined literal ratio. The adapter records `first_observation`, `zero_increment` or `ratio_undefined` rather than inventing a perfect zero angle. A sequence with non-increasing time is identified instead of being silently reordered into a new physical history.

A change in the axis alone changes the raw residual by the opposite angle. A common positive scaling of both increments preserves the ratio, and simultaneous reversal also preserves the literal ratio while reversing the actual directed motion. These are inherited operator laws, not new navigation-sensor guarantees.

### 31.2 Literal J/K and the explicit feedback profile

The retained equation is

$$q^+ = (J \wedge \neg q) \vee (\neg K \wedge q). \quad (182)$$

SRK-R1 binds J and K to supplied expressions that read the old state and literal ASA output, then carries  $q^+$  into the next step. The full recurrence, expression input and execution semantics are given in Section 18.

The default observation adapter now invokes CGK-R1 and publishes its actual carried state. The earlier hold event remains available through the explicit legacy option. CGK-R1 supplies new circle-plus, encoder and mechanical-operator definitions in Section 10; the unavailable original M2 forms are not claimed to have been recovered.

## 32 Paired supports and the canonical UGTS event

This section records the retained observation-only adapter, selected with `-legacy-hold`. The default coupled handoff is specified in Section 10; it retains residual diagnostics without using this earlier admission rule to override its equations.

U normalizes the two spherical supports as

$$f_L(x) = \|x - c_L\| - R_L, \quad f_R(x) = \|x - c_R\| - R_R, \quad (183)$$

with a profile selecting union, intersection or either-with-tag. The overlap flag is retained separately. [U, p.3]

SATNAV-R1 uses the union of two *declared local processing domains* for its worked event adapter. With supplied deterministic position-radius allowance  $u_p$ , its per-sphere margin is

$$m_j = R_j - \|x - c_j\| - u_p. \quad (184)$$

Admission means at least one margin is nonnegative. This is a geometric/query policy for the supplied local frame, not a claim that the true receiver error is bounded by  $u_p$ . Its origin is explicitly recorded. [N09]

The corresponding handoff is implemented in order:

| UGTS stage     | Satnav realization                                                                      |
|----------------|-----------------------------------------------------------------------------------------|
| Support        | Candidate lies within at least one declared local support under the supplied allowance. |
| Compatibility  | Fixed frame and increasing unwrapped time; supported estimator-clock convention.        |
| Guard          | Residual interval compared with a named metre-domain budget.                            |
| Verified event | Candidate satisfies those computational/model conditions with residual redundancy.      |
| Transition     | Publish an observation record only. No external control command.                        |
| Lineage        | Source/profile IDs, full time, source keys and the preceding record digest.             |

For the demo,  $u_p = 5$  m, the two support radii are 10,000 m, and centres are (0, 0, 0) and (100, 100, 0) ENU metres. These are labelled worked parameters, not values extracted from the attached prototype or a real site survey.

The phrase `POSITION_CANDIDATE_VERIFIED` means verified under this declared computational handoff. It does not supply certified GNSS integrity or prove the physical correctness of the upstream observations.

### 33 Finite cones and translational sweeps retained

For the source's slant length  $T > 0$  and half-angle  $0 < \alpha_c < \pi/2$ ,

$$h = T \cos \alpha_c, \quad R = T \sin \alpha_c. \quad (185)$$

With apex  $c$ , unit axis  $a$  and query  $x$ , set

$$z = a \cdot (x - c), \quad q = \|(x - c) - za\|. \quad (186)$$

The exact primitive relation is the signed meridian distance to the triangle  $(-R, h), (R, h), (0, 0)$ , with negative interior and zero boundary. The implementation computes distances to the three boundary segments and uses the source's inside test  $0 \leq z \leq h, q \leq z \tan \alpha_c$ . [U, pp. 2–3]

These are optional UGTS query helpers in the archive. A finite cone is not silently substituted for the satellite pseudorange sphere or a physical antenna response. The default receiver solve uses the actual distance-to-satellite equations already stated.

For a fixed-orientation cone translated along a segment of length  $L$ ,  $n \geq 2$  uniform samples and sample minimum  $m_n$ , retain

$$m_n - \frac{L}{2(n-1)} \leq \inf_{u \in [0,1]} d_C(x - s(u)) \leq m_n. \quad (187)$$

The source derives this using the one-Lipschitz distance response to translation and the nearest sample spacing. It is not a general exact SDF for arbitrary rotated sweeps. [U, p. 3]

#### 33.1 Numerical and nesting distinction

The helper evaluates this analytic interval in ordinary floating arithmetic. It does not implement directed rounding or a proven machine-error enclosure. The interval's *width* decreases with sample spacing. Intervals from arbitrary nonnested sample grids need not be nested, so the implementation does not claim that stronger property. [N07]

The source's constraint-release and hinge-torque mechanisms remain separately typed geometric/mechanical profiles. Removing a constraint is not an automatic improvement to satellite observability, nor does the existence of a hinge model supply a physical receiver trajectory. They are not inserted into the default position estimator.

## 34 Jitter intervals and the explicit source correction

U defines  $f_j = f + \epsilon_j \sigma$  with  $\sigma \in \{-1, +1\}$ , then prints the equality

$$[f_j - \epsilon_j, f_j + \epsilon_j] = [f - \epsilon_j, f + \epsilon_j]. \quad (188)$$

That equality is generally false for nonzero  $\epsilon_j$ . This edition preserves its source location and explicitly separates the two interpretations rather than silently changing the supplied page. [U, p.5][N06]

### 34.1 Two valid, different interval statements

The perturbation family obeys

$$f_j \in [f - \epsilon_j, f + \epsilon_j]. \quad (189)$$

If  $f_j$  is the available perturbed observation and  $|f_j - f| \leq \epsilon_j$ , then

$$f \in [f_j - \epsilon_j, f_j + \epsilon_j]. \quad (190)$$

Both are valid enclosure statements, but the interval centres differ. The helper returns both named intervals and a test verifies that they are not falsely equated. The source's example centred around  $f$  is treated as the perturbation-family interval.

No jitter is added to the GNSS code observations or to solved ECEF coordinates. The deterministic bit can be metadata or a separate query perturbation, not a method for manufacturing measurement precision or removing multipath.

### 34.2 Metre-domain guard in the legacy event adapter

For the largest absolute post-fit residual  $m = \max_i |r_i|$  and a *supplied* absolute residual perturbation allowance  $u_r$ , form

$$[\max(0, m - u_r), m + u_r]. \quad (191)$$

A budget  $B$  is met when the upper endpoint is at most  $B$ , exceeded when the lower endpoint is greater than  $B$ , and otherwise unresolved. This is a conditional model classification. The demo uses  $u_r = 0.1$  m and  $B = 20$  m as stated fixture parameters. It does not identify nominal  $\sigma_i$  with a hard worst-case bound.

## 35 Two exact 64-bit key layouts

U's field widths and layouts are retained exactly:

| Field            | Bits | Contiguous range | Domain                       |
|------------------|------|------------------|------------------------------|
| Log-radius index | 20   | 63:44            | Declared $\rho \in [-20, 0]$ |
| Chart angle      | 18   | 43:26            | Periodic $[0, 2\pi)$         |
| Modular tick     | 14   | 25:12            | $0, \dots, 16383$            |
| Hoop phase       | 12   | 11:0             | Periodic $[0, 2\pi)$         |

The contiguous word is

$$K_C = (q_\rho \ll 44) \vee (q_\theta \ll 26) \vee (q_X \ll 12) \vee q_{\phi_h}. \quad (192)$$

The Morton word emits source bits in MSB round-robin order

$$\rho_{19}, \theta_{17}, X_{13}, \phi_{11}, \rho_{18}, \theta_{16}, X_{12}, \phi_{10}, \dots \quad (193)$$

until each source field is exhausted. They are intentionally different encodings. [U, pp. 7–8]

### 35.1 The source's exact golden pair

For (949111, 0, 1920, 227), the new tests recover

```
contiguous: e7b77000007800e3
Morton:    88823bb88099128b
```

Both decode to the same quantized tuple. Random round trips and prefix-refinement bounds are tested separately from the new positioning arithmetic.

### 35.2 Address identity is not state identity

The 64-bit key has exactly  $2^{64}$  possible words. A key does not encode arbitrary clock bias, full linear time, Up, formal covariance, correction products or lineage. Those remain separate full fields in the event record. The 12-bit field is the named hoop phase, not a simultaneously hidden arbitrary matrix and grammar payload. [U, pp. 8–9][N05]

The address construction follows the position estimate; it is not used to quantize satellite coordinates or code ranges before the receiver solve. A matching key is an index match, not proof that two entire navigation events are identical.

## 36 Quantization, error and explicit origin handling

SATNAV-R1 declares nearest-node radial quantization over  $[-20, 0]$ :

$$h_\rho = \frac{20}{2^{20} - 1}, \quad q_\rho = \text{round}_+ \left( \frac{\rho + 20}{20} (2^{20} - 1) \right). \quad (194)$$

Angles use nearest periodic nodes with upward half ties, while the source's bit widths and range remain unchanged. This tie/quantization choice is an explicit specialization, not an unquoted algorithm recovered from an unavailable source ZIP. [N05]

For an angular width  $b$ ,  $h_\theta = 2\pi/2^b$ . A nearest-node quantization gives  $|\delta\rho| \leq h_\rho/2$  and circular  $|\delta\theta| \leq h_\theta/2$  within the chosen range. Using the exact source radial step, a conservative planar error bound is

$$\|\hat{x} - x\| \leq r_h(e^{h_\rho/2} - 1) + 2r_h e^{h_\rho/2} \sin(h_\theta/4). \quad (195)$$

It follows from adding a radial displacement and an angular chord at the enlarged radius. The bound scales with radius; log packing is not uniform physical resolution. [U, pp. 4, 8–9]

### 36.1 Default diagnostic chart

The demo chooses  $r_0 = 100000$  m and a 0.1 m origin core. It keys only values with  $-20 \leq \rho \leq 0$ . These choices define an indexed diagnostic domain, not the receiver's positioning accuracy or a geographic route. Out-of-range or core values keep the full position and a chart status without wrapping them into a new position.

### 36.2 No equal-information compression claim

U explicitly audits the ratios 192/64, 32/1 and 512/12 as nominal width ratios between records with different semantics. This specialization does not reintroduce a claim that one address is equivalent to all original floating information. [U, pp. 8–11]

The packed key is useful for lookup and local indexing after the relevant state exists. Position estimation, corrected observations, uncertainty and event verification remain real computational and storage costs.

## 37 Linear time, winding and separate topology profiles

For reference tick  $X_0$ , period  $P > 0$  and full linear tick  $X$ ,

$$u = \frac{X - X_0}{P}, \quad w = \lfloor u \rfloor, \quad \psi = u - w \in [0, 1). \quad (196)$$

The source's example  $X = 1920$ ,  $X_0 = 1135$ ,  $P = 256$  yields  $w = 3$  and  $\psi = 0.06640625$ . The new helper reproduces that value, including floor-division behavior for negative ticks. [U, p. 5]

The satnav key stores only the 14-bit tick residue, while the event retains full tick, GPST coordinate, origin/period and winding. A period of 16384 one-second ticks repeats its residue after 4 h 33 min 4 s. It does not identify events across those repetitions without the additional time/lineage fields.

### 37.1 The source half-turn is not the reflective quotient

U's literal odd radial-wrap bundle applies

$$(\rho, \theta, \phi_h, o) \mapsto (\rho', \theta + \pi, -\phi_h, -o). \quad (197)$$

Its separately named reflective Klein profile applies

$$(\rho, \theta, \phi_h, o) \mapsto (\rho', \pi - \theta, -\phi_h, -o). \quad (198)$$

The second reverses angular orientation; the first is a rotation on the base circle with a changed internal state. The archive retains independent helpers and tests, matching the distinction in U's Figure 4. [U, pp. 5–6]

Neither rule is applied to the GNSS receiver's ECEF position, a satellite orbit or the physical Earth. A local index experiment may select either profile explicitly; it does not make a geographical discontinuity an actual displacement. [N08]

The source's finite grammar and radix refinement remain local state/index mechanisms. Their existence does not turn one receiver solution into an unbounded tree or prove that every possible trajectory is retrieved in constant time.

## 38 GPU kernel: one independent solve per thread

The kernel takes a bounded batch of immutable epoch metadata and a structure-of-arrays observation buffer. One thread evaluates one full independent four-state solution. Each thread owns its QR workspace, iterative estimate, status and residual record. There is no cross-epoch update or unordered reduction of position states.

```
const size_t e = size_t(blockIdx.x) * blockDim.x + threadIdx.x;
if (e < count) {
    output[e] = solve_epoch(epochs[e], observations, e, config);
}
```

The launch uses 128 threads per block. Repeated jobs can supply thousands of independent epochs; a single live low-rate receiver does not by itself provide enough parallel work to establish a GPU speed advantage. This kernel is a batch/replay specialization.

### 38.1 Numerical implementation

The actual shared implementation is in `include/satnav_core.hpp`. It performs the retained ASA arithmetic, row construction, QR update, convergence and final covariance/residual calculation. The CUDA translation unit invokes that implementation with its device reader. The CPU uses the same word and numerical function with a plain double reader.

An independent Python implementation uses separate matrix construction and Householder QR. It is not merely calling back into the shared C++ function. The two verification levels therefore distinguish device/host conformance from an additional numerical reference comparison.

### 38.2 Potential performance costs remain visible

Independent threads may finish in different iterations, producing normal warp imbalance. QR uses FP64 arithmetic and square roots; no tensor-core speed claim is made. The source keeps full precision where satellite-range subtraction requires it, rather than forcing coordinates into float textures or one-bit words.

The build disables fused contraction for the shared review profile and does not enable fast math. The acceptance checks use named tolerances, not a claim of bit-identical floating evaluation across every compiler and processor.

## 39 Texture and global input paths

For  $N$  independent epochs, channel  $c \in [0, 31]$  and observation field  $f \in [0, 5]$ , flatten the input as

$$\text{index}(e, c, f) = (6c + f)N + e. \quad (199)$$

At a fixed channel/field, neighboring epoch threads therefore read neighboring values. The six fields are satellite  $x, y, z$ , raw code, additive correction and nominal sigma.

### 39.1 Exact FP64 bit transport in the texture path

A CUDA linear texture resource uses an int2 channel. Each texel carries the 64 raw bits of one double; the device reconstructs the value using the low and high 32-bit words. The read uses an integer `tex1Dfetch`; there is no interpolation, normalized-coordinate lookup or conversion of ranges into float values. [H4][N10]

```
const int2 bits = tex1Dfetch<int2>(texture, integer_index);
const double value = __hiloInt2Double(bits.y, bits.x);
```

The global path reads the same double buffer directly. Both paths use the same solver and output record. Equality of decoded inputs is a data-representation property; it does not guarantee a texture speedup for this access pattern.

### 39.2 Ownership and publication

Device buffers and events use local ownership wrappers. A texture object is destroyed before its backing buffer. Each launched epoch has one writer. The program checks CUDA allocation, copies, launch and completion before publishing downloaded result files. A failed device call is an error, not a CPU fallback labelled as a GPU success.

The first chunk receives an untimed warmup. Device events measure subsequent kernel intervals and their sum is reported. CSV parsing, allocations, texture construction, host packing, uploads, downloads and the Python event adapter are outside that sum. A CPU loop interval and a full application latency are therefore distinct quantities.

## 40 The RTX 5070 Ti Laptop target and VRAM plan

NVIDIA lists the **RTX 5070 Ti Laptop GPU** with a **12 GB GDDR7** memory configuration. NVIDIA's CUDA table lists the GeForce RTX 5070 Ti family at compute capability 12.0; the build selects `sm_120` and matching virtual code. CUDA 12.8 introduced compiler support for `SM_120`. These are supporting implementation facts, not equations supplied by the UGTS prototype. [H1–H3]

The program's `-probe` path records the actual device name, major/minor capability, driver/runtime versions, total/free memory and linear-texture limit. This edition's targeted binary reports a mismatching capability rather than assuming that a marketing label guarantees the compiled architecture.

### 40.1 Payload accounting

The executed CPU build reports 72 bytes per epoch metadata record and 376 bytes per result. The fixed observation representation is  $32 \cdot 6 \cdot 8 = 1536$  bytes per epoch, so the tested ABI payload is

$$M_{\text{payload}}(N) = N(72 + 376 + 1536) = 1984N \text{ bytes.} \quad (200)$$

The 256-epoch demo therefore uses 507,904 payload bytes. A maximum chunk of 65,536 epochs uses 124 MiB before a separate 64 MiB planning margin. These figures do not count the entire driver/context or all host copies.

Admission uses the actual free/total bytes, the user budget and reserve. The defaults are a 512 MiB budget and 1536 MiB total-device reserve. After checking that each subtraction is valid, capacity is bounded by

$$B_{\text{usable}} = \min(B_{\text{user}}, 0.7F, T - R_{\text{reserve}}) - 64 \text{ MiB.} \quad (201)$$

The integer implementation rounds the free-memory fraction downward, caps each chunk at 65,536 epochs and checks texture/index limits. Another process may consume memory after probing, so actual allocation errors still propagate.

The package does not fill all 12 GB to demonstrate activity. Useful payload depends on the batch; automatically leaving space for other laptop processes is a resource plan, not a claim of permanent texture-cache residency. No operating-system ring-0 driver or executable-code-in-texture mechanism is part of the CUDA interface.

## 41 Actual input and result files

The executable reads two CSV files with exact headers. The source-to-domain model is recorded separately in `profile.json`; prepared real data must supply its own correction/product provenance.

```
epochs.csv:
epoch_id,t_rx_gpst_s,x0_m,y0_m,z0_m,b0_m,
asa_mask,na_mask,boundary_mask

observations.csv:
epoch_id,channel,satellite_id,sx_rx_m,sy_rx_m,sz_rx_m,
code_m,add_correction_m,sigma_m,ready
```

The display above wraps the headers for readability; the actual files use one header line each. `docs/INPUT.md` defines their exact names, ranges and physical meaning. The reader rejects malformed headers, duplicate channels, nonfinite fields and invalid sigmas before passing data to a backend.

### 41.1 Results

`solutions.csv` retains one row per input epoch: source/time identity, solver and fit status, iteration count, channel count, all four ASA fields, position/clock, formal diagonal covariance and residual statistics. `residuals.csv` records each active post-fit channel when the solver converges.

For a failed solve, estimate fields may contain the starting guess or a partial iterate. The status is authoritative; those fields must not be published as a valid fix. Zero variances in an unevaluated result do not mean a perfectly known position. The parser and status record prevent that interpretation without hiding the original calculation state.

`run.json` records the backend, timing scope, selected numerical settings and actual device report where available. `ugts_events.jsonl` and `ugts_summary.json` are produced by the explicit post-processing step. They do not replace the original solution or residual files.

A new output directory is required for each run. Completed runs can be compared without silently overwriting earlier evidence. Corruption of a source file or a changed result is visible in the delivered integrity manifest or retained event chain.

## 42 Ordered coupled records and source lineage

A CGK-R1 event binds the original solution and residuals, modeled position and velocity, executed J/K transition, complete native trace row, both compact keys, full time and prior digest. A condensed structural sketch is:

```
{
  "version": "3.6.1.7", "profile": "CGK-R1",
  "trajectory_id": 0, "epoch_id": 0, "input_index": 0,
  "event": "COUPLED_OBSERVATION_ADVANCED",
  "observation": {"solver_record": {}, "epoch_record": {}},
  "residual_records": [], "formal_variance_m2": {},
  "modeled": {"enu_m": [], "ecef_m": [],
    "velocity_enu_m_s": [], "q": 0, "hoop_phi_rad": 0},
  "jk": {"before": 0, "j": 0, "k": 0, "after": 0},
  "native_state": {"complete_executed_trace_row": "..."},
  "chart": {"up_m": 0, "time_gpst_s": 0, "winding": 0},
  "packing": {"contiguous_hex": "...", "morton_hex": "..."},
  "lineage": {"sources": {}, "native": {}},
  "previous_hash": "...", "record_hash": "..."}
}
```

This is a structural sketch; the delivered exports contain full numeric arrays, source records, statuses and hashes. Unavailable observations produce explicit prediction events; a failed mechanical advance produces a failure event. The modeled state is never presented as the original receiver measurement.

The chain uses a domain-tagged SHA-256 of canonical local JSON and retains the prior head. It binds the emitted sequence to a retained reference. It is not a signature, a satellite-signal authentication primitive or proof that an upstream observation was physically true.

Author attribution is included as supplied: Tom Klootwijk, with the identifier and date in the release metadata. A full profile/source digest remains separate from the 64-bit coordinate key or a one-bit event field. This preserves the base engine's identity/state distinction and U's requirement that uncertainty and lineage survive packing. [A, retained M2 project contract] [U, pp.8–10]

### 43 The retained deterministic satnav batch

The shipped demo has 256 artificial receiver epochs. The anchor is a declared WGS84 location at 52° latitude, 5° longitude and 100 m ellipsoidal height. Receiver points are generated from a local ENU path, with a known clock bias near 75,000 m. Satellite directions and ranges are constructed fixtures, not a downloaded constellation or an observation session from the user's laptop.

The first 128 cases are noiseless model data. The next ordinary cases add independent Gaussian fixture noise with their specified sigmas. The final six cases deliberately exercise failure and edge conditions. Corrections are labelled artificial additive test values, not an atmosphere or ephemeris model.

| Outcome                                 | Observed count / result |
|-----------------------------------------|-------------------------|
| Converged numerical epochs              | 253 of 256              |
| Too few selected satellites             | 1                       |
| Whole-word absorbed                     | 1                       |
| Rank-deficient geometry                 | 1                       |
| Converged within chosen residual budget | 251                     |
| Gross-outlier residual alert            | 1                       |
| Exactly four, no residual redundancy    | 1                       |

The exact-four and 32-channel cases verify those interface boundaries. The outlier case adds an artificial 200 m error to one code observation. It converges to an incorrect candidate with a large residual alert, demonstrating why convergence and navigation reliability are not the same field.

The earlier observation-only geometry-support adapter emits 251 verified candidate records, three no-solution records and two unresolved candidates under its declared conditions. The exact-four case remains unresolved because no redundancy is present; the outlier remains unresolved because the chosen residual criteria are exceeded. CGK-R1 instead advances 253 observation-driven and three prediction-only states; residual classifications remain diagnostics beside the original solutions.

These are finite results of a known model. The demo is useful for reviewing signs, units, frame adapters and implementation agreement. It is not a real-world accuracy measurement or a GPU speed result.

## 44 Independent results and error measures

The C++ solve is compared against a separately written Python Householder solver. The comparison checks epoch identity, all mask fields, status/fit, position and clock, formal covariance and residual statistics, yielding **4,828 checked comparisons** over the 256 epochs.

| Cross-reference quantity              | Observed maximum                          |
|---------------------------------------|-------------------------------------------|
| Absolute state-component difference   | $1.210719347 \times 10^{-8} \text{ m}$    |
| Formal covariance diagonal difference | $6.394884622 \times 10^{-14} \text{ m}^2$ |

The comparison tolerance for each state component is  $10^{-4} \text{ m}$ . Small algorithmic agreement does not make the underlying physical inputs exact; it shows that the two implementations solve the same declared finite problem to closely matching values.

| Fixture class | Epochs | 3D position RMSE                | Maximum error                   |
|---------------|--------|---------------------------------|---------------------------------|
| Noiseless     | 128    | $3.61 \times 10^{-9} \text{ m}$ | $7.84 \times 10^{-9} \text{ m}$ |
| Noisy         | 122    | 1.6017 m                        | 3.3883 m                        |
| Gross outlier | 1      | 147.0987 m                      | 147.0987 m                      |
| Exactly four  | 1      | $1.23 \times 10^{-8} \text{ m}$ | $1.23 \times 10^{-8} \text{ m}$ |
| 32 channels   | 1      | $5.71 \times 10^{-9} \text{ m}$ | $5.71 \times 10^{-9} \text{ m}$ |

Errors above are measured only against the synthetic generating truth. The nanometre- scale noiseless figures reflect arithmetic roundoff in matched artificial data, not an achievable GNSS hardware accuracy. The noisy fixture demonstrates the selected noise model; a real receiver's error includes its actual correction/model limitations.

Formal covariance, predicted geometry quality and empirical truth error are different quantities. The package keeps them separate, following the distinction made in the positioning-error reference. [G5]

## 45 Prior release verification record

This section retains the 3.6.1.5 preparation record. Archived 3.6.1.6 evidence is given in the SRK-R1 validation section; historical 3.6.1.7 checks appear in the coupled implementation validation section. Retained 3.6.1.8 evidence appears in Section 9.9. The following unexecuted CUDA statements refer to the prior preparation environment only.

| Category                                | Recorded result                        |
|-----------------------------------------|----------------------------------------|
| CPU C++ build and CTest                 | Passed                                 |
| C++ explicit checks                     | 105,559 passed                         |
| Python UGTS/geodesy tests               | 18 passed                              |
| Command-line scenarios                  | 11 passed                              |
| Independent solver comparisons          | 4,828 passed                           |
| CPU Address/UndefinedBehavior Sanitizer | CTest passed                           |
| CUDA compilation                        | not_run: nvcc unavailable              |
| CUDA execution / memory sanitizer       | not_run: no accessible NVIDIA GPU      |
| Windows native build                    | not_run in preparation                 |
| Live GNSS observation validation        | Not performed; synthetic fixtures only |

The C++ count includes all 65,536 possible 16-bit population-count cases and 10,000 independent random word-mask comparisons. It also checks the full JK truth table, QR identities, covariance, rank rejection and the retained whole-word rule. A large assertion count is not a count of separate receivers tested.

The Python suite checks both key codecs against the source's golden pair and 5,000 random tuples, all prefix lengths for the reference key, quantization bounds, full winding, two separate topologies, cone/sweep relations, the jitter-interval distinction, geodetic/ENU round trips and literal OTAN2 semantics.

The CLI scenarios include malformed IDs/channels, negative sigma, NaN input, invalid back-end/options, existing-output protection, a normal run and an iteration-limit run. These prior outcomes remain in the preserved sibling releases, with earlier source test claims kept separate from these actual executions.

In the prior preparation, the device validation script was invoked and recorded its unavailable-tool result. Later releases supply separate actual laptop execution records. No general CUDA speedup or texture-residency claim is made in this edition.

## 46 Build, run and verify on the requested laptop

Use a CUDA-supported host compiler, CMake 3.24 or newer, and CUDA Toolkit 12.8 or newer for the requested SM120 target. The installed NVIDIA driver must support the chosen runtime/-toolchain. This specifies a minimum architecture-capable toolkit, not a claim that 12.8 is the newest available release. [H2, H3]

```
cmake -S . -B build_gpu -DSATNAV_ENABLE_CUDA=ON \  
  -DCMAKE_BUILD_TYPE=Release  
cmake --build build_gpu --config Release --parallel 2  
ctest --test-dir build_gpu -C Release --output-on-failure  
./build_gpu/satnav --probe  
./build_gpu/satnav --input examples/demo --out run_gpu \  
  --backend cuda --read texture --verify  
python tools/verify_run.py --input examples/demo --run run_gpu  
python tools/ugts_handoff.py --input examples/demo --run run_gpu
```

On a Windows multi-configuration build, use the following executable path:

```
build_gpu/Release/satnav.exe
```

Run configuration from the matching x64 compiler developer terminal so that CMake can locate the host toolchain.

### 46.1 Complete device comparison

```
python tools/validate_gpu.py --sanitizer
```

The script records compilation, device probing, texture/global runs, shared CPU comparisons, independent Python comparisons and requested memcheck results. It uses fresh run directories. Missing required tooling produces a separate unavailable status and exit code 3. A real build/test error remains a failed status.

### 46.2 CPU-only review

```
cmake -S . -B build_cpu -DCMAKE_BUILD_TYPE=Release  
cmake --build build_cpu --config Release --parallel 2  
./build_cpu/satnav --input examples/demo --out run_cpu  
python tools/verify_run.py --input examples/demo --run run_cpu  
python -m unittest discover -s tests -p "test_*.py" -v
```

The retained SATNAV reference uses the Python standard library. The coupled reference additionally requires NumPy, installed from `requirements.txt`.

## 47 Integration decisions and source issue register

| ID  | Decision                        | Effect                                                                                                                                                               |
|-----|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N01 | Corrected-code satnav model     | The GNSS observation equation and solver are explicit new additions, not attributed to U.                                                                            |
| N02 | Full receiver state retained    | Solve in 3D ECEF plus clock; use a separate local log chart and retain Up and provenance.                                                                            |
| N03 | Independent QR methods          | Shared CPU/CUDA Givens implementation and independent Python Householder solve; named rank and convergence policy.                                                   |
| N04 | Fixed channel-to-bit mapping    | One epoch is one at-most-32-channel word. Boundary absorption stays whole-word.                                                                                      |
| N05 | Source key layouts preserved    | Exact contiguous/Morton pair retained; quantization ties named; key does not replace full state.                                                                     |
| N06 | Two jitter intervals separated  | Explicit source equality issue documented; perturbation-family and measurement-centred intervals are not silently equated.                                           |
| N07 | Bounded sweep interpretation    | Translation-only analytic interval; floating arithmetic and nonnested grids not promoted to stronger certificates.                                                   |
| N08 | Topology does not rewrite Earth | Bundle/reflection helpers remain separate local contracts, not GNSS propagation rules.                                                                               |
| N09 | Computational event semantics   | Declared support/residual policies produce observation records, not a protection level, spoof-proof fix or actuator command.                                         |
| N10 | Recorded hardware evidence      | CUDA was unexecuted in 3.6.1.5 preparation; current device builds, replay and memcheck results are recorded separately.                                              |
| N11 | Explicit literal feedback       | SRK-R1 composes ASA/NA and JK with supplied Boolean expressions and synchronous state carry.                                                                         |
| N12 | Complete geometric coupling     | CGK-R1 defines circle-plus, predicate encoders, state-dependent mechanics and eigenmatrix; the actual handoff executes and publishes the coupled state.              |
| N13 | Live receiver positioning       | LIVE-GPS-L1-R1 decodes real GPS observations and streamed ephemerides, applies explicit physical corrections and solves the remote receiver position during capture. |

These decisions define the explicit profiles of aTOMos 3.6.1 with the retained UGTS prototype integrated at explicit interfaces. They do not discard the user's operator vocabulary or pretend that all missing GNSS physics were already encoded in a name.

A later variant can extend the estimator with new states, observable types or correction adapters. It must identify the changed observation model and preserve a projection or compatibility test to this profile where appropriate. An unreported inter-system bias, new residual exclusion rule or changed topology is not a transparent optimization.

## 48 Source and artifact inventory

**U.** *UGTS\_KC\_3\_6\_2\_SCLP\_Tom\_Klootwijk(1).pdf*. The file has 19 physical pages and a 16-page numbered main/appendix sequence. The source’s referenced 13-page precursor and software ZIP were not supplied. The inherited source manifest records its PDF SHA-256 as

```
6987490a0a7b7ed76b0091057a2e69ab831fee786434a0b4219782923f5ea580
```

The U page references in this formalization point to the printed main numbers. For example, its key table on printed p.8 is physical PDF page 11. The relevant figure, table and jitter-equation pages were inspected as rendered pages as well as text.

**A.** aTOMos v3.6.1 / M2 source and code contract in this project conversation. The prior archive was not mounted in this turn; the specialization preserves the written ASA/JK/OTAN2 contracts rather than claiming a binary patch to an unavailable artifact. The actual new source is included and tested to the extent recorded above.

| Delivered file family                        | Purpose                                                                                                                                                                                           |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| include/, src/, cuda/<br>python/             | Shared solver, executable and real CUDA backend source.<br>Independent solution and source-derived UGTS/geodesy helpers.                                                                          |
| examples/demo/<br>tests/, results/<br>tools/ | Prepared artificial epochs, observations, truth and profile.<br>Reproducible checks and the actual recorded outcomes.<br>Build/review, independent comparison, event adapter and integrity tools. |
| docs/                                        | This editable formalization, input contract and source-decision register.                                                                                                                         |
| source/manifest.json<br>SHA256SUMS.txt       | Exact source availability and fingerprint record.<br>Delivered-byte integrity manifest.                                                                                                           |

Source font files and proprietary binaries are not distributed. The current uploaded UGTS reference may be retained separately for source comparison; its equations and selected mechanisms are integrated into this volume instead of concatenating unrelated older transcripts into the kernel package.

## Primary implementation references

Recorded by the 3.6.1.5 preparation, 14 September 2026. These external references support the satnav model and hardware contract; they are separate from the user's source corpus.

- G1 ESA Navipedia, J. Sanz Subirana, J.M. Juan Zornoza and M. Hernández-Pajares, [Code Based Positioning \(SPS\)](#). Four-state model, correction terms, linearized geometry and iterative position update.
- G2 ESA Navipedia, [Satellite Coordinates Computation](#). Emission-time satellite state and reception-time frame treatment.
- G3 ESA Navipedia, [Transformations between ECEF and ENU coordinates](#). Local frame conversion and angular conventions.
- G4 ESA Navipedia, [Parameters adjustment](#). Linearized residual and least-squares framework.
- G5 ESA Navipedia, [Positioning Error](#). Formal covariance, predicted geometry and measured error distinguished.
- G6 NGA Office of Geomatics, [World Geodetic System 1984](#). Ellipsoid/reference-system definition and constants used in the coordinate utility.
- H1 NVIDIA, [GeForce RTX 50 Series Laptops](#). RTX 5070 Ti Laptop memory specification.
- H2 NVIDIA, [CUDA GPU Compute Capability](#). GeForce RTX 5070 Ti family listed at compute capability 12.0.
- H3 NVIDIA, [CUDA 12.8 Features Archive](#). Introduction of SM\_120 compiler support.
- H4 NVIDIA, [CUDA Runtime API: Texture Object Management](#). Linear resource and element-read contract.